



COFFEE MAKER/WATER BOILER ASSEMBLY

**PART NUMBER
3510-0052-01**

COMPONENT MAINTENANCE MANUAL WITH ILLUSTRATED PARTS LIST

ORIGINALLY ISSUED 15 FEBRUARY 1993

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Rev 002

ALERT

THIS COMPONENT MAINTENANCE MANUAL IDENTIFIES REPLACEMENT PARTS AND REPAIR PROCEDURES SPECIFICALLY ENGINEERED FOR BEVERAGE MAKER PRODUCTS MANUFACTURED BY B/E AEROSPACE, INC., COMMERCIAL AIRCRAFT PRODUCTS GROUP. B/E PARTS ARE MANUFACTURED UNDER STRINGENT QUALITY CONTROL PROCESSES. USING NON-B/E PARTS MAY JEOPARDIZE YOUR WARRANTY AND RESULT IN DEGRADED PRODUCT PERFORMANCE. ALL MODIFICATIONS SHOULD BE COORDINATED IN ADVANCE TO AVOID THIS POSSIBILITY.

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REVISION NUMBER 002 DATED 07 MAY 2013

HIGHLIGHTS

This Component Maintenance Manual (CMM) with Illustrated Parts List (IPL) has been reviewed for technical changes resulting from B/E Service Bulletins, B/E Service Information letters, configuration changes and part improvements. Any existing Temporary Revisions (TRs) against this CMM have been incorporated with this revision.

This CMM and IPL have been completely revised. Please replace the old copies of this manual with this Revision Number 002 dated 07 May 2013. A description of the changes follows.

SECTION AND PAGES	CHANGE DESCRIPTION
Title Page, T-1	Updated to reflect current revision.
Alert Page, A-1	Added alert page to reflect current revision.
Record of Revisions, ROR-1	Updated Record of Revisions to reflect current revision.
Record of Temporary Revisions, RTR-1	Updated to reflect current revision.
Service Bulletin List, SBL-1	Updated to reflect current revision.
Service Information Letter List, SIL-1	Updated to reflect current revision.
List of Effective Pages, LEP-1 and LEP-2	Updated List of Effective Pages to reflect current revision.
Table of Contents, TOC-1	Updated Table of Contents to reflect current revision.
Introduction, INTRO-1 and INTRO-2	Updated to reflect current maintenance concept.
Description and Operation, 1 thru 4	Updated to reflect current revision.
Testing and Fault Isolation, 1001 thru 1016	Updated to reflect current maintenance concept.
Schematics and Wiring Diagrams, 2001 thru 2006	Updated to reflect current maintenance concept.
Disassembly, 3001 thru 3008	Updated to reflect correct disassembly order and item numbers to reflect IPL changes in Revision Number 002.
Cleaning, 4001 and 4002	Updated to reflect current revision.
Inspection/Check, 5001 thru 5004	Updated to reflect current revision.

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SECTION AND PAGES	CHANGE DESCRIPTION
Repair, 6001 and 6002	Updated to reflect current revision.
Assembly, 7001 thru 7008	Updated to reflect correct assembly order and item numbers to reflect IPL changes in Revision Number 002.
Special Tools, Fixtures, and Equipment, 9001 thru 9004	Updated to reflect current revision.
Illustrated Parts List, 10001 thru 10066	Added procedures, alpha/numeric index, illustration, and detailed parts list to reflect current revision.

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RECORD OF TEMPORARY REVISIONS

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SERVICE BULLETIN LIST

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INTRODUCTION

1. General

- A. This manual has been prepared to fulfill the requirements of ATA iSpec 2200. It is intended to provide such data as is necessary for an approved repairer to return an unserviceable coffee maker/water boiler PN 3510-0052-01 to a serviceable condition.
- B. All procedures established in this manual have been performed and verified acceptable by trained personnel at B/E Aerospace, Inc. A competent shop mechanic who is unfamiliar with these units should, with a minimum amount of effort and tools, be able to isolate the problems of a nonfunctional unit and restore it to service.
- C. Procedures are provided for the full range of maintenance and include all the work functions that could or may be done to the unit throughout its entire service life.
- D. A competent shop mechanic should use this manual first to test and isolate a faulty component. The mechanic should then go directly to the applicable manual subject to effect a repair in the specific problem area without disturbing other areas.

2. Abbreviations

The following abbreviations may be used in this manual:

ABBREVIATION	MEANING
AC	Alternating Current
ASSY	Assembly
° C	Degree Celsius
CAGE	Commercial and Government Entity
CCW	Counter Clockwise
CW	Clockwise
CM	Centimeter
EFF	Effectivity
° F	Degree Fahrenheit
FIG	Figure
HZ	Hertz
IN	Inch
IPL	Illustrated Parts List
KPAG	Kilo Pascals Gauge

ABBREVIATION	MEANING
LBS	Pound
LG	Long
LKG	Locking
NHA	Next Higher Assembly
NO	Number
PSIG	Pounds Per Square Inch Gauge
QTY	Quantity
RF	Reference
SB	Service Bulletin
TPN	True Part Number
TTL REQ	Total Required
V	Vendor
VAC	Volt(s) Alternating Current

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DESCRIPTION AND OPERATION

1. Description

- A. Coffee Maker/Water Boiler PN 3510-0052-01 is manufactured at B/E Aerospace, Inc., Commercial Aircraft Products Group in Lenexa, Kansas.
- B. Coffee Maker/Water Boiler part number PN 3510-0052-01 is designed for use in aircraft galleys (refer to Figure 1). The coffee maker/water boiler is easily removable from the permanently installed base assembly by a single release pin. The coffee maker/water boiler is of modular design, utilizing cast aluminum-magnesium construction for maintainability and reliability. The front casting serves as a framework for the housing platen, and electrical controls. The rear casting serves as a framework for the housing, hot water tank, plumbing controls, and provides close dimensional controls for accurately locating automatic disconnects to the aircraft electrical system, water supply and drain.
- C. The coffee maker/water boiler uses soft packaged coffee (pillow pack) placed in a cup assembly which serves as the brew chamber. Heated water sprays into the pillow pack. The brewing takes place as the hot water soaks through the coffee. The brewed coffee then flows down into the server below. The grounds are retained within the pillow pack. When brewing is completed, the cup assembly can be withdrawn and the old pillow pack discarded.
- D. The water is heated in a single tank by three 900 watt immersion type heaters mounted in the bottom of the tank.
- E. A 35 watt hot plate, mounted in the base of the coffee maker/water boiler, will maintain brewed coffee or tea at 170° F (76° C) to 190° F (87° C) for 30 minutes.
- F. Temperature of brewed coffee or tea in the server will be 165° F (74° C) to 185° F (85° C) from initial brew and 175° F (79° C) to 195° F (90° C) from subsequent brews with aircraft water supply at 62° F (16° C) to 72° F (22° C).
- G. Manual control of coffee maker/water boiler functions is provided by a set of controls on the front panel. These controls are push button type with LED indicators.

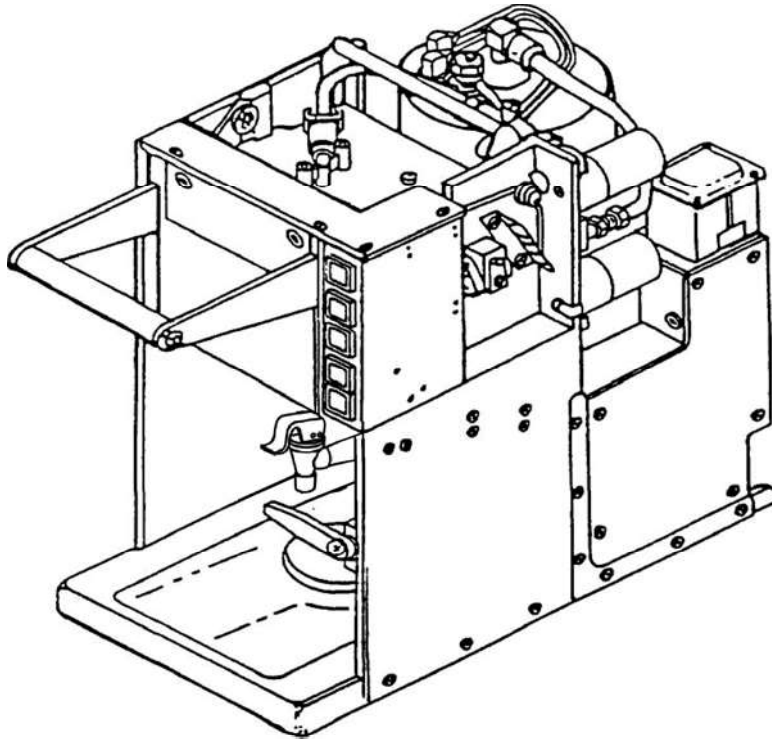


Figure 1
Coffee Maker/Water Boiler Assembly

2. Operation

NOTE: The server must be in place and the brew handle in the down position to activate the coffee maker/water boiler BREW switch S2.

- A. 115VAC, 400 Hz, three phase potential is required at P1, Pins B, C, and D for operation of the coffee maker/water boiler. Phase A is applied to transformer T1 on PCB 3530-0059-05 through J5, F1, P3-10. S1, P3-8 and to primary of T1 when S1 is energized. 23 VAC from the secondary of T1 is applied to CR2 through CR5, bridge rectifier, on PCB 61-1675-11 through J2-B and D. The bridge rectifier converts the 23 VAC to 23 VDC, unregulated and this DC potential is then applied to J2-H. The 23 VDC, unregulated is then applied from J2-H to Q1 on PCB 3530-0059-05 which provides two regulated outputs to be used by electrical components on the coffee maker/water boiler. The first output, P4-11, lights the ON/OFF switch S1 indicator. The second output supplies 23 VDC to the solenoid valve assembly and to heater relay K1.
- B. When ON/OFF switch S1 is in the ON position, 23 VDC is applied to the brew solenoid assembly, to heater relay K1, and to the logic circuits of printed circuit board assembly 61-1675-11. If the water level in the tank is low, the NO WATER indicator will illuminate until the tank is filled.
- C. When power is applied to the logic circuits, with the brew handle down and the server in place, the brew latch may be armed by momentarily depressing the BREW switch S2.

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- D. When the temperature in the water tank reaches the set point (approximately 195° F (90° C)), the brew latch is completed, powering the brew solenoid assembly open and allowing water to flow through the brew head. Also, at this time, the TEA light will illuminate and remain on as long as water temperature is 195° F (90° C). The water flow will continue until the liquid level reaches the tip of the liquid level sensor assembly U1, or the server is removed. Completing a circuit across the liquid level sensor tip, either by liquid or other means, will cause the logic circuits to reset, closing off the brew solenoid. A time delay causes the BREW light to remain on for approximately 15 ±5 seconds after resetting the logic circuits. The TEA cycle may also be reset by cycling the ON/OFF switch S1 through the OFF position.
- E. Power may be applied to platen heater HR4 by placing both the ON/OFF switch S1 and the HOT PLATE ON switch S4 in the ON position. With the server in place, heater HR4 will maintain coffee temperature at 170° F (76° C) to 190° F (87° C) for approximately 30 minutes.
- F. In the event of failures disabling the electronic safety features, control circuits prevent coffee maker/water boiler operation and opens fuse F1 on PCB 3510-0059-05 under the following conditions: Internal tank temperature below 32° F (0° C), internal tank temperature above 235° F (113° C).
- G. Additional safety features provided in the coffee maker/water boiler design includes a resettable thermal switch, pressure relief valve, low water sensing, BREW light time delay and hot water flow stoppage when the brew handle is lifted.

3. Operating Instructions

A. Brew Coffee

- (1) Press ON/OFF switch.
- (2) Place empty server in coffee maker/water boiler.
- (3) Place coffee pack in cup, seam down. Install on cup shelf under brew head.
- (4) Pull brew handle down to locked position.
- (5) Press COFFEE BREW switch.
- (6) When COFFEE BREW light goes out coffee is ready.
- (7) At the completion of the brew cycle, raise the brew handle and remove the brew cup and dispose used coffee grounds.
- (8) Return the brew cup and server to the coffee maker/water boiler. Pull brew handle to the locked position to retain the cup.

B. Brew Tea

- (1) Press ON/OFF switch.
- (2) Place tea bags in server and place server in coffee maker/water boiler.
- (3) Place empty brew cup on shelf under brew head.
- (4) Pull brew handle down to locked position.

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- (5) Press TEA switch.
- (6) When the TEA light goes out the tea is ready.

4. Leading Particulars

A. Refer to Table 1 for a list of leading particulars on the Coffee Maker/Water Boiler Assembly.

Table 1
Leading Particulars

ITEM	CHARACTERISTIC
Operating Voltage	115/200 VAC 400 Hz 3 Phase
Weight	18 lbs (8.2 kg)
Wattage	2760 watts total
Phase A	960 watts
Phase B	900 watts
Phase C	900 watts
Dimensions (Maximum)	
Length (handle up)	17.12 in (43.48 cm)
Width	6.31 in (16.02 cm)
Height	12.00 in (30.48 cm)
Recovery Time (Maximum)	
Brew cycle	2 min 5 sec $\pm$ 25 sec
Total time (repeat brew cycle)	3 min, 45 sec

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TESTING AND FAULT ISOLATION

1. General

- A. TESTING AND FAULT ISOLATION section covers the test equipment, bench setup, functional checkout procedure and fault isolation procedures required to return an inoperative Coffee Maker/Water Boiler to service.
- B. Completely read and thoroughly understand the entire TESTING AND FAULT ISOLATION section. Become thoroughly familiar with the test sequences and their interactive correlations before performing any tests.
- C. The test equipment and materials required for testing of the Coffee Maker/Water Boiler are listed in Table 1001.

NOTE: Equivalent substitutes may be used for listed items.

Table 1001
Test Equipment and Materials

NOMENCLATURE	PARTS/SPEC NO	SOURCE (CAGE)
Card Extension Adapter	60-2344-1	Commercially Available
Circuit Breaker, 2 Amps	None	Commercially Available
Digital Thermometer and Thermocouple	None	Commercially Available
Jumper for J5 and J6 tied together	None	Commercially Available
Megohmmeter	Associated Research Model 201	Commercially Available
Milliohmmer	QuadTech Model 1880	Commercially Available
Rail Assembly	416-0201-15	B/E Aerospace, Inc. (V63367)
Resistance Bridge	None	Commercially Available
Server	PS-3374	Commercially Available
Screw Aluminum Adapter	None	Commercially Available
Stop Watch	None	Commercially Available
Test bench, consisting of flow meter, adjustable water supply with pressure gauge, power supply, 3 ammeters and voltmeter.	None	Commercially Available
Test Harness	None	Commercially Available
Water hose with thermocouple	None	Commercially Available
10K Ohm Potentiometer, 1 Watt minimum	3701H-1-103	Bourns Instruments Inc. (V32997)

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- D. Figure 1001 defines a standard coffee maker/water boiler test bench configuration. This is the minimum test setup required to functionally check a coffee maker/water boiler. All test and fault isolation procedures assume this setup and corresponding data availability.

NOTE: The water should be regulated (while flowing) to 26 psig.

2. Testing (refer to Figures 1001 thru 1004)

NOTE: When testing a unit, each test must be performed and the procedures must be followed in the sequence listed below.

NOTE: If any of the tests fail, refer to Fault Isolation Table 1003.

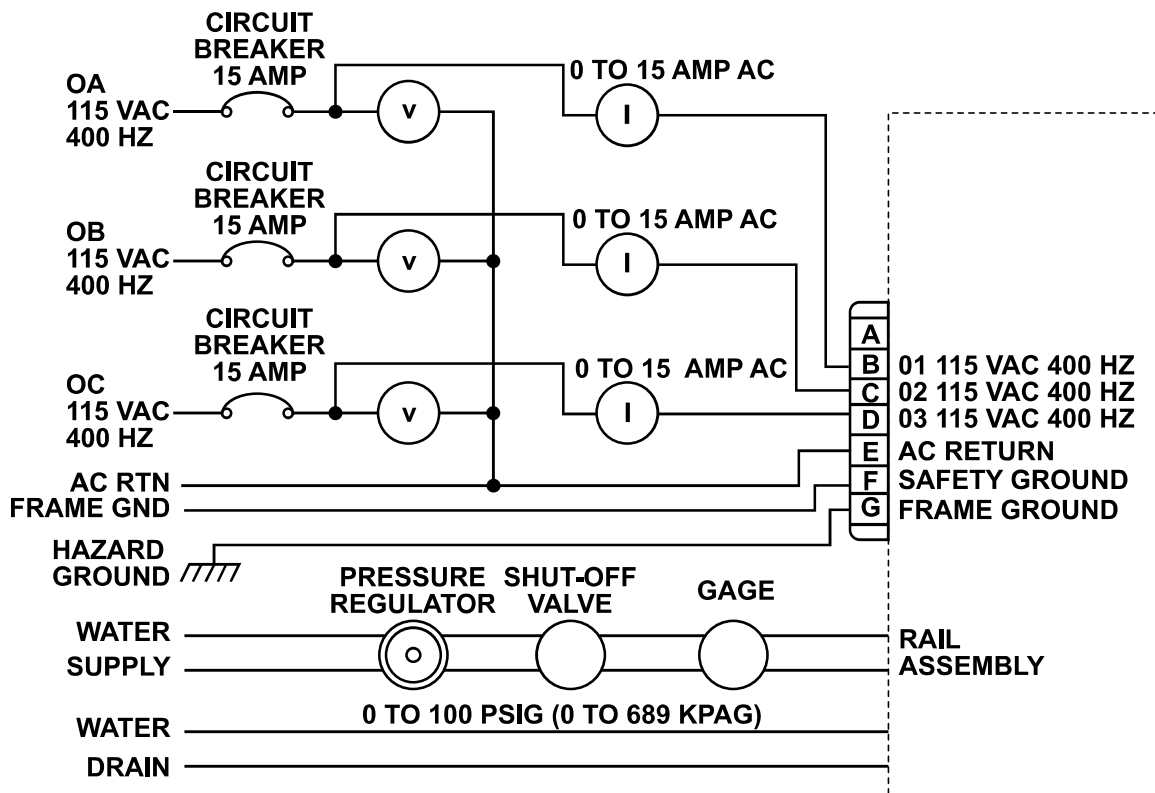


Figure 1001
Test Set-up

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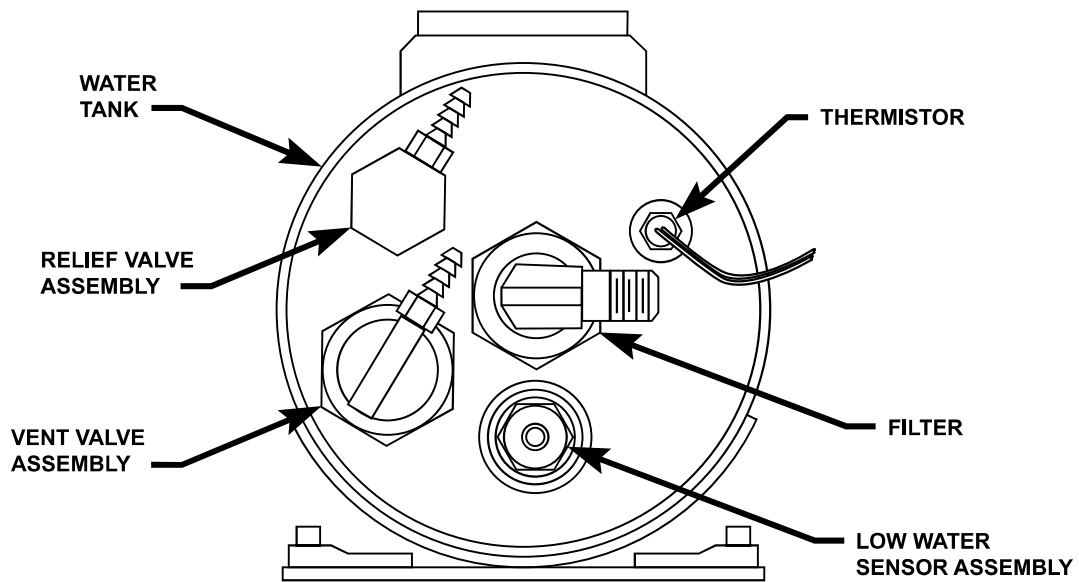


Figure 1002
Top End Configuration

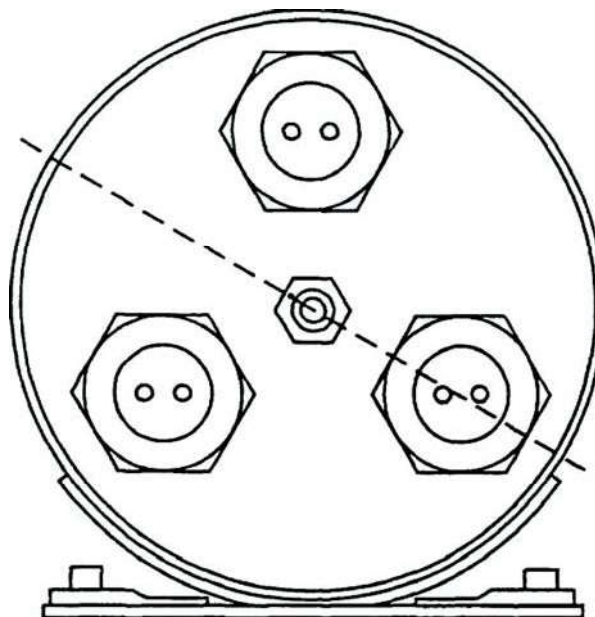


Figure 1003
Bottom End Cap Configuration

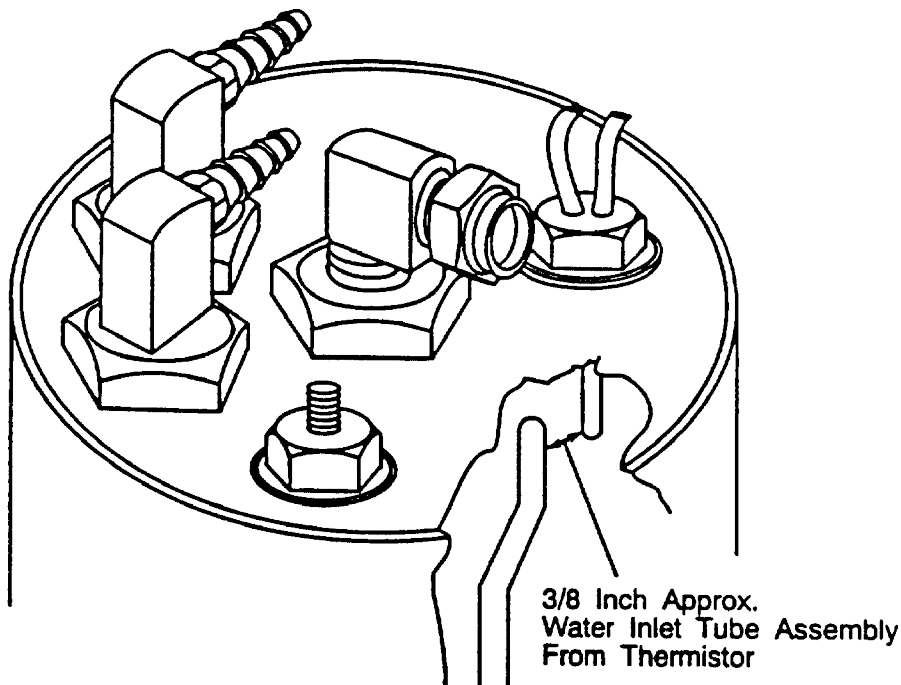


Figure 1004
Tube Assembly Alignment

A. Dielectric Test (To be done only when a complete rewire of unit is done)

NOTE: Some components of the water boiler have had separate dielectric tests performed by the vendor, as noted on their stock code sheets or specification control drawings. These components are

Solenoids (S8 and S9)

Switches (S1 through S4)

Transformer (T1)

Heaters (HR1 through HR4)

Printed circuit boards 3530-0059-05 and 61-1675-11

Some of these items may be retested during the coffee maker/water boiler dielectric test.

- (1) Remove printed circuit boards 3510-0059-05 and 61-1675-11.
- (2) Insert jumper plugs J3 and J4 into P3 and P4 respectively and short the pins on J3 and J4 together, except J4, pin 12, jumper J5 to J6.
- (3) On the power module assembly, remove relay K1 from the relay socket. Jumper relay K1 socket, pins 1 and 3, 5 and 7, 9 and 11.
- (4) On the tank assembly, disconnect U1-1 (+) at CP1. Insulate wire and tie back.

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- (5) Place brew handle up. Remove switch caps and lamps from S1, S2, S3, S4, and DS1 and make sure switches are off.
- (6) Tie pins B, C, D1 and E of P1 together. Connect the Hipot test set between the terminating point of P1, pins B, C, D, and E, and the coffee maker/water boiler housing. Increase the voltage of the Hipot test set in increments of 250 to 500 volts per second until 1500 volts are applied across test connection. Maintain the voltage at the 1500 volt level for 60 seconds. Current flow shall not exceed a maximum of 2.0 milliamperes during the test period.
- (7) Remove Hipot test set and connect megohmmeter between the terminating point of P1, pins B, C, D, and E and the coffee maker/water boiler housing. The megohmmeter indication shall be greater than 200 megohms.
- (8) Remove test equipment and jumpers from P1, B, C, D, and E.
- (9) On the solenoid valve assembly, attach connector CP1 of the liquid level sensor. On power module assembly (K1), remove jumper from relay socket and install relay K1. Remove jumper plugs J3 and J4, and J5 to J6. Install printed circuit boards 3530-0059-05 and 61-1675-11 and insert connectors.

B. Insulation check (Resistance Check)

- (1) Connect a resistance bridge between P1-G and unanodized surface of brew cup shelf. The measured resistance shall not be greater than 1.7 milliohms.
- (2) Connect a resistance bridge between P1-G and front of platen. The measured resistance shall not be greater than 2.5 milliohms.
- (3) Connect a resistance bridge between P1-G and ground stud SG2 in the power module assembly. The measured resistance shall not be greater than 1.4 milliohms.
- (4) Remove test equipment.

C. Test Equipment Interface

- (1) Connect the coffee maker/water boiler to an external 115/200 VAC, 400 Hz, 3-phase source.
- (2) Connect ammeters in each phase of the 3-phase AC source.
- (3) Connect an external water supply using a flowmeter, regulator, and cutoff valve. Adjust the water supply pressure to 24 psig, +5/-0 psig.
- (4) Place server in coffee maker/water boiler.

D. Leak Test

- (1) Make sure that the coffee maker/water boiler is configured as in para 2C(3).
- (2) Remove the fitting from relief valve assembly (refer to IPL Figure 6) to permit access to adjusting cap; close or tighten adjusting cap.
- (3) Apply 24 psig water pressure input to the coffee maker/water boiler and allow the tank to fill normally.

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- (4) Increase the water pressure to the coffee maker/water boiler to 125 psig and allow unit to remain pressurized for five minutes. The water boiler shall not leak when subjected to 125 psig water pressure applied for five minutes.
- (5) Decrease the coffee maker/water boiler water input pressure to 80 psig. Open the adjusting cap assembly until the relief valve begins to operate. Return the water pressure to 100 psig. The relief valve shall begin to open at 80 psig minimum and fully open at 100 psig maximum.
- (6) Disconnect the water supply and drain unit. The vent valve shall operate properly.

NOTE: The vent valve allows the tank to fill completely and also allows draining the tank when required. If these two functions are satisfactory, the vent valve is functioning properly.

E. Water Supply Temperature Test

- (1) If connected, disconnect coffee maker/water boiler water supply.
- (2) Using thermocouple and thermometer, measure the water supply temperature. The temperature of the water shall be between 62° F (17° C) minimum and 72° F (22° C).

NOTE: The temperature established in this paragraph 2.E.(2) is the norm for the remaining paragraphs of the testing procedure.

F. ON/OFF and NO WATER Lamp Test

- (1) Use the test harness to connect a power supply to the power module assembly using the test harness.
- (2) Connect test equipment as in paragraphs 2.C.(1) and 2.C.(2).
- (3) If connected, remove the water supply to the coffee maker/water boiler.

NOTE: The coffee maker/water boiler tank must be drained before proceeding.

- (4) At the coffee maker/water boiler, depress ON/OFF switch to the ON position. The ON/OFF indicator lamp shall illuminate blue, the NO WATER indicator lamp shall illuminate red, and the heaters shall not activate (low water check).
- (5) Depress coffee maker/water boiler ON/OFF switch to OFF.

G. Coffee Maker/Water Boiler Fill Test

- (1) Connect test equipment as in 2.C.(1) thru 2.C.(4).
- (2) Verify unit fills by observing the flowmeter.
- (3) Depress coffee maker/water boiler ON/OFF switch to ON.
- (4) The NO WATER lamp shall extinguish at the end of the fill cycle and the coffee maker/water boiler shall not leak.
- (5) Depress coffee maker/water boiler ON/OFF switch to OFF.

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H. Preheat Time Test

- (1) Connect test equipment as in 2.C, paragraphs (1) through (4).

NOTE: Preheat time is the elapsed time from heaters ON to heaters OFF.

- (2) Depress coffee maker/water boiler ON/OFF switch to ON.

NOTE: The NO WATER lamp indicator must be OFF during entire test.

- (3) Using a stopwatch, measure the elapsed time between ON/OFF switch to ON and zero amperes current on phase B as indicated on ammeter. The elapsed time shall be 3 minutes, 30 seconds maximum.

J. Heater Operation Test

- (1) Connect test equipment as in 2.C.(1) thru 2.C.(4).

NOTE: If unit has been in use, TEA must be dispensed until heaters cycle on.

- (2) Activate the ON/OFF and HOT PLATE ON switches. With both the hot plate and heaters activated, the coffee maker/water boiler amperes shall be as follows:

Phase A - 8.5 +0.4/ -0.9 amperes

Phase B - 7.8 +0.4/ -0.8 amperes

Phase C - 7.8 +0.4/ -0.8 amperes

- (3) Depress HOT PLATE switch to OFF.
- (4) Place a thermocouple in the server and lower the brew handle.
- (5) Depress BREW switch. The flowmeter shall indicate the beginning of the brew cycle and the BREW lamp shall illuminate white.
- (6) Raise brew handle. The flowmeter shall indicate a brew cycle interrupt.
- (7) Lower the brew handle. The flowmeter shall indicate continuing brew cycle.
- (8) Measure the temperature of the water in the server. The temperature shall be between 165° F (74° C) and 185° F (85° C).
- (9) Allow the coffee maker/water boiler to operate through one brew cycle as indicated on the flowmeter. Using a stopwatch, measure the brew cycle time and light delay time. The cycle time shall be 2 minutes, 5 seconds ± 25 seconds. The delay cycle time shall be 15 ± 5 seconds minimum.

NOTE: The brew cycle time is measured by observing the flowmeter indicating water flow during the brew cycle. The delay time is determined by the flowmeter showing no water flow and BREW lamp extinguishing.

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- (10) Measure the coffee maker/water boiler brew cycle recovery time. The recovery time shall be 4 minutes 55 seconds maximum.

NOTE: The recovery time is elapsed time until the heaters deactivate as observed on the ammeters.

- (11) Remove, empty, and reinstall server.
- (12) Depress BREW switch to initiate a subsequent brew cycle. Measure the temperature if the water in the server. The temperature shall be between 175° F (79° C) and 195° F (90° C).

K. Tea Cycle Test

- (1) Connect test equipment as in 2C, paragraphs (1) through (4).

NOTE: If unit has been in operation, draw two servers of water from faucet to ensure unit is starting from ambient temperature.

- (2) Place a thermocouple in server and lower brew handle.
- (3) Activate the TEA switch to start the tea cycle. The TEA lamp shall illuminate green and the BREW lamp shall be extinguished.
- (4) Raise the brew handle. The TEA cycle shall stop. Lower the brew handle, the TEA cycle shall continue.
- (5) Measure the temperature of the water in the server. The temperature shall be between 165° F (74° C) and 185° F (85° C).
- (6) Remove, empty, and reinstall the server.
- (7) Start a stopwatch and press TEA switch simultaneously. Allow the unit to operate through one complete cycle, including recovery cycle, noting temperature and times as follows:
- (a) Server temperature shall be between 175° F (79° C) and 195° F (90° C).
 - (b) Cycle time shall be 2 minutes, 5 ± 25 seconds.
 - (c) Cycle delay time shall be 15 ± 5 seconds minimum.
 - (d) Recovery cycle shall be 4 minutes, 55 seconds maximum.

NOTE: The cycle time begins with TEA lamp illumination and ends with no water flow on flowmeter. Cycle delay is the time from no flow on flowmeter until TEA lamp extinguishes. Recovery cycle is indicated by TEA lamp extinguishing and a zero current indication on Phase B ammeter.

L. Hot Plate Test

- (1) Connect test equipment as in 2C, paragraphs (1) through (4).
- (2) Activate the ON/OFF and HOT PLATE ON switches. The HOT PLATE ON lamp shall illuminate orange.
- (3) At the coffee maker/water boiler, set the ON/OFF switch to OFF.
- (4) Remove power at the external power source and remove test equipment.

3. Coffee Maker/Water Boiler Alignment and Adjustments

A. Inlet Water Tube Adjustment

- (1) Water temperature must be verified during the brew cycle by the following method, after checking the positioning of the water inlet tube assembly as defined below.
- (2) Drain water from tank assembly.
- (3) Loosen tube assembly (refer to Figure 1002) from elbow on filter assembly. Remove filter assembly for visual inspection from the top of the tank.
- (4) Invert coffee maker/water boiler. Inspect the water inlet tube assembly position as shown in Figure 1003. The fitting must be tight and aligned midway between the heaters.
- (5) Return coffee maker/water boiler to its proper upright position. Verify that upper end of tube assembly is aligned with thermistor and approximately 3/8 inch from the thermistor (refer to Figure 1004)

NOTE: If tube assembly is not in the position as described above, deflect the top of the tube as required.

- (6) After the adjustment of the tube assembly has been made, reinstall the filter assembly and tube assembly. Then perform the following test to confirm specified heater "turn-on-time" and water temperature:
 - (a) Install a water supply which is regulated to $24 \pm 5 / -0$ psig.
 - (b) Position a thermocouple, thermistor, mercury or fast acting dial thermometer in the server so that incoming water will strike directly on the temperature device. If the coffee maker/water boiler has been preheated or in service prior to test, two full containers of water should be drawn from the faucet to ensure the cycle starts with a cold tank of water.
 - (c) With server in position, depress ON/OFF switch to ON and depress BREW switch.
 - (d) Brew cycle will start as soon as water is at correct temperature. During brew cycle, the water temperature measured in the server shall be $185^{\circ} \text{ F } (85^{\circ} \text{ C}) \pm 5^{\circ} \text{ F } (-15^{\circ} \text{ C})$.

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- (e) If server temperature exceeds the above values, further adjustment will have to be made by minor rotation of the water inlet tube assembly. A clockwise rotation (as viewed in Figure 1004) will decrease the server temperature and a counterclockwise rotation will increase temperature. After adjustment repeat paragraphs 3.A.6.(a) through 3.A.6.(d).

B. Circuit Card Assembly Adjustment

NOTE: Adjustments to the circuit card should be made only if components have been replaced or if the water inlet tube assembly adjustment or thermistor replacement fails to accomplish the desired water temperature.

- (1) Anchor the coffee maker/water boiler securely on a work bench. The input power source shall be 115 VAC, 400 Hz, 3-phase with an AC voltmeter and an AC ammeter installed in each phase.
- (2) Install a water supply equipped with a regulator and gauge. The water supply shall be regulated to 24 +5/ -0 psig.
- (3) Remove cover assembly from the power control module (IPL Figure 4) and remove circuit card assembly. Plug card extension adapter (refer to Table 1001) into connector J2 and insert circuit card assembly into the other end of adapter
- (4) Replace thermistor leads at butt splices BS1 and BS2 with a calibrated 10K ohm potentiometer, Bourns part number 3701S-1-103 or equivalent.

CAUTION: DURING REMAINING PARAGRAPHS, CARE SHOULD BE TAKEN NOT TO ALLOW THE HEATER TO REMAIN ON FOR ANY EXTENDED PERIOD OF TIME BECAUSE THE TEMPERATURE IN THE TANK WILL BECOME TOO HIGH.

NOTE: Adjust the reference potentiometer to 10K ohm.

- (5) Apply 115 VAC, 400 Hz power to Phase A only and observe ammeter. If test setup prevents connection of Phase A only, disconnect two heaters to accomplish this test. Current of approximately 10 amperes indicates the coffee maker/water boiler is energized and heater circuit is on.
- (6) Slowly reduce the reference potentiometer to 7.3K ohm. The heater should turn off between 7.5K ohm and 7.3K ohm. If not, adjust potentiometer R44 on the circuit card assembly to turn the heater off within this range.
- (7) Increasing the resistance of the reference potentiometer to 10K ohm should turn the heater back on. Turn power OFF.
- (8) Remove calibrated 10K ohm potentiometer and reinstall thermistor leads. Remove circuit card extension adapter and reinstall circuit card assembly

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4. Bench Test

NOTE: The following test procedures are to be performed using card extension adapter PN 60-2344-1. Refer to Figure 1001 for electrical and water connections.

- A. Refer to Table 1002 for operating conditions.
- B. Anchor the coffee maker/water boiler securely on a work bench. The input power source shall be 115 VAC, 400 Hz, 3-phase with an ac voltmeter and an ac ammeter installed in each phase
- C. Install a water supply equipped with a regulator and gauge. The supply shall be regulated to 24 + 5/ -0 psig. Install a water drain tube.
- D. Remove cover assembly from the power control module (IPL Figure 4) and remove circuit card assembly. Plug the card extension adapter into connector J2 and insert circuit card assembly into other end of the card extension adapter.
- E. Apply the 115 VAC, 400 Hz, 3-phase power and observe each phase of ammeter. Currents of approximately 10 amperes per phase indicate the coffee maker/water boiler is energized and heater assemblies are drawing proper current to heat the water in the tank. Zero current indicates the ON/OFF switch is in the OFF position.
- F. Place the ON/OFF switch to the ON position with the HOT PLATE ON switch in the OFF position. Simulate over temperature operation by placing a 3.3K ohm resistor between test points K and P on the card extension adapter. Fuse F1 on printed circuit board 3530-0059-05 will open, removing the load from all three phases of input power. Replace fuse F1.

NOTE: The Fuse F1 can be temporarily replaced by a 2A circuit breaker (refer to Table 1001) for this step .

- G. Install a server in the coffee maker/water boiler. Place the ON/OFF switch in the ON position and HOT PLATE ON switch in the OFF position.
- H. Perform the checks described in Table 1002.

Table 1002
Operating Conditions

MODE OF OPERATION	Operating Conditions					
	I	II	III	IV	V	VI
ON/OFF Switch ON	♦	♦	♦	♦	♦	♦
Hot Plate						♦
Water Heaters ON	♦	♦		♦	♦	
Brew Switch Depressed		♦				
Brew Light ON		♦	♦	♦		
Server full, liquid touching water level sensor and brew light off					♦	

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Table 1002 (Continued)
Operating Conditions

MODE OF OPERATION	Operating Conditions					
	I	II	III	IV	V	VI
No water in tank						♦

5. Fault Isolation

It is recommended that circuit card assembly PN 61-1675-11 and printed circuit board assembly PN 3530-0059-05 be replaced in lieu of repair if these units fail. Refer to Table 1003 for Fault Isolation of component on Coffee Maker/Water Boiler assembly.

Table 1003
Fault Isolation

FAULT	PROBABLE CAUSE	TEST	CORRECTION
Pressing ON/OFF switch S1, indicator does not illuminate (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Defective lamp. Defective ON/OFF switch S1.	Depress HOT PLATE ON switch S4 to determine if voltage is present. Check with ohmmeter.	Replace S1 lamp if HOT PLATE ON indicator illuminates. Replace ON/OFF switch S1.
Pressing COFFEE BREW switch S2, indicator does not illuminate (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Defective lamp. Defective COFFEE BREW switch S2.	Determine if voltage is present across S2, L1 to L2. Check with ohmmeter.	Replace S2 lamp if voltage is present. Replace COFFEE BREW switch S2.
Pressing TEA switch S3, indicator does not illuminate (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Defective lamp. Defective TEA switch S3.	Determine if voltage is present across S3, L1 to L2. Check with ohmmeter.	Replace S3 lamp if voltage is present. Replace TEA switch S3.
Pressing HOT PLATE ON switch S4, indicator does not illuminate (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Defective lamp. Defective HOT PLATE ON switch S4.	ON/OFF switch S1 indicator illuminates when depressed. Check with ohmmeter.	Replace S4 lamp. Replace HOT PLATE ON switch S4.

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Table 1003 (Continued)
Fault Isolation

FAULT	PROBABLE CAUSE	TEST	CORRECTION
NO WATER indicator DS1 does not illuminate with tank empty and ON/OFF switch S1 depressed (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Defective lamp. Tank level sensor U2 defective. Logic circuits Faulty.	Determine if voltage is present at DS1, 1 to 2. Replace PCB 61-1675-11 with known functional unit and check.	Replace DS1 lamp if voltage is present. Replace tank level sensor U2. Replace PCB 61-1675-11.
Will not brew coffee when pressing switch S2 COFFEE BREW. TEA indicator is illuminated (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	SCR2 or associated circuit defective on PCB 3530-0059-05. Relay K1 on PCB 3530-0059-05 defective. Faulty logic card on PCB 61-1675-11. Q1 and associated circuit on PCB 3530-0059-05 defective.	Check circuit components D1, R1, R4 and SCR2 on PCB 3530-0059-05 with ohmmeter. Check with ohmmeter. Replace PCB 61-1675-11 with known functional PCB. Replace PCB 3530-0059-05 with known functional PCB.	Replace PCB 3530- 0059-05 or defective circuit components. Replace PCB 3530- 0059-05 or relay K1. Replace PCB 61-1675-11. Repair and/or replace PCB 3530-0059-05.
NO WATER indicator remains ON (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Tank does not fill. Defective logic circuits on PCB 61-1675-11. Tank level sensor U2 defective.	Make sure the tank is not filled with water. Replace PCB 61-1675-11 with known functional PCB.	Replace vent valve assembly on tank assembly. Replace PCB 61-1675-11. Replace tank level sensor U2.
Coffee overflows server during brew cycle (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Defective sensor assembly U1. Defective logic circuits on PCB 61-1675-11. Defective solenoid valve assembly.	Install sensor assembly known to be operational and recheck. Replace PCB 61-1675-11 with known functional PCB. Raise handle assembly; water continues to flow.	Replace sensor assembly U1. Replace PCB 61-1675-11. Replace solenoid valve assembly.

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Table 1003 (Continued)
Fault Isolation

FAULT	PROBABLE CAUSE	TEST	CORRECTION
Will not make TEA when pressing switch S3, TEA. TEA indicator is illuminated (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Defective circuit components associated with SCR3 on PCB 3530-0059-05.	Check with ohmmeter.	Replace defective circuit components or replace PCB 3530-0059-05.
	Relay on PCB 3530-0059-05 defective.	Check with ohmmeter.	Replace PCB 3530-0059-05 or relay K1.
	Faulty logic card on PCB 61-1675-11.	Replace PCB 61-1675-11 with known functional PCB.	Replace PCB 61-1675-11.
No indicators illuminate with S1 ON/OFF switch depressed (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	F1 on PCB 3530-0059-05 open.	Check with ohmmeter.	If F1 is open, replace fuse.
	Q1 or associated circuitry defective.	Check with ohmmeter.	Replace PCB 3530-0059-05 or SCR 1.
	Faulty logic card PCB 61-1675-11.	Replace PCB 61-1675-11 with known functional PCB.	Replace PCB 61-1675-11.
Heater assemblies will not operate (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Open thermistor RT1.	Check thermistor with ohmmeter.	Replace the thermistor RT1 if open, replace fuse.
	Open thermal switch(es).	Check thermal switch(es) with ohmmeter.	Replace PCB 3530-0059-05 or SCR 1.
	Defective relay K1 on power module assembly.	Check relay with ohmmeter.	Replace PCB 61-1675-11.
Coffee Maker/Water Boiler Assembly has abnormal power consumption (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	One defective heater assembly.	Check resistance of each heater element with an ohmmeter. Reading should be less than 17.2 ohms. Resistance between element and case should be more than 500 ohms.	Replace defective heater.
	Defective relay K1 on power module assembly.	Check relay coil and contacts with ohmmeter.	Replace relay K1 on power module assembly.
	One section of thermal switch assembly S6 defective.	Check with ohmmeter.	Replace thermal switch assembly S6.

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Table 1003 (Continued)
Fault Isolation

FAULT	PROBABLE CAUSE	TEST	CORRECTION
Coffee Maker/Water Boiler Assembly temperature improper during brew cycle.	Misaligned tube assembly.	Refer to TESTING AND FAULT ISOLATION section.	Perform alignments as required.
	Defective thermistor RT1.	Check thermistor with ohmmeter.	If open circuit, replace thermistor.
	Defective logic circuits, PCB 61-1675-11.	Install PCB 61-1675-11 known to be operational and recheck.	Replace PC board.
Coffee Maker/Water Boiler Assembly TEA cycle out of tolerance.	Improper water pressure.	Check water pressure.	Water pressure should be between 20 psig to 40 psig.
	Excessive scale buildup.	Inspect water lines and filter assembly.	Clean water lines and filters as required (refer to CLEANING section).
Hot plate heater temperature incorrect (refer to Figure 2002, SCHEMATICS AND WIRING DIAGRAMS section).	Defective switch S3.	Check with ohmmeter.	Replace switch.
	Defective heater assembly HR4.	Check element resistance. Resistance should be approximately 250 ohms.	Replace heater assembly.
Coffee Maker/Water Boiler Assembly leaks.	Loose fittings.	High pressure check. (refer to TESTING AND FAULT ISOLATION section, Leak Test).	Secure compression fittings.
	Cracked or corroded bushing on tank.	High pressure check. (refer to TESTING AND FAULT ISOLATION section, Leak Test).	Replace tank assembly.
	Corroded tank.	High pressure check. (refer to TESTING AND FAULT ISOLATION section, Leak Test).	Replace tank assembly.

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SCHEMATICS AND WIRING DIAGRAMS

1. General

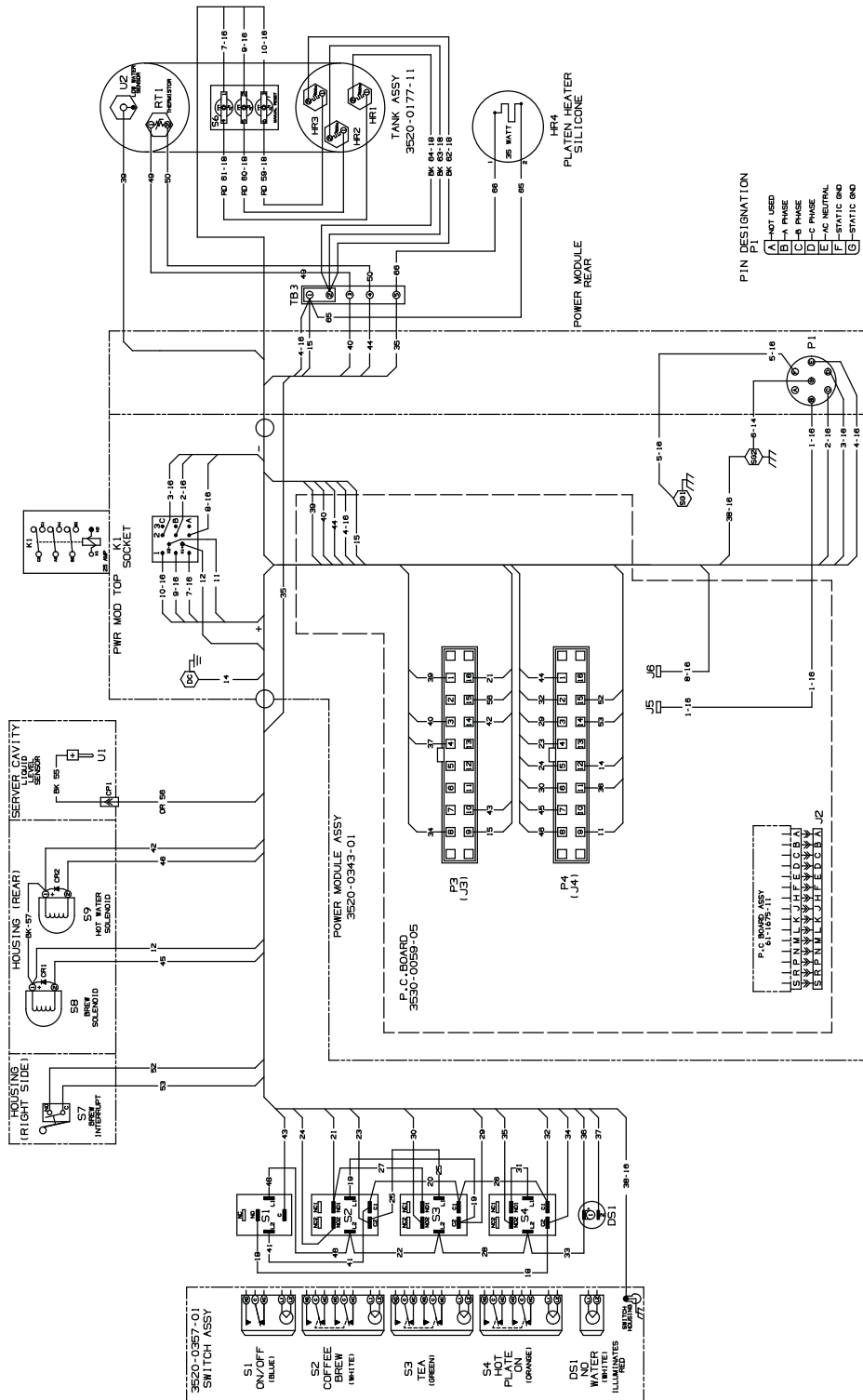
A. This section contains the schematics and wiring diagrams for the Coffee Maker/Water Boiler assembly.

2. Wiring Diagrams

A. The coffee maker/water boiler PN 3510-0052-01, Figure 2001 shows the system wiring diagram. Figure 2002 shows the Electric schematic for a coffee maker/water boiler Printed Circuit Board Assembly.

B. Refer to Table 2001 for the wire run list.

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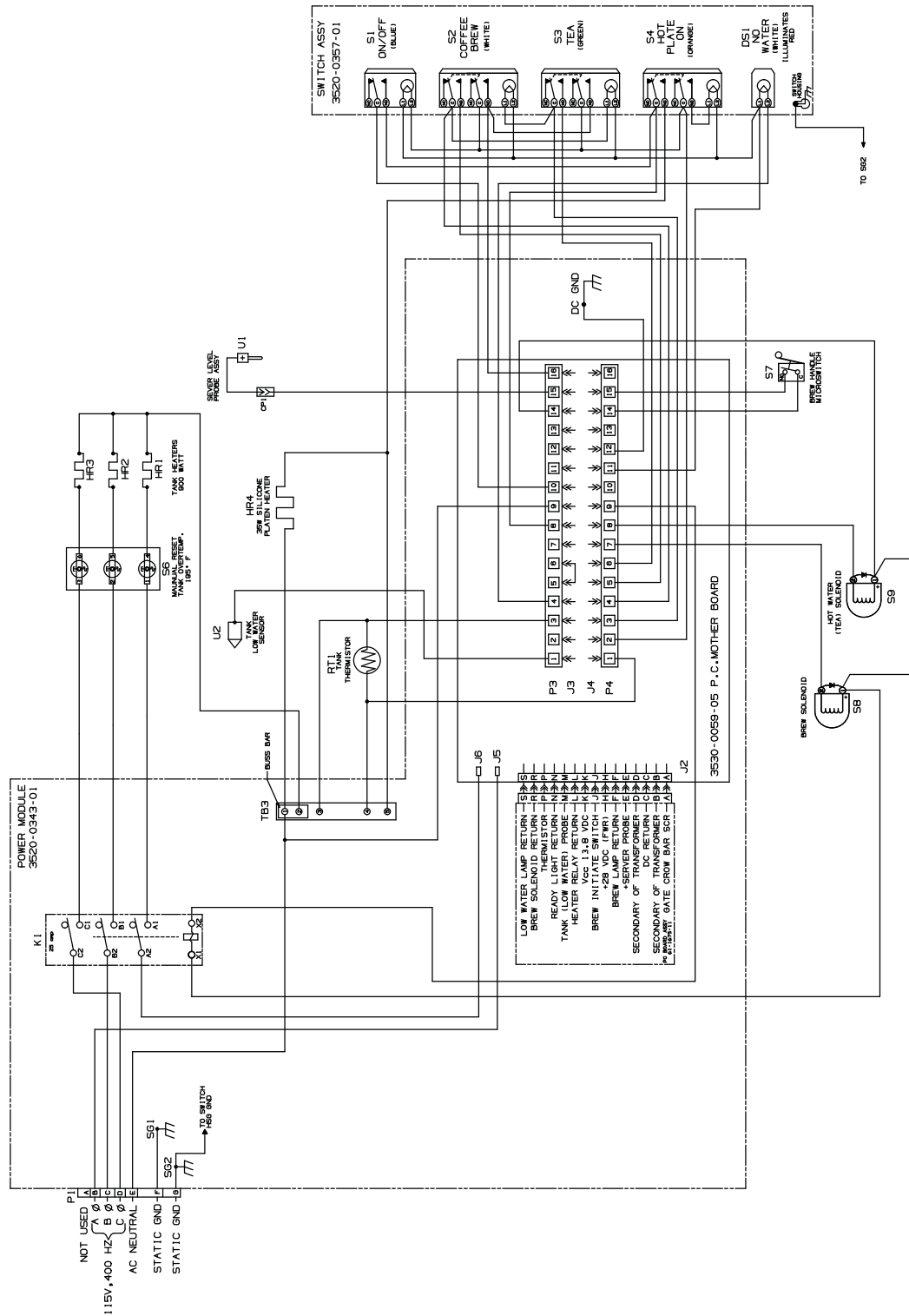


Figure 2002
Schematic Diagram

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Table 2001
Wire Run List

WIRE NO.	FROM	TO	COLOR	LENGTH	GAUGE	B/E AEROSPACE PN
1	P1-B	J5	WHITE	3.2	16	WI-142
2	P1-C	K1-B2	WHITE	7.5	16	WI-142
3	P1-D	K1-C2	WHITE	8.0	16	WI-142
4	P1-E	TB3-1	WHITE	11.0	16	WI-142
5	P1-F	SG1	WHITE	5.0	16	WI-142
6	P1-G	SG2	WHITE	1.8	14	WI-50106
7	K1-A1	S6-1	WHITE	7.0	16	WI-142
8	K1-A2	J6	WHITE	7.0	16	WI-142
9	K1-B1	S6-2	WHITE	7.0	16	WI-142
10	K1-C1	S6-3	WHITE	6.5	16	WI-142
11	K1-X2	P4-9	WHITE	7.8	22	WI-585
12	K1-X1	S8-1 (+)	WHITE	14.0	22	WI-585
13			NOT USED	4.5		
14	DC GND	P4-12	WHITE	8.0	20	WI-349
15	TB3-1	P3-9	WHITE	11.0	22	WI-585
16			NOT USED			
17			NOT USED			
18	S1-NO	S4-C2	WHITE	5.3	22	WI-585
19	S2-L1	S3-C2	WHITE	3.0	22	WI-585
20	S2-C1	S3-C1	WHITE	3.0	22	WI-585
21	S2-NO1	P3-16	WHITE	19.0	22	WI-585
22	S2-L2	S3-L2	WHITE	3.0	22	WI-585
23	S2-C2	P4-4	WHITE	18.0	22	WI-585
24	S2-NO2	P4-5	WHITE	18.6	22	WI-585
25	S3-L1	S2-C2	WHITE	3.0	22	WI-585
26	S3-C1	S4-C1	WHITE	3.0	22	WI-585
27	S3-NO1	S2-NO1	WHITE	3.0	22	WI-585

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Table 2001 (Continued)
Wire Run List

WIRE NO.	FROM	TO	COLOR	LENGTH	GAUGE	B/E AEROSPACE PN
28	S3-L2	S4-L2	WHITE	3.0	22	WI-585
29	S3-C2	P4-3	WHITE	17.2	22	WI-585
30	S3-NO2	P4-6	WHITE	17.4	22	WI-585
31	S4-L1	S4-NO1	WHITE	3.0	22	WI-585
32	S4-C1	P4-2	WHITE	17.5	22	WI-585
33	S4-L2	DS1-1	WHITE	3.0	22	WI-585
34	S4-C2	P3-8	WHITE	16.2	22	WI-585
35	S4-NO2	TB3-5	WHITE	19.8	22	WI-585
36	DS1-1	P4-11	WHITE	16.5	22	WI-585
37	DS1-2	P3-4	WHITE	16.8	22	WI-585
38	SWITCH HSG	SG2	WHITE	19.5	16	WI-142
39	P3-1	U2	WHITE	16.0	22	WI-585
40	P3-3	TB3-3	WHITE	13.0	22	WI-585
41	S1-L2	S2-C1	WHITE	3.0	22	WI-585
42	P3-14	S9-1 (+)	WHITE	15.5	20	WI-349
43	P3-10	S1-C	WHITE	19.1	22	WI-585
44	P4-1	TB3-4	WHITE	14.0	22	WI-585
45	P4-7	S8-2	WHITE	17.0	22	WI-585
46	P4-8	S9-2	WHITE	15.0	22	WI-585
47			NOT USED			
48	S1-L1	S2-L2	WHITE	3.0	22	WI-585
49	RT1-1	TB3-3	WHITE		22	WI-585
50	RT1-2	TB3-4	WHITE		22	WI-585
51			NOT USED			
52	S7-NO	P4-15	WHITE	18.0	20	WI-349
53	S7-C	P4-14	WHITE	18.0	20	WI-349
54			NOT USED			

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Table 2001 (Continued)
Wire Run List

WIRE NO.	FROM	TO	COLOR	LENGTH	GAUGE	B/E AEROSPACE PN
55	U1 +	CP1	BLACK	18.0	20	WI-256
56	CP1	P3-15	ORANGE	13.5	20	WI-307
57	S8-1 (+)	S9-1 (+)	BLACK		22	WI-423
58			NOT USED	10.0	18	
59	HR3-1	S6-6	RED	10.0	18	WI-194
60	HR2-1	S6-5	RED	10.0	18	WI-194
61	HR1-1	S6-4	RED	10.0	18	WI-194
62	HR3-2	TB3-2	BLACK	10.0	18	WI-197
63	HR2-2	TB3-2	BLACK	10.0	18	WI-197
64	HR1-2	TB3-2	BLACK	10.0	18	WI-197
65	HR4-2	TB3-1	WHITE	10.0	22	WI-585
66	HR4-1	TB3-5	WHITE	10.0	22	WI-585
67			NOT USED			
68			NOT USED			
NOTE: Length in inches for reference only.						

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DISASSEMBLY

1. General

- A. Prior to disassembly carefully read all instructions applicable to the item of equipment undergoing maintenance.
- B. Test each unit in accordance with the procedures in TESTING AND FAULT ISOLATION section. Refer to Fault Isolation section to determine the condition of the unit and the most probable cause of any malfunction. That will determine the extent of disassembly required to effect necessary repairs without complete teardown of the unit.
- C. Refer to the Illustrated Parts (IPL) for disassembly and assembly relationships. The number in parentheses after the noun name of a part refers to the item number in the IPL figure.
- D. The tools required for the disassembly of the Coffee Maker/Water Boiler assembly are listed in the Table 3001.

NOTE: Equivalent substitutes may be used for listed items.

2. Procedure

WARNING: ELECTRICAL SHOCK MAY OCCUR WHEN REMOVING RELAY K1 FROM HOUSING. SHUT OFF COFFEE MAKER/WATER BOILER MAIN POWER BEFORE REMOVING RELAY.

CAUTION: ALL DISASSEMBLY IS TO BE ACCOMPLISHED WITH THE COFFEE MAKER/WATER BOILER REMOVED FROM THE RAIL (I.E., POWER AND WATER SUPPLY ARE OFF) AND THE WATER TANK DRAINED.

A. Coffee Maker/Water Boiler (Refer to IPL Figure 1)

- (1) Remove brew cup assembly (10) from coffee maker/water boiler subassembly (5). Remove nut (25), screw (30), and spacer plate (20) from brew cup (15).

B. Coffee Maker/Water Boiler Subassembly (Refer to IPL Figure 2)

NOTE: Tag all unsoldered or disconnected wires required during disassembly to facilitate assembly of coffee maker/water boiler. Refer to Figure 2001, Figure 2002 and Table 2001 for wire number identification and point-to-point wiring of coffee maker/water boiler and subassemblies.

NOTE: Remove tie wraps as required to release wiring harness during the disassembly process.

- (1) Remove screws (260) and mounting brace plate (255).
- (2) Disconnect water supply tube assembly (85) (consisting of items 90 through 105) from drain valve assembly (165) and tank assembly (50). Remove ferrule (100), hex nut (105), and fitting (95) from tube (90).

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- (3) Remove screw (455), washer (460), and nut (465) from connector support (445). Remove screws (450) and connector support (445) from housing assembly (735). Remove hex nut (480), lock washer (475), and alignment pin guide (470). Remove water shutoff plug (170), o-ring (175), and drain valve assembly (165). Refer to Paragraph 2.L for disassembly of drain valve assembly (165).
- (4) Remove screws (130), thermal switch cover (120), and standoffs (125) from tank assembly (50). Remove screws (145), washers (150), and hex nuts (155), freeing ring terminals (140) from thermal switch assembly (135). Remove ring terminals (140) from leads if replacement is required. Remove attaching hardware and wire assembly (160) from thermal switch assembly (135). Remove thermal switch assembly (135). Refer to Paragraph 2.K for disassembly of thermal switch assembly (135).
- (5) Remove screws (430) and rail guide (425). Remove screws (440) and rail guide (435).
- (6) Remove screw (520), lock washer (525), washer (510), and clamp (515) from platen assembly (485). Remove butt splices (540) and tubing (535) from silicone rubber heater (530) leads. Remove screws (490), lock washers (495), and platen assembly (485). Remove screen assembly (420) from platen assembly (485). Remove lock nut (550), heater retainer (545), and silicone rubber heater (530). Remove blind rivets (505) and placard (500) if replacement is required.
- (7) Unsolder wires #12, #42, #45, and #46 on solenoid valve assembly (40). Unsolder wires #52 and #53 on micro switch assembly (385). Remove hex nut (80), washers (70 and 75), and ring terminal (65) from tank assembly (50). Remove ring terminal (65) from lead if replacement is required. Disconnect wire #55 and CP1 on housing assembly (735). Disconnect wire #40 at butt splice BS1 and wire #44 at butt splice BS2.
- (8) Remove screws (35) and power module cover (30) from power module assembly (20). Remove wire ties (730). Disconnect connectors P3 and P4, J5 and J6 in the power module assembly (20). Remove butt splice (200, IPL Figure 4) from SG2 ground. Slide wire harness and grommet (90, IPL Figure 3) from power module assembly (20, IPL Figure 2). Remove screw (15) and switch housing stud (10) and free switch assembly (5) from housing assembly (735). Remove screw (135, IPL Figure 3), lock nut (140, IPL Figure 3), washers (125 and 130, IPL Figure 3), and switch assembly (5). Refer to Paragraph 2.C for disassembly of switch assembly (5).
- (9) Remove terminal board cover (75, IPL Figure 4). Remove hex nuts (115, IPL Figure 4), lock washers (110, IPL Figure 4), flat washers (105, IPL Figure 4), and disconnect ring terminals (95 and 100, IPL Figure 4) and ring terminals (415). Remove ring terminals (415) from leads if replacement is required. Remove power module assembly (20). Refer to Paragraph 2.D for disassembly of power module assembly (20).
- (10) Remove screws (270) and front plate assembly (265) (consisting of items 275 through 295). Remove screws (290) and washers (295) and separate instruction plate (285), front plate (275) and brace (280).
- (11) Remove screws (390), washers (395), and micro switch assembly (385). Refer to Paragraph 2.N for disassembly of micro switch assembly (385).
- (12) Remove screws (55) and tank assembly (50). Remove tubing (60) from tank assembly (50). Refer to Paragraph 2.H for disassembly of tank assembly (50).

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- (13) Remove clamp (200) from tea tube assembly (185) and disconnect formed tube assembly (195) from solenoid valve assembly (40). Remove clamp (200) from formed tube assembly (195).
- (14) Remove clamp (790) and faucet tube (785).
- (15) Disconnect tube assembly (85, IPL Figure 10) from solenoid assembly (40). Remove screws (45) and solenoid valve assembly (40). Refer to Paragraph 2.G for disassembly of solenoid valve assembly (40).
- (16) Release tension on tension springs (640). Remove screws (405), lock washers (410), and spring bracket (400).
- (17) Remove screw (665), handle bushing (625), washer (635), tension springs (640), spring retainer (645), and lock washer (650). Remove screw (630), handle bushing (625), guide screws (620), spring bushing (655), carrying handle bushing (660), and carrying handle (615).
- (18) Remove screws (190) and tea tube assembly (185).
- (19) Remove screws (310), handle washers (315), bushings (320), flat washers (325), and nuts (330). Remove handle assembly (305) and brew head assembly (180) from housing assembly (735). Remove screws (380), handle (375), and brew head assembly (180) from handle assembly (305). Remove screw (350) from arm assembly (335) (consisting of items 340 through 350), handle bushing (300), guide bushing assembly (345), and arm (340). Remove screw (370) from arm assembly (355) (consisting of items 360 through 370, handle bushing (300), guide bushing assembly (365), and arm (360). Refer to Paragraph 2.M for disassembly of brew head assembly (180).
- (20) Remove screw (680), pivot pin (675), and lever locking assembly (670) (consisting of items 685 through 690). Remove retainer ring (705), pin (700), pin (695), lever (685), and fastener (690). Remove helical spring (710) from housing assembly (735).
- (21) Remove screws (745), faucet handle yoke (765) and faucet bracket assembly (740) with attached items. Remove hex nuts (750) and internal lock washers (755) and separate faucet bracket assembly (740), faucet tube assembly (780), and faucet body (770). Release and remove cap and handle assembly (760). Remove aerator (775) from faucet body (770).
- (22) Remove screw (605), washer (600), nut (610), wire clamp (595), and sensor wire assembly (575). Remove screw (585) ring terminal (580) and lock nut (590).
- (23) Remove screw (560), nut (570), and sensor wire assembly (555).
- (24) Remove screw (225) and guide rod (220).
- (25) Remove screw (235) and guide rod (230).
- (26) Remove screws (720), lock washers (725), flat washers (727), and probe assembly (715).
- (27) Remove screws (210), lock washers (215), and cup retainer (205).
- (28) Remove support block (240) and quarter turn retainer (250).

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C. Switch Assembly (Refer to IPL Figure 3)

- (1) Refer to Paragraph 2.B for removal of switch assembly from coffee maker/water boiler assembly.

NOTE: Tag all unsoldered or disconnected wires required during disassembly to facilitate assembly of coffee maker/water boiler. Refer to Figures 2001 and 2002 for wire number identification and point-to-point wiring of coffee maker/water boiler and assemblies.

NOTE: Remove tie wraps as required to release wiring harness during the disassembly process.

- (2) Remove switch caps (10 through 30) and lamps (35) from switch bezels (60).
- (3) Unsolder leads on switches (40 through 50) and indicator (55).
- (4) Remove screw (135), nut (140), washers (125 and 130), and terminal (120).
- (5) Remove screw (110), nut (115), clamp washer (105), and clamp (100).
- (6) Remove wire harness, grommets (90), and tie down straps (95). Remove connectors (P3, P4) (85) only if replacement is required.
- (7) Remove nuts (80), washers (70 and 75), switches (40 through 50), indicator (55), switch mounting brackets (65), and switch bezels (60).

D. Power Module Assembly (Refer to IPL Figure 4)

- (1) Refer to Paragraph 2.B for removal of power module assembly from coffee maker/water boiler subassembly.

NOTE: Tag all unsoldered or disconnected wires required during disassembly to facilitate assembly of coffee maker/water boiler. Refer to Figure 2001, Figure 2002 and Table 2001 for wire number identification and point- to-point wiring of coffee maker/water boiler and subassemblies.

NOTE: Remove tie wraps as required to release wiring harness during the disassembly process.

NOTE: Item numbers 75, 105, 110, 115, and 200 were removed during disassembly of coffee maker/water boiler subassembly.

- (2) Remove PC Board (170).
- (3) Remove screws (165), washers (160), lock nuts (155), flat washers (150), spacers (145), insulator (135), PC Board (140), and insulator (130).
- (4) Remove screw (125) and spacer (120).
- (5) Remove screws (85), lock nuts (90), and terminal board (80). Remove ring terminals (95 and 100) from leads if replacement is required.

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- (6) Remove screw (180), flat washers (185), ring terminal (175), lock washer (190), and lock nut (195). Remove terminal (175 from lead if replacement is required.
- (7) Remove screw (210), flat washers (215), ring terminal (205), flat washer (220), and lock nut (225). Remove ring terminal (205) from lead if replacement is required.
- (8) Remove screw (235), flat washers (240), ring terminals (230), lock washer (245), and lock nut (250). Remove ring terminals (230) from leads if replacement is required.
- (9) Remove pin (255) and guides (260) from housing assembly (70).
- (10) Remove attaching hardware and connector P1 (270) and grommet (265). Remove spade terminal (35) and mini PV receptacle (40) from leads if replacement is required.

E. Printed Circuit Board, Part No. 3530-0059-05 (Refer to CMM 25-33-35).

- (1) Refer to Paragraph 2.D for removal of printed circuit board from coffee maker/water boiler subassembly.
- (2) The printed circuit board assembly will not be disassembled in its entirety. Refer to REPAIR section for replacement of parts.

F. Printed Circuit Board, Part No. 61-1675-11 (Refer to CMM 25-33-36)

- (1) Refer to Paragraph 2.D for removal of printed circuit board from coffee maker/water boiler subassembly.
- (2) The printed circuit board assembly will not be disassembled in its entirety. Refer to REPAIR section for replacement of parts.

G. Solenoid Valve Assembly (Refer to IPL Figure 5)

- (1) Refer to Paragraph 2.B for removal of solenoid assembly from coffee maker/water boiler subassembly.
- (2) Remove fitting nut (110) and ferrule (105).
- (3) Remove fitting (100), flow washer (95), retaining ring (90), and fitting (85).
- (4) Remove valve (30), and fittings (15 and 20).
- (5) Remove screw (10) and support (5) from valve (30) (consisting of items 35 through 75). Remove fitting (25). Remove hex nut (70), nameplate (65), housing strap (60), coil assembly (75), plunger guide assembly (55), spring (50), plunger assembly (45), o-ring (40), and body detail (35).

H. Water Tank Assembly (Refer to IPL Figure 6)

- (1) Refer to Paragraph 2.B for water tank assembly removal from coffee maker/water boiler subassembly.
- (2) Remove low water sensor assembly (140) from tank welded assembly (170). Refer to Paragraph 2.J for disassembly of low water sensor assembly (140).

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- (3) Remove thermistor assembly (55) from tank welded assembly (170).
- (4) Remove elbow (20) from filter assembly (25) (consisting of items 30 through 50). Remove filter assembly (25) from tank welded assembly (170). Remove screw (30), lock washer (32), retainer (35), filter screen (40), Water jet (45), and o-ring (50).
- (5) Remove fitting assembly (5) (consisting of items 10 and 15) from vent valve assembly (60). Remove gasket (15) from fitting (10). Remove vent valve assembly (60) (consisting of items 65 through 80) from tank welded assembly (170). Separate body subassembly (75) from housing (80) and remove o-ring (70).
- (6) Remove relief valve assembly (95) (consisting of items 105 through 135) from tank welded assembly (170). Remove fitting assembly (135). Loosen lock nut (110) and remove valve body (130), lock nut (110), valve assembly (125), spring rest (120), spring (115), and adjusting cap (105) from relief valve subassembly (100).
- (7) Remove tube assembly (155) (consisting of items 160 and 165) from tank welded assembly (170). Remove elbow (165) from tube (160) if replacement is required.
- (8) Remove heater assemblies (145) from tank welded assembly (170).

J. Low Water Sensor Assembly (Refer to IPL Figure 7)

- (1) Refer to Paragraph 2.J for removal of low water sensor from tank assembly.
- (2) Remove low water sensor (55).

K. Thermal Switch Assembly (Refer to IPL Figure 8)

- (1) Refer to Paragraph 2.B for removal of thermal switch assembly from coffee maker/water boiler subassembly.
- (2) Remove lock nuts (15) and thermal switches (10) from mounting plate assembly (5).

L. Drain Valve Assembly (Refer to IPL Figure 9)

- (1) Refer to Paragraph 2.B for removal of drain valve assembly from coffee maker/water boiler subassembly.
- (2) Unscrew stem assembly (10) from valve body (5). Remove o-rings (15 and 20).

M. Brew Head Assembly (Refer to IPL Figure 10)

- (1) Refer to Paragraph 2.B for removal of brew head assembly from coffee maker/water boiler subassembly.
- (2) Remove clip (90), tube assembly (85), and flow washer (120).
- (3) Remove nuts (20), retainer assembly (15), and gasket (10).
- (4) Remove screws (70), nuts (80), washers (75), and brew fitting (65).

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- (5) Remove screws (45 and 60), washers (35, 40, 50, and 55), spring bushing (45B), screw (45C) and guides (25 and 30) from brew head (5).

N. Switch Assembly (Refer to IPL Figure 11)

- (1) Refer to Paragraph 2.B for removal of switch assembly from coffee maker/water boiler subassembly.
- (2) Remove screw (supplied with item 25), washers (30), and nut (supplied with item 25) from switch subassembly (5). Separate switch actuator (25) and mounting plate (20) from switch (10).
- (3) Remove terminal strips (15) from switch leads if replacement is required.

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CLEANING

1. General

CAUTION: TURN ALL POWER TO COFFEE MAKER/WATER BOILER AND COMPARTMENT OFF. EXCESSIVE WATER MAY CAUSE A SHORT CIRCUIT. USE A DAMP CLOTH ONLY.

CAUTION: DO NOT USE ANY SOLVENTS, DEGREASERS, DETERGENTS, AEROSOL SPRAY, ETC., AS THESE HARM SENSORS SWITCHES AND PLASTIC PARTS. THE ONLY ACCEPTABLE CLEANSER IS A CLOTH MOISTENED WITH WARM WATER.

NOTE: The Coffee Maker/Water Boiler should be completely cleaned and the brew head lubricated every 2000 operating hours.

A. The materials required for cleaning the Coffee Maker/Water Boiler are listed in Table 4001.

NOTE: Equivalent substitutes may be used for listed items.

Table 4001
Cleaning Materials

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Air	Oil-free air	Oil-Free Compressor
Brush, Soft-bristled	None	Commercially Available
Cleaner	Mild Soap Solution, Liquid Detergent	Commercially Available
Cloth, Cleaning, Low-Lint,	None	Commercially Available
Distilled Water	None	Commercially Available
Isopropyl Alcohol	Approximate 90% Isopropyl Alcohol Solution	Commercially Available
Protective Gloves	None	Commercially Available
Sulfamic Acid-based cleaner or other agent that acts upon calcium, lime and rust deposits	Approximately 5% Sulfamic Acid Solution, LimeAway, vinegar, or agent that removes calcium, lime and rust deposits	Commercially Available

2. Procedure

A. Coffee Maker/Water Boiler Assembly (refer to IPL Figure 1)

WARNING: ALLOW WATER TANK TO COOL BEFORE DRAINING.

- (1) Clean electrical parts with isopropyl alcohol (refer to table 4001).
- (2) Wash brew cup of any contaminated surfaces with a cleaner (refer to table 4001). Rinse with distilled water (refer to table 4001).
- (3) Clean tube assemblies by running a flexible wire brush (refer to table 4001) through the interior. Rinse with distilled water (refer to table 4001).

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- (4) Clean stained surfaces with a cleaning compound (refer to table 4001). Rinse with distilled water (refer to table 4001).
- (5) Clean sensor probes of sensor assembly (140, IPL Figure 6) with a clean damp cloth (refer to table 4001).
- (6) Scale build-up due to operation with hard water may be removed by soaking the affected parts in a solution of 5% sulfamic acid (refer to table 4001). Rinse with distilled water (refer to table 4001).
- (7) To remove rust or oxidation from stainless steel parts, passivate per Federal Specification MIL-F-14072, finish E300, or equivalent.

B. Flushing Procedure (Optional) (Per SIL 10240)

NOTE: The following flushing procedure (using vinegar or CLR) should be performed every 12 months at a minimum, more frequently if a buildup of calcium or mineral deposits occur.

- (1) Fill Coffee Maker tank with Calcium Lime Rust (CLR). CLR should be diluted with 8 parts water, 1 part CLR. Carefully follow instructions on CLR container.
- (2) Circulation of CLR throughout the system should be accomplished as if making multiple pots of coffee.
- (3) Leave CLR in Coffee Maker for the minimum amount of time required to clean the Coffee Maker.
- (4) Immerse brew cup and filter screen (screen not applicable in all brew cups) in CLR for the same period of time.
- (5) Drain CLR from Coffee Maker.
- (6) Flush CLR out of Coffee Maker with water/baking soda solution. Flush out once with water/baking soda solution, then with just plain water until all traces of CLR have been removed.
- (7) Rinse brew cup with water.
- (8) After entire system is thoroughly flushed, operate unit to verify all features are working properly.

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INSPECTION/CHECK

1. General

- A. A line operational check has been added to enable the Coffee Maker/Water Boiler to be checked while still installed on an aircraft.

2. Procedure

- A. Coffee Maker/Water Boiler Subassembly (refer to IPL Figure 2)

NOTE: Visually inspect that all parts are present and secure.

- (1) Perform high pressure leak test as described in TESTING AND FAULT ISOLATION section.
- (2) Inspect tube assembly for evidence of deterioration.
- (3) Check guide rods (220 and 230) for straightness.
- (4) Inspect springs for distortion.
- (5) Check brew handle assembly (305) for free operation and check switch assembly (5) for proper actuator movement when brew head guides (25, IPL Figure 10 and 30, IPL Figure 10) is raised and lowered.
- (6) Check molded sensor assembly (140, IPL Figure 6) for damaged probes and defective wiring.
- (7) Check all electric wires for damaged insulation.
- (8) Check rail assemblies for straightness.
- (9) Check support block (240) for distortion.
- (10) Check faucet assembly for proper operation.
- (11) Check cup retainer (205) for free operation.
- (12) Check lock down lever assembly (670) for free operation.

- B. Control Switch Assembly (refer to IPL Figure 3)

- (1) Inspect switch caps (10 thru 30) for cracks and illegible nomenclature.
- (2) Inspect switches (40, 45 and 50) for corroded or damaged contacts and terminals.
- (3) Visually check LEDs (35) for discoloration and open filaments.

- C. Power Module Assembly (refer to IPL Figure 4)

- (1) Inspect connectors for pin distortion and security of connections.
- (2) Inspect relay socket (5) for corrosion and burned or overheated wires.

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- (3) Inspect wire connections for good solder joints and good workmanship.
- (4) Check wire terminal attaching screws for tightness.
- (5) Check connector P1 (270) for damaged pins.
- (6) Check terminal block for damaged terminals.
- (7) Check wire routing. All wires shall be secured and out of the way as much as possible.
- (8) Inspect guides (260) for straightness.

D. Printed Circuit Board PN 3530-0059-05 (refer to CMM 25-33-35)

E. Printed Circuit Board PN 61-1675-11 (refer to CMM 25-33-36)

F. Solenoid Valve Assembly (refer to IPL Figure 5)

- (1) Check fittings for leaks and corrosion.
- (2) Check valves (30) for discoloration due to overheating.

G. Tank Assembly (refer to IPL Figure 6)

- (1) Inspect filter screen (40) for clogging or corrosion.
- (2) Inspect water jet (45) for clear water passages.
- (3) Check relief valve assembly (95) for proper operation.
- (4) Refer to TESTING and FAULT ISOLATION section and check thermistor assembly (55) for proper operation.
- (5) Check tank assembly for cracks and corrosion.

H. Low Water Sensor Assembly (refer to IPL Figure 7)

- (1) Check low water sensor (55) for corrosion.

J. Thermal Switch Assembly (refer to IPL Figure 8)

- (1) Check thermal switches (10) for signs of overheating.
- (2) Check mounting hardware for placement and security.

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K. Drain Valve Assembly (refer to IPL Figure 9)

- (1) Check stem assembly (10) for corrosion.
- (2) Check o-rings (15, 20) for deterioration.
- (3) Check valve body (5) for sediment and corrosion.

L. Brew Head Assembly (refer to IPL Figure 10)

- (1) Check guides (25, 30) for free operation.
- (2) Check tube assembly (85) for evidence of leaks and corrosion.
- (3) Check fittings for corrosion and cleanliness.

M. Switch Assembly (refer to IPL Figure 11)

- (1) Check actuator (25) for bending and switch actuation.
- (2) Check leads and terminals for cracks and wear.
- (3) Check hardware for missing or incorrect parts.

N. Recommended Periodic Maintenance

NOTE: It is imperative that relief valve be checked for proper operation every six months and/or 1500 flight hours, or every shop visit.

- (1) Daily (on aircraft)
 - (a) Thoroughly clean coffee maker/water boiler, sensor assembly, brew head cavity and switch panel.
- (2) Weekly
 - (a) Clean, inspect and replace damaged components such as probe assembly.
- (3) 1500 flight hours or six months (whichever comes first)
 - (a) Completely flush entire coffee maker/water boiler water system.
 - (b) Clean or replace float tip on vent valve and thoroughly clean vent valve housing.
 - (c) Inspect and clean relief valve. Replace relief valve seat, if necessary. Recalibrate relief valve and seal lock nut.
 - (d) Tighten all external screws.

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- (4) 3000 flight hours or 12 months (whichever comes first)
 - (a) Perform 1500 hour maintenance procedures per Paragraph 2.N.(3).
 - (b) Visually check brew head gasket; replace if necessary.
 - (c) Check Probe assembly for cracks; replace if necessary.
 - (d) Check brew head fitting; replace as required.
 - (e) Inspect and clean tank heaters, platen heaters, solenoid valves and low water sensor.
- (5) 6000 flight hours or 24 months (whichever comes first)
 - (a) Perform 1500 and 3000 hour maintenance procedures per Paragraphs 2.N.(3) and 2.N.(4).
 - (b) Visually check relay; replace if necessary.
 - (c) Visually check platen heater; replace if necessary.
 - (d) Check solenoid valves; overhaul or replace if necessary.
 - (e) Check tank heaters; replace if necessary.

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REPAIR

1. General

CAUTION: B/E AEROSPACE, INC. DOES NOT RECOMMEND THE FOLLOWING PROCEDURES BE UTILIZED BY OTHER THAN THE MANUFACTURER. THE FOLLOWING INFORMATION IS SUPPLIED AT THE REQUEST OF OPERATING CUSTOMERS. IT IS RECOMMENDED THAT MATERIAL REQUIRING THIS LEVEL OF REPAIR BE RETURNED TO THE FACTORY. B/E AEROSPACE, INC. ACCEPTS NO LIABILITY FOR THE PERFORMANCE OF THE EQUIPMENT WHEN REPAIRS ARE EFFECTED BY THE CUSTOMER UTILIZING THIS INFORMATION.

- A. All parts found defective or inoperative in TESTING AND FAULT ISOLATION section and INSPECTION/CHECK section should be replaced or repaired.
- B. Standard parts such as screws, nuts, and washers should be replaced rather than repaired for economy purposes.
- C. The materials required for the repair of the Coffee Maker/Water Boiler are listed in Table 6001.

NOTE: Equivalent substitutes may be used for the listed items.

Table 6001
Repair Materials

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Alpha soldering chemical	810T	Commercially Available
Conformal Coating	PC-18 CM-238	Dexter Corporation Hysol Division (V7H439) B/E Aerospace, Inc. (V63367)

2. Procedure

A. General

- (1) All plastic and rubber parts are non-repairable and should be replaced if defective.
- (2) All anodized parts can be refinished by iriditing per Federal Specification MIL-C-5541, and painting with flat black paint per Federal Specification MIL-E-5557, Type 3.

B. Printed Circuit board Assemblies

- (1) Use a suitable heat sink when soldering solid state components. After soldering, clean printed circuit board with Alpha soldering chemical (refer to Table 6001) and apply a uniform (0.001 to 0.006 inch thick) coating of Hysol (refer to Table 6001) Standard to repaired surface.

WARNING: REPLACEMENT OF ELECTRICAL COMPONENTS ON PRINTED CIRCUIT BOARD ASSEMBLY WILL REQUIRE FUNCTIONAL TEST AND POSSIBLE RECALIBRATION. REFER TO TESTING AND FAULT ISOLATION SECTION. COFFEE MAKER/WATER BOILER COULD OVERHEAT AND CAUSE INJURY IF INSTRUCTIONS ARE NOT FOLLOWED.

(2) If handle is defective, remove by drilling out rivets. Attach new handle with two new rivets.

C. Thermal Switch Assembly

(1) If thermal switch is defective, repair should not be attempted; the unit should be replaced.

D. Parts Replacement

(1) Replace all parts determined defective during TESTING AND FAULT ISOLATION and INSPECTION/CHECK sections.

E. Anchor Nuts

(1) Replacement of anchor nuts is as follows:

(a) Drill out rivets securing anchor nut.

(b) Attach replacement anchor nut with two new rivets.

F. Threaded inserts (helicoils) can be replaced as follows:

(1) Free top end of the insert by bending toward center of hole with a sharp instrument.

(2) Grasp free end with needle-nosed pliers and extract insert by turning counterclockwise.

(3) Refer to IPL Figure 1 for part numbers and locations of replacement inserts.

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ASSEMBLY

1. General

- A. Refer to the figures in the Illustrated Parts List and assemble the Coffee Maker/Water Boiler Assembly in accordance with the following procedures.
- B. The equipment and materials required for assembly of the Coffee Maker/Water Boiler are listed in Table 7001.

NOTE: Equivalent substitutes may be used for listed items.

Table 7001
Assembly Materials and Equipment

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Sealant, white	#738 - white AD-053	Dow Corning Corporation (V5D028) B/E Aerospace, Inc. (V63367)
Loctite (Blue)	242 AD-300	Loctite Corporation (V05972) B/E Aerospace, Inc. (V63367)
Loctite (Red)	262 AD-156	Loctite Corporation (V05972) B/E Aerospace, Inc. (V63367)
Conformal coating	1B31 CM-50021	Columbia Chase Corporation (V99109) B/E Aerospace, Inc. (V63367)
Conformal coating	PC-18 CM-238	Dexter Corporation Hysol Division (V7H439) B/E Aerospace, Inc. (V63367)
Tubing, heat shrink	M23053/8-004-C TU-310	Raychem Corporation (V06090) B/E Aerospace, Inc. (V63367)
Tubing	TENT001TTV03512 TU-379	Acme Plastics (1EN65) B/E Aerospace, Inc. (V63367)
Tubing, heat shrink	RNF-100 3/16 TU-448	Raychem Corporation (V06090) B/E Aerospace, Inc. (V63367)
Tubing, heat shrink	RNF-100 1/4 TU-4705	Raychem Corporation (V06090) B/E Aerospace, Inc. (V63367)
Tape, teflon (1/4 inch wide)	MS-STR-4 TT-7454	Crawford Fitting (V1YJV3) B/E Aerospace, Inc. (V63367)

2. Procedure

- A. Anchor nuts are permanently attached to various assemblies in the coffee maker/water boiler subAssembly. If they become damaged, they may be replaced by drilling out attaching rivets and riveting new anchor nuts where used in the coffee maker/water boiler subassembly.
- B. Switch Assembly (refer to IPL Figure 11)
 - (1) If removed, replace terminal strips (15) on switch leads.

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- (2) Assemble switch (10), mounting plate (20), switch actuator (25), and secure with lock washers (30).
- (3) Refer to Paragraph M for installing switch assembly on coffee maker/water boiler subassembly.

C. Brew Head Assembly (IPL Figure 10)

- (1) Install guides (25 and 30) on brew head (5) using screws (45 and 60), spring bushing (45B), screw (45C) and washers (35, 40, 50, and 55).
- (2) Install retainer assembly (15) and gasket (10), securing with nuts (20).
- (3) Install brew fitting (65) on brew head (5) using screws (70), washers (75), and nuts (80).
- (4) Assemble tube assembly (85). Install flow washer (120) with the concave side facing into brew fitting (65) and attach tube assembly (85) to brew fitting (65) using clip (90).
- (5) Refer to Paragraph 2.N for installing brew head assembly in coffee maker/water boiler subassembly.

D. Drain Valve Assembly (refer to IPL Figure 9)

- (1) Install o-rings (15 and 20) on stem (10). Screw stem (10) into valve body (5).
- (2) Refer to Paragraph 2.N for installing drain valve assembly in coffee maker/water boiler subassembly.

E. Thermal Switch Assembly (refer to IPL Figure 8)

- (1) Attach thermal switches (10) on mounting plate assembly (5) and secure with stop nuts (15).
- (2) Refer to Paragraph 2.N for installing thermal switch assembly on coffee maker/water boiler subassembly.

F. Low Water Sensor (refer to IPL Figure 7)

- (1) Assemble low water sensor (55).
- (2) Refer to Paragraph 2.G for installing low water sensor on tank assembly.

G. Water Tank Assembly (refer to IPL Figure 6)

- (1) Install heater assemblies (145) into tank welded assembly (170).
- (2) Install tube assembly (155) (consisting of items 160 and 165). Attach elbow (165) to tube (160) and install in tank welded assembly (170). Refer to TESTING AND FAULT ISOLATION section for tube assembly (155) alignment.
- (3) Assemble relief valve subassembly (100) (consisting of items 105 through 135) using valve body (130), valve assembly (125), spring rest (120), spring (115), lock nut (110), adjusting cap (105), and fitting assembly (135). Install relief valve assembly (95) into tank welded assembly (170). Refer to TESTING AND FAULT ISOLATION section for tank top end cap configuration.

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- (4) Assemble vent valve assembly (60) (consisting of items 65 through 80). Install vent valve assembly (60) into tank welded assembly (170). Assemble fitting assembly (5) (consisting of items 10 and 15) and install into vent valve assembly (60).
- (5) Assemble filter assembly (25) (consisting of items 30 through 50) using Water jet (45), filter screen (40), retainer (35), lock washer (32), screw (30), and o-ring (50). Install filter assembly (25) into tank welded assembly (170). Install elbow (20) into filter assembly (25).
- (6) Install thermistor assembly (55) into tank welded assembly (170).
- (7) Install low water sensor assembly (140) into tank welded assembly (170).
- (8) Refer to Paragraph 2.N for installing tank assembly on coffee maker/water boiler subassembly.

H. Solenoid Valve Assembly (refer to IPL Figure 5)

- (1) Assemble solenoid valves (30) (consisting of items (35 through 75) using body detail (35), o-ring (40), plunger assembly(45), coil assembly (75), nameplate (65), housing strap (60), and hex nut (70). Attach valve (30) to fitting (25).
- (2) Install connection fittings (15 and 20).
- (3) Attach support (5) to valve (30) using screws (10).
- (4) Install fitting (85), retaining ring (90), flow washer (95), and fitting (100).
- (5) Install ferrule (105) and fitting nut (110).

NOTE: THE ROUNDED EDGE OF FLOW WASHER (95) SHOULD BE INSERTED INTO FITTING (85) FIRST. THE FLAT SIDE WOULD BE CLOSEST TO FITTING (100).

- (6) Refer to Paragraph 2.N for installing solenoid assembly on coffee maker/water boiler subassembly.

J. Printed Circuit Board, PN 611675-11 (refer to CMM 25-33-36)

- (1) The printed circuit board does not require a complete assembly. Replace only those parts found defective during TESTING AND FAULT ISOLATION section.
- (2) Refer to Paragraph 2.L for installing printed circuit board in power module assembly.

K. Printed Circuit Board, PN 3530-0059-05 (refer to CMM 25-33-35)

- (1) The printed circuit board does not require a complete assembly. Replace only those parts found defective during Testing/Fault Isolation.
- (2) Refer to Paragraph 2.L for installing printed circuit board in power module assembly.

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L. Power Module Assembly (refer to IPL Figure 4)

NOTE: Refer to SCHEMATICS AND WIRING DIAGRAMS section, Figure 2001, Figure 2002 and Table 2001, for wire number identification and point-to-point wiring of the coffee maker/water boiler.

NOTE: Install tie wraps as needed to maintain the wiring layout as required during assembly.

NOTE: Item numbers 75, 105, 110, 115, and 200 will not be assembled at this level. Items will be installed during assembly of the coffee maker/water boiler subassembly.

- (1) Attach mini PV receptacle (40) and spade terminal (35) to leads. Install grommet (265) and connector (270).
- (2) Insert guides (260) and pin (255) in housing assembly (70).
- (3) Install relay socket (5) in the housing assembly (70) with bolts and nuts supplied with the relay socket. Install relay (65) in relay socket (5).
- (4) If removed, attach ring terminal (230) to lead. Attach ring terminal (230) to ground using screw (235), flat washers (240), lock washer (245), and locknut (250).
- (5) If removed, attach ring terminal (205) to lead. Attach ring terminal (205) to ground using screw (210), flat washers (215), lock washers (220), and lock nut (225).
- (6) If removed, attach ring terminal (175) to lead. Attach ring terminal (175) to ground using screw (180), flat washers (185), lock washers (190), and lock nut (195).
- (7) If removed, attach ring terminals (95 and 100) to leads. Install terminal board (80) using screws (85) and locknuts (90).
- (8) Install spacer (120) using screw (125).
- (9) Install insulator (130), PC board (140), and insulator (135) using spacers (145), screws (165), washers (160), flat washers (150), and lock nuts (155).
- (10) Install PC board (170).
- (11) Refer to Paragraph 2.N for installing power module assembly in coffee maker/water boiler subassembly.

M. Switch Assembly (refer to IPL Figure 3)

NOTE: Refer to SCHEMATICS AND WIRING DIAGRAMS section, Figure 2001, Figure 2002 and Table 2001, for wire number identification and point-to-point wiring of the coffee maker/water boiler.

NOTE: Install wire ties as needed to maintain the wiring layout as required during assembly.

- (1) Install switches (40, 45, and 50) and indicator (55) using bezels (60), switch mounting brackets (65), washers (70 and 75), and nuts (80).

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- (2) Replace connectors (85) if required and install wire harness using tie down straps (95) and grommets (90).
- (3) Secure wire harness with clamp (100), washer clamp (105), screw (110), and nut (115).
- (4) Attach terminal (120) to ground using screw (135), washers (125 and 130), and nut (140).
- (5) Solder leads on switches (40, 45, and 50) and indicator (55).
- (6) Install lamps (35) and switch caps (10 through 30) in bezels (60).
- (7) Refer to Paragraph 2.N for installing switch assembly in coffee maker/water boiler subassembly.

N. Coffee Maker/Water Boiler Subassembly (refer to IPL Figure 2)

NOTE: Refer to SCHEMATICS AND WIRING DIAGRAMS section, Figure 2001, Figure 2002 and Table 2001, for wire number identification and point-to-point wiring of the coffee maker/water boiler.

NOTE: Install wire ties as needed to maintain the wiring layout as required during assembly.

- (1) Install sensor wire assembly (555) using screw (560) and nut (570).
- (2) Install sensor wire assembly (575) and wire clamp (595), using screw (605), washer (600), and nut (610). Install ring terminal (580). Secure with screw (585) and lock nut (590).
- (3) Insert helical spring (710) in housing assembly (735). Assemble lever locking assembly (670) (consisting of items (685 through 705), using lever (685), pin (695), pin (700), and retainer ring (705). Install lever locking assembly (670) in housing assembly (735) and secure with pivot pin (675) and screw (680).
- (4) Assemble handle assembly (305) (consisting of items 310 through 380). Assemble arm assembly (335) (consisting of items 340 through 350) using arm (340), guide bushing assembly (345), handle bushing (300), and screw (370). Assemble arm assembly (355) (consisting of items 360 through 370), using arm (360), guide bushing assembly (365), handle bushing (300), and screw (370). Install brew head (180) between arm assemblies (335 and 355) and attach handle (375) to arm assemblies (335 and 355) using screws (380). Install brew head assembly (180), and handle assembly (305) in housing assembly (735) and secure items in place with screws (310), handle washers (315), bushings (320), flat washers (325), and nuts (330).
- (5) Install tea tube assembly (185) using screws (190).
- (6) Install cup retainer (205) using screws (210) and lock washers (215).
- (7) Install probe assembly (715). Secure with screw (720) and lock washer (725).
- (8) Install guide rod (230) using screw (235).
- (9) Install guide rod (220) using screw (225).
- (10) Install carrying handle (615) using guide screws (620), handle bushing (625), and screw (630). Install tension springs (640) and carrying handle bushing (660), using screw (665), handle bushing (625), washer (635), spring retainer (645), and lock washer (650).

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- (11) Install spring bracket (400) using screws (405), and lock washers (410). Set tension on springs (640) using spring bracket (400).
- (12) Install solenoid valve assembly (40), using screws (45). Attach tube assembly (85, IPL Figure 10) to solenoid assembly (40, IPL Figure 2).
- (13) Attach fitting (114) to tube assembly (112). Install tube assembly (112) between solenoid valve assembly (40) and faucet tube (785) and secure with clamp (790).
- (14) Install formed tube assembly (195) between solenoid valve assembly (40) and tea tube assembly (185) and secure with clamp (200).
- (15) Attach tubing (60) to tank assembly (50) vents. Install tank assembly (50) on housing assembly (735) routing tubing (60), as required. Secure tank assembly (50), using screws (55).
- (16) Install micro switch assembly (385) using screws (390) and washers (395).
- (17) Assemble front plate assembly (265) (consisting of items 275 through 295). Use screws (290) and washers (295), and attach instruction plate (285), and front plate (275) to brace (280). Install front plate assembly (265) to housing assembly (735), using screws (270).
- (18) Install power module assembly (20) on housing assembly (735) using screws (25). If required, attach ring terminals (415) to lead. Connect ring terminals (415), wire assembly (555), and ring terminals (95 and 100, IPL Figure 4) to terminal board TB3 on power module assembly using flat washers (105), lock washers (110), and hex nuts (115).
- (19) Attach terminal (120, IPL Figure 3) to switch assembly (5, IPL Figure 2) using screw (135, IPL Figure 3), washers (125 and 130), and lock nut (140). Attach switch assembly (5, IPL Figure 2) to housing assembly (735) using switch housing stud (10) and screw (15). Slide wire harness and grommet (90, IPL Figure 3) into power module assembly (20, IPL Figure 2). Attach butt splice (200, IPL Figure 4) to lead from SG2 ground. Attach connectors P3 and P4, J5 and J6 to plugs and jacks in the power module assembly (20, IPL Figure 2). Secure harness with wire ties (730). Install cover (30) in power module assembly (20) using screws (35).
- (20) Connect wire #44 at butt splice BS2 and wire #40 at butt splice BS1. Connect wire #67 at CP1 on housing assembly (735). If removed, attach ring terminal (65). Attach ring terminal (65) to tank assembly (50) using washers (70 and 75) and hex nut (80). Solder wires #52 and #53 on micro switch assembly (385). Solder wires #12, #42, #45, and #46 on solenoid valve assembly (40).
- (21) Attach aerator (775) and cap and handle assembly (760) to faucet body (770). Attach faucet tube (780) and faucet body (770) to faucet bracket assembly (740) using internal lock washers (755) and hex nuts (750). Attach faucet handle yoke (765) and faucet bracket assembly (740) to housing assembly (735) using screws (745).
- (22) Attach placard (500) on platen assembly (485) using blind rivet (505). Attach screen assembly (420) on platen assembly (485) using captive fastener. Attach silicone rubber heater (530) and heater retainer (545) to platen assembly (485), lock nut (550). Attach platen assembly (485) to housing assembly (735) using lock washers (495) and screws (490). Slide tubing (535) over heater leads and attach butt splices (540) to leads. Secure heater leads to platen assembly (485) using clamp (515), washer (510, lock washer (525) and screw (520).
- (23) Install rail guide (425) using screws (430). Install rail guide (435) using screws (440).

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- (24) Attach wire assembly (160) to thermal switch assembly (135) with supplied hardware. If removed, attach ring terminals (140) to leads. Attach ring terminals (140) to thermal switch assembly (135) using screws (145), washers (150), and nuts (155). Attach thermal switch cover (120) using standoffs (125) and screws (130) to tank assembly (50).
- (25) Attach drain valve assembly (165) to connector support (445) using water shutoff plug (170) and o-ring (175). Attach alignment guide pin (470) to connector support (445) using lock washer (475) and hex nut (480). Attach connector support (445) to housing assembly (735) using screws (450). Install screw (455), washer (460), and nut (465) in connector support (445).
- (26) Attach mounting brace plate (255) to housing assembly (735) using screws (260).
- (27) Attach support block (240) on housing assembly (735). Attach quarter turn retainer (250) to support block (240).
- (28) Refer to Paragraph 2.N for coffee maker/water boiler final assembly.

P. Coffee Maker/Water Boiler (refer to IPL Figure 1)

- (1) Assemble brew cup assembly (10) (consisting of items 20 through 30). Install spacer plate (20) in brew cup (15) and secure with screws (30) and nuts (25).
- (2) Refer to TESTING AND FAULT ISOLATION section for procedures to test the coffee maker/water boiler subassembly following assembly.

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SPECIAL TOOLS, FIXTURES, AND EQUIPMENT

1. General

- A. All work described in this manual may be conducted in any normal work area with ambient environmental and physical conditions and whose environment has been cleaned for coffee maker/water boiler assembly repair.
- B. Special tools, fixtures, and equipment required to perform the work function listed in all paragraphs of this manual are listed in Table 9001.

NOTE: Equivalent substitutes may be used for listed items.

Table 9001
Special Tools, Fixtures, and Equipment List

NOMENCLATURE	PARTS/SPEC NO	SOURCE (CAGE)
Air	Oil-free air	Oil-Free Compressor
Alpha soldering chemical	810T	Commercially Available
Brush, Soft-bristled	None	Commercially Available
Card Extension Adapter	60-2344-1	Commercially Available
Cleaner	Mild Soap Solution, Liquid Detergent	Commercially Available
Cloth, Cleaning, Low-Lint,	None	Commercially Available
Conformal coating	1B31	Columbia Chase Corporation (V99109)
	CM-50021	B/E Aerospace, Inc. (V63367)
Conformal coating	PC-18	Dexter Corporation Hysol Division (V7H439)
	CM-238	B/E Aerospace, Inc. (V63367)
Digital Thermometer and Thermocouple	None	Commercially Available
Distilled Water	None	Commercially Available
Isopropyl Alcohol	Approximate 90% Isopropyl Alcohol Solution	Commercially Available
Jumper for J5 and J6 tied together	None	Commercially Available
Loctite (Blue)	242 AD-300	Loctite Corporation (V05972) B/E Aerospace, Inc. (V63367)
Loctite (Red)	262 AD-156	Loctite Corporation (V05972) B/E Aerospace, Inc. (V63367)
Megohmmeter	Associated Research Model 201	Commercially Available

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Table 9001 (Continued)
Special Tools, Fixtures, and Equipment List

NOMENCLATURE	PARTS/SPEC NO	SOURCE (CAGE)
Milliohm meter	QuadTech Model 1880	Commercially Available
Protective Gloves	None	Commercially Available
Rail Assembly	416-0201-15	B/E Aerospace, Inc. (V63367)
Resistance Bridge	None	Commercially Available
Screw Aluminum Adapter	None	Commercially Available
Sealant, white	#738 - white AD-053	Dow Corning Corporation (V5D028) B/E Aerospace, Inc. (V63367)
Server	PS-3374	Commercially Available
Stop Watch	None	Commercially Available
Sulfamic Acid-based cleaner or other agent that acts upon calcium, lime and rust deposits	Approximately 5% Sulfamic Acid Solution, LimeAway, vinegar, or agent that removes calcium, lime and rust deposits	Commercially Available
Tape, teflon (1/4 inch wide)	MS-STR-4 TT-7454	Crawford Fitting (V1YJV3) B/E Aerospace, Inc. (V63367)
Test bench, consisting of flow meter, adjustable water supply with pressure gauge, power supply, 3 ammeters and voltmeter.	None	Commercially Available
Test Harness	None	Commercially Available
Tubing	TENT001TTV03512 TU-379	Acme Plastics (1EN65) B/E Aerospace, Inc. (V63367)
Tubing, heat shrink	M23053/8-004-C TU-310	Raychem Corporation (V06090) B/E Aerospace, Inc. (V63367)
Tubing, heat shrink	RNF-100 3/16 TU-448	Raychem Corporation (V06090) B/E Aerospace, Inc. (V63367)
Tubing, heat shrink	RNF-100 1/4 TU-4705	Raychem Corporation (V06090) B/E Aerospace, Inc. (V63367)
Water hose with thermocouple	None	Commercially Available
10K Ohm Potentiometer, 1 Watt minimum	3701H-1-103 or None	Bourns Instruments Inc. (V32997) Commercially Available

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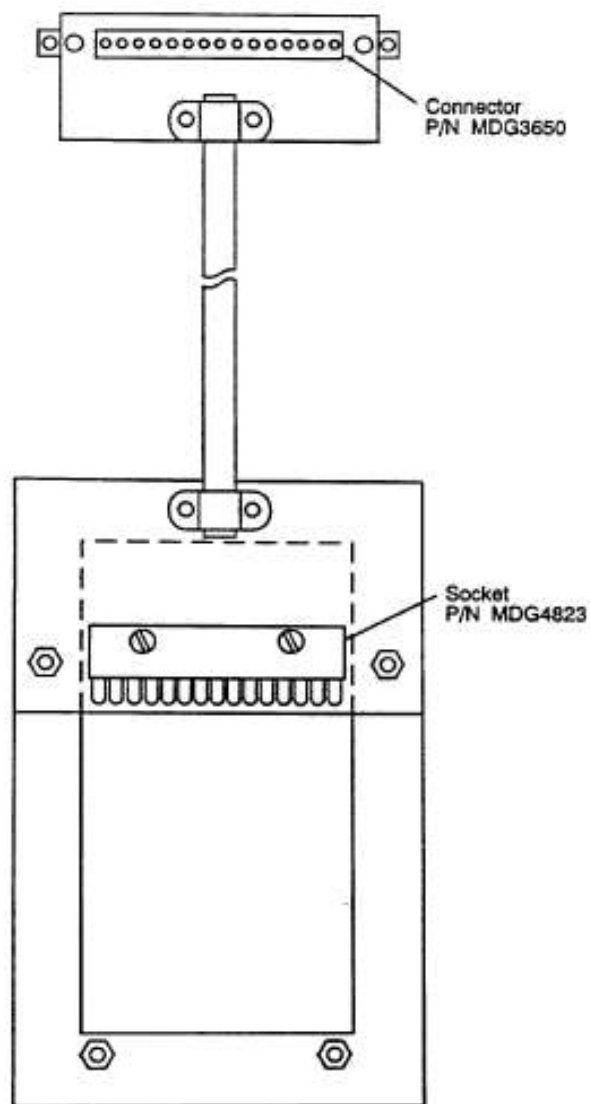


Figure 9001
Card Extension Adaptor - PN 60-2344-1

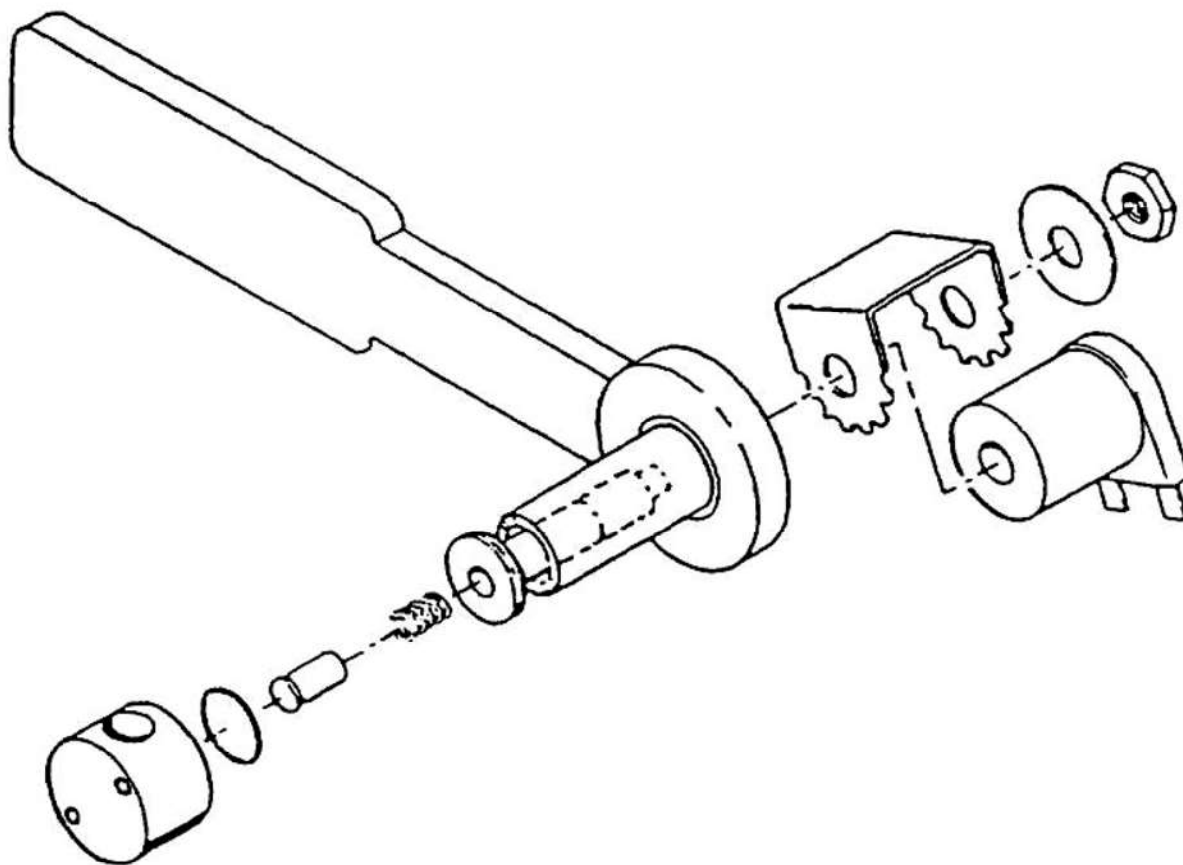


Figure 9002
Socket Spanner - PN 3525-0052-01

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ILLUSTRATED PARTS LIST

1. Introduction

- A. The ILLUSTRATED PARTS LIST is intended for use in provisioning, storing, and issuing replaceable parts for the component, and in identification of new and reclaimed parts.

2. Vendor's Parts

- A. Vendor's parts used in the unit and which are not altered by B/E Aerospace, Inc are listed in the Detailed Parts List by the vendor's part number, vendor's description of the part, and vendor's code listed in parentheses following the description. Vendors' code symbols used in this publication are taken from the CAGE Cataloging Handbook, and consist of the applicable code symbol preceded by the Letter "V." Table 10001 shows a numerically arranged list of vendor codes used in this publication.

Table 10001
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V00779	Tyco Electronics Corporation 2800 Fulling Mill Road Building 38 Middletown, PA 17057
V02288	Allied Controls Inc. 150 East Aurora Street Suite 3 Waterbury, CT 06708
V02570	Swagelok Company 29500 Solon Road Solon, OH 44139
V02697	Parker-Hannifin Corporation 2360 Palumbo Drive Lexington, KY 40509
V04713	Freescall Semiconductor Inc 2100 East Elliot Road Tempe, AZ 85284
V05972	Loctite Corp 1001 Trout Brook Xing Rocky Hill, CT 06067
V06090	Tyco Electronics Corporation 300 Constitution Drive Menlo Park, CA 94025
V09922	Souriau Usa Inc 25 Grumbacher Road York, PA 17406
V0AFE8	Transtar Metals Corporation 14400 South Figueroa Gardena, CA 90248

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Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V0BED2	BAE Systems Imaging Solutions Inc 1801 Mccarthy Boulevard Milpitas, CA 95035
V0DTW8	Texas Instruments, Inc. 11816 Borman Drive Saint Louis, MO 63146
V0GD60	National Semiconductor Corporation 1111 West Bardin Road Arlington, TX 76017
V0J4G8	Panasonic Industrial Company Affil of Matsushita Electric Corporation of America 2 Panasonic Way Secaucus, NJ 07094
V0KHJ5	Precision Metal Industries Inc. 1408 Sw 8th Street Pompano Beach, FL 33069
V0MT56	Britech Industries/Scully Enterprises, Inc. 14839 South West 110 Terrace Miami, FL 33196
V0WT32	Dupont Electronic Materials Inc Highway 686 Km 2 Manati, PR 00674
V12060	Diodes, Inc. 3050 East Hillcrest Drive Westlake Village, CA 91362
V12405	Henkel Loctite Corp 15051 East Don Julian Road City Of Industry, CA 91746
V14099	Semtech Corporation 200 Flynn Road Camarillo, CA 93012
V14726	Electro-Term Inc. Crystal Park-Building Pompano Beach, FL 33064
V14936	General Semiconductor Inc 10 Melville Park Road Melville, NY 11747
V1EN65	Acme Plastics 6437 B General Green Way Alexandria, VA 22312

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Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V1KA24	Automatic Coax and Cable, Inc. DBA Automatic Cable 4060 St. Johns Parkway Sanford, FL 32771
V1MQ07	Semiconductor Components Industries LLC DBA Semi Conductor Components Industries Division Corporate Headquarters 5005 East McDowell Road Phoenix, AZ 85008
V1YJV3	Crawford Fitting Co Ltd 29500 Solon Road Solon, OH 44139
V22526	FCI USA LLC 825 Old Trail Road Etters, PA 17319
V24444	General Semiconductor Industries, Inc. Sub. of Square D Co. 2001 W. 10th Place Tempe, AZ 85281
V25184	Precision Rubber Products Corporation 104 Hartman Drive Lebanon, TN 37087
V25281	Oetiker Inc. 6317 Euclid Suite Marlette, MI 48453
V27264	Molex Incorporated 2222 Wellington Court Lisle, IL 60532
V30161	Aavid Thermalloy LLC 67 Primrose Drive Laconia, NH 03246
V30528	Hollingsead International, Inc. 21583 Baltimore Avenue Georgetown, DE 19947
V30682	FCI, Inc. 4661 Giles Road Cleveland, OH 44135
V30780	Parker-Hannifin Corporation 3885 Gateway Boulevard Columbus, OH 43228
V31433	Kemet Electronics Corporation 2835 Kemet Way Simpsonville, SC 29681

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Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V32997	Bourns Inc. 1200 Columbia Avenue Riverside, CA 92507
V43990	C.A. Norgren Co. 5400 South Delaware Street Littleton, CO 80120
V4J413	Replaced by V02697
V50088	St. Microelectronics, Inc. 1310 Electronics Drive Carrollton, TX 75006
V50948	Parker-hannifan Corp. 220 Roberts Cut Off Rd Fort Worth, TX 76114
V51814	Smalley Steel Ring Company 555 Oakwood Road Lake Zurich, IL 60047
V52607	Actron Manufacturing Inc. 1841 Railroad Street Corona, CA 92280
V55680	Nichicon America Corporation 927 East State Parkway Schaumburg, IL 60195
V56289	Sprague Electric co. World Headquarters 678 Main Street Sanford, ME 04073
V56637	RCD Components, Inc. 520 East Industrial Park Drive Manchester, NH 03109
V56878	SPS Technologies LLC 301 Highland Avenue Jenkintown, PA 19046
V59993	International Rectifier Corporation 233 Kansas Street EL Segundo, CA 90245
V5D028	Dow Corning Corporation 3901 S Saginaw Rd Midland, MI 48686
V63367	B/E Aerospace Inc. 10800 Pflumm Road Lenexa, KS 66215

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Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V70485	Atlantic India Rubber Works Inc. 1437 Kentucky Rt 1428 Hagerhill, KY 41222
V71771	Cory Components Inc. 2201 Rosecrans Avenue El Segundo, CA 90245
V75915	Littelfuse Inc. 8755 West Higgins Road Suite 500 Chicago, IL 60631
V76854	Electro Switch Corporation 2010 Yonkers Road Raleigh, NC 27604
V77342	Tyco Electronics Corporation 8010 Piedmont Triad Parkway Greensboro, NC 27409
V78189	ITW Shakeproof Automotive Products 1201 St. Charles Road Elgin, IL 60120
V79074	Varflex Corporation 512 West Court Street Rome, NY 13440
V79963	Zierick Manufacturing Corporation 131 Radio Circle Mount Kisco, NY 10549
V7H439	Replaced by V12405
V7T184	Nte Electronics Inc Sub of Solid State Inc 44 Farrand Street Bloomfield, NJ 07003
V83131	Olson Mfg Company 620 South Broadway Avenue Albert Lea, MN 56007
V83330	Dialight Corporation 1501 State Route 34 South Farmingdale, NJ 07727
V83803	Peerless Electronics Inc 700 Hicksville Road Suite 100 Bethpage, NY 11714
V85480	Brady United States 6555 West Good Hope Road Milwaukee, WI 53223

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Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V87924	Meyer Company The DBA Tomlinson Industries 13700 Broadway Avenue Cleveland, OH 44125
V8Z410	Ledtronics, Inc. 23105 Kashiwa Ct Torrance, CA 90505
V91929	Honeywell International Inc. 315 East Stephenson Street Freeport, IL 61 032
V94222	Southco Inc. 210 North Brinton Lake Road Concordville, PA 19331
V95124	Beatrice Foods Company Wrightway Manufacturing Company Division 1050 Central Avenue University Park, IL 60466
V98410	ETC-Molex, Inc. 4650 62nd Avenue Pinellas Park, FL 33781
V99109	Chase Corporation Dba Royston Laboratories Div Humiseal Division 128 First Street Pittsburgh, PA 15238
V99699	DRI Relays Inc. 55 Engineers Road Hauppauge, NY 11788

3. Use of the Illustrated Parts List

A. When the part number is known:

Turn to the Numerical Index and locate the part number. Figure and item numbers on the illustration where the part appears are listed in the column to the right of the part number. Turn to the illustration and locate the item number. The corresponding item number in the accompanying Detailed Parts List will give part number, description, assembly relationship, quantity required for that particular application, and whether or not the part appears in the illustrations.

B. When the part number is not known:

Look through the figures and identify the part by appearance. Note the figure number and the item number. The corresponding Fig-Item number in the Detailed Parts List shows the part number, description, assembly relationship, effectivity code, and the number of units per assembly.

4. Material Arrangement

A. Numerical Index

The Numerical Index is an alphanumeric listing of all the part numbers in the Detailed Parts List. Each part number is listed together with its corresponding item number, ILLUSTRATED PARTS LIST figure number, and the total required number of units per assembly. The figure-item number is composed of two parts. To the left of the dash is the figure number. It is the number of the figure in the ILLUSTRATED PARTS LIST that shows the part. To the right of the dash is the item number. It is the number that identifies a part in a figure.

B. Illustrations

The ILLUSTRATED PARTS LIST illustrations interrelate to the Numerical Index and the Detailed Parts List. The separate illustrations pictorially describe recommended disassembly/assembly sequence and support DISASSEMBLY and ASSEMBLY as well as the Numerical Index and the Detailed Parts List of this manual.

C. Detailed Parts List

(1) Fig-Item

The Detailed Parts List is arranged numerically by item numbers. These are the same item numbers listed in the Numerical Index. In the Detailed Parts List, the figure number portion of the figure-item number is listed only once per column. Immediately below a figure number are the item numbers that appear on that figure. A dash in front of an item number means the corresponding part is not shown in that or any other figure. Each item number is listed together with its corresponding part number, nomenclature (description, assembly relationship), effectivity code, and units per assembly.

(2) Part Number

The part number column contains the original manufacturer's part number. When standard parts are used, the standard part number is listed in this column. When B/E Aerospace, Inc. is not the original manufacturer the vendor's part number is shown in the part number column, and the CAGE vendor number and equivalent B/E Aerospace, Inc. part number are given in the nomenclature column.

(3) Airline Part Number

This column is left blank for airline internal use.

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(4) Nomenclature

(a) Indenture System

The indenture system shows the relationship of parts and assembly to next higher assembly or installations, as follows:

1 2 3 4 5 6 7

Unit (Product which this manual describes)

- Detail parts for assembly into the unit
- Subassembly
- Attaching parts for subassembly
- * * *

- • Detail parts for subassembly
- • Sub-subassembly
- • Attaching parts for sub-subassembly
- * * *
- • • Detailed parts for sub-subassembly

(b) Attaching Parts

Attaching parts are captioned ATTACHING PARTS and are listed immediately following the parts attached. The * * * symbol follows the last item of the attaching parts group.

(c) Vendor Code

Parts manufactured by companies other than B/E Aerospace, Inc. are identified by an appropriate vendor's CAGE code following the nomenclature, and are preceded by the letter "V." Standard parts such as AN, MS, etc., are not identified by a vendor code.

(5) Effectivity Code

When two or more units (main assembly) are listed in the same ILLUSTRATED PARTS LIST, a code ("A," "B," "C," etc.) is assigned to each main unit. All subcomponents that are peculiar to a particular unit are identified by the same effectivity code letter as the unit. If parts are common to all units, the effectivity code column is left blank.

(6) Units per assembly

Quantities specified in the "UNITS PER ASSEMBLY" column are the total number of each part required per assembly or subassembly and are not necessarily the total used per the complete unit.

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5. Parts Replacement Data

Interchangeability relationship between parts is identified in the 'NOMENCLATURE' column. A list of terms used and their definition is as follows.

TERM	ABBREVIATION	DEFINITION
Alternate	ALT	Part fully meets functional and structural specifications, but differs either in overall external dimensions, connection installation, and/or mounting provisions, and may require additional rework or modifications to install in a specific application.
Optional	OPT	The part is fully interchangeable in form, fit, and function with other item numbers shown.
Preferred	PRFD	Part is preferred over the other optional parts shown.
Replaced By	REPLD BY	Part is replaced by and is interchangeable with item number listed in notation.
Replaces	REPLS	Part replaces and is interchangeable with number listed in notation.
Superseded By	SUPSD BY	Part is replaced by and is not interchangeable with item number listed in notation.
Supersedes	SUPSDS	Part replaces and is not interchangeable with item number listed in notation.
Service Bulletins	PRE SB POST SB SB RWK	Identifies when a service bulletin (SB) has an effect on a part. PRE shows the condition of the part before the work in the service bulletin is done; POST shows the condition after the work is done; RWK shows that the part is reworked by the service bulletin.

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6. Numerical Index

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
A35581-4R		2	250	1
AN960-10L		4	190	1
		4	220	1
		4	245	1
AN960-6L		3	130	1
AN960-8L		4	105	4
AN960C6		7	40	1
AN960C6L		2	70	2
AN960D10L		4	185	2
		4	215	2
		4	240	2
AN960D6		2	565	DEL
AN960D6L		3	125	1
B1870BN		2	540	2
B1870BNT		2	-540A	2
B1871BNT		4	-200	1
BB8141		4	-35A	2
BB8141-1		4	35	2
CDC06-16S1P		4	270	1
CO6505-01		4	271	1
CO6505-03		4	-272	6
CO6505-05		4	273	1
COMMERCIAL		2	-45A	2
		2	60	2
		2	-510A	1
		2	-600A	1
		4	-255	1
		4	-255A	1
		4	-270A	1
		6	-85	DEL
F4NY-125BK		2	515	1
F4NY-250BK		2	595	1
		3	100	1
F4NY250-0		2	-595A	1
		3	-100A	1
FF200OCW028B		3	-35A	5
JE5		11	25	1
KR-90693-1		4	65	1
KR-90693-2		4	-65A	1
M35338-136		10	55	2
M83536/33-003		4	65B	1
MS15795-706		4	-15	DEL
MS15795-803		4	160	2
MS15795-805		10	50	3
MS15795-807		2	325	2
		2	460	1
		10	35	2
MS16625-4031		5	90	1
MS16633-4012		2	705	1

- ITEM NOT ILLUSTRATED

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS18029-1S5		4	-75A	1
MS18029-6S4		4	75	1
MS20604AD3W1		2	505	2
MS21042L3		4	195	1
		4	225	1
		4	250	1
MS21043-04		8	15	6
MS21043-06		2	570	1
		2	590	4
		2	610	1
		4	155	2
		10	-20A	4
		10	80	2
MS21209CO810		2	-736	1
MS21209CO815		2	-737	7
MS24693C1		2	190	2
MS24693C24		2	15	1
MS24693C25		2	35	6
		2	260	4
		2	270	4
		4	125	1
MS24693C26		3	110	1
		3	135	1
MS24693C27		2	745	2
MS24693C31		10	70	2
MS24693C48		2	25	6
		2	225	2
		2	350	1
		2	370	1
		2	430	3
		2	440	3
		2	450	5
		2	680	1
MS24693C49		2	-235A	2
		2	245	DEL
		5	10	2
MS24693C5		4	85	2
MS24693C53		2	310	2
		2	665	1
MS25226-2-2		4	-275	1
MS25237-387		3	35	5
MS27212-1-5		4	-80A	1
MS27212-6-4		4	80	1
MS3367-4-9		2	730	14
		3	95	5
MS35338-134		11	30	2
MS35338-135		2	215	2
		2	295	2
		2	-725A	2
MS35338-136		2	75	1
		2	150	6

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS35338-136		2	395	2
		2	410	2
		2	495	8
		2	525	1
		2	725	2
		4	-55	DEL
		6	-32	1
		7	45	1
		8	-30	
		10	-55A	2
MS35338-137		10	40	2
MS35338-138		2	635	1
MS35338-41		4	-110A	5
MS35338-42		4	110	4
MS35649-262		4	-115A	5
MS35649-264		2	80	1
		2	155	6
MS35649-282		7	50	1
		8	-35	DEL
		4	115	4
		4	-150	2
MS51859-3		2	290	2
MS51957-14		2	210	2
MS51957-15		4	165	2
		2	405	2
MS51957-26		2	490	8
		2	585	4
		6	30	1
		8	-25	
MS51957-27		2	145	6
		2	520	1
MS51957-28		2	-585A	4
		2	720	2
		4	-60	DEL
MS51957-29		10	45B	1
		2	390	2
		4	-10	DEL
MS51957-30		10	-60A	2
		2	-560A	1
		2	-605A	1
MS51957-33		2	130	4
MS51957-43		2	235	2
MS51957-45		10	-45A	2
MS51957-46		2	45	2
		2	-55A	8
		2	455	1
MS51958-61		2	380	2
MS51958-68		2	630	1
NAS1149CN616R		2	-70A	2
NAS1149D0316H		4	-185A	2
		4	-215A	2

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
NAS1149D0316H		4	-240A	2
NAS1149F0332P		4	-190A	1
		4	-220A	1
		4	-245A	1
NAS1149FN616B		4	-105A	5
NAS1189E06P5L		1	30	2
NAS1801-3-7		4	180	1
		4	210	1
NAS1801-3-8		4	235	1
NAS620C6		2	-727	2
		10	75	2
S1138-702-005		9	-15A	1
S11382-010		9	-20A	1
		10	-95A	2
S11382-115		6	-50A	1
		6	-70A	1
		6	-150A	3
S11382-110		2	-175A	1
SS405-3		2	95	2
		2	115	1
		10	115	
TENT001TTV03512		2	-60A	2
TYPE HO, SIZE 9		2	535	1
TYPE PFA		2	-110	DEL
VHL33QPAS02		5	-90A	1
W4NY125-0		2	-515A	1
WA-120		2	510	1
		2	600	1
		3	105	1
X1903SNT		4	205	1
XR1891SNT		4	-175A	1
XR1902SNT		4	100	4
XR1903SNT		4	230	2
XR26343N		2	65	1
		2	140	3
		2	580	4
XR26344N		2	415	3
		4	-95	1
XR26345N		4	175	1
CL1-2107-267		4	5A	1
005-3077		9	15	1
006-3077		7	15	1
010-3077		6	-90	DEL
		7	30	1
		9	20	1
		10	95	2
02-09-1202		4	-30	DEL
02-09-1206		4	-25	DEL
0500-0243		7	55	1
100KB		2	775	1
1011088		2	770	1
110-3077		2	175	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
115-3077		6	50	1
115-3077		6	70	1
		6	150	3
1410-01		6	115	1
1410-02-523CC7		6	-115A	1
1416-6		3	120	1
1497-03		6	125	1
1497-06-523CC7		6	-125A	1
1534-01		6	120	1
1534-05-523CC7		6	-120A	
16700983		2	-200A	1
		2	-790	1
		10	-110A	1
1724-02		2	755	2
1726-03		2	475	1
1N4007		5	-80	1
1SE3-3		11	10	1
2-1174-144		3	75	5
2-2718-620		3	80	5
22FT832		2	330	2
		2	465	1
2903-26012		3	-90A	2
		4	-265A	1
2903Z6012		3	90	2
		4	265	1
3-3767-638		3	60	5
349-0544-11		2	715	1
349-0551-1		3	70	5
3510-0052-01		1	-1	RF
352-0036-3		2	135	1
		8	-1	RF
352-0134-1		2	615	1
352-0183-1		8	-20	DEL
352-0282-1		3	65	5
352-0980-1		2	120	1
352-0981-1		2	125	4
352-0983-1		8	10	3
352-0984-1		8	5	1
3520-0177-07		2	50	1
		6	-1	RF
3520-0177-11		2	-50A	1
		6	-1A	RF
3520-0215-01		2	-335	1
3520-0215-02		2	-355	1
3520-0215-03		2	340	1
3520-0215-04		2	360	1
3520-0216-01		2	-305	1
3520-0228-01		10	-85	1
3520-0228-03		10	105	1
3520-0235-05		2	20	1
		4	-1	RF

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0236-01		2	185	1
3520-0241-03		2	5	1
		3	-1	RF
3520-0244-01		4	-70A	1
3520-0244-03		4	70	1
3520-0275-01		1	5	1
		2	-1	RF
3520-0276-01		2	40	1
		5	-1	RF
3520-0314-01		2	-135A	1
		8	-1A	RF
3520-0319-01		2	780	1
3520-0343-01		2	-20A	1
		4	-1A	RF
3520-0357-01		2	-5A	1
		3	-1A	RF
3520-0523-01		6	-60A	
3530-0059-05		4	140	1
3530-0069-01		2	285	1
3530-0070-01		2	500	1
3540-0001-01		6	-60	1
3575-0010-01		1	20	1
3575-0021-01		4	45	DEL
3575-0185-01		5	100	1
3575-0189-01		10	65	1
3575-0191-01		3	-65A	5
3575-0388-01		1	25	2
3575-0432-01		5	85	1
3575-0584-01		6	80	1
3575-0604-01		4	120	1
		4	145	2
3575-0608-01		2	195	1
3575-0610-01		4	130	1
3575-0611-01		4	135	1
3575-0633-01		2	660	1
3575-0634-01		10	45C	1
3575-0647-01		6	45	1
3575-0650-01		2	-670	1
3575-0650-03		2	685	1
3575-0692-01		2	220	1
3575-0693-01		2	230	1
3575-0694-01		2	240	1
3575-0893-01		2	-785	1
3575-1251-07		6	-145A	3
374-0872-5		3	25	1
4-2CBZ-SS		5	25	1
		6	20	
4-2CBZSSCD960053		5	-25A	1
4-2FBZSSC		5	15	1
4-4-4JBZSSC		2	-114	1
4-BZSSC		2	105	2

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
405-0044-13		3	5	1
405-0215-1		5	5	1
405-0437-3		5	20	1
411-0676-3		2	640	2
411-0686-1		2	655	1
411-0692-1		2	300	2
411-0693-1		10	10	1
411-0695-1		2	375	1
411-0701-1		2	275	1
411-0709-1		10	15	1
411-0715-1		2	280	1
411-0720-1		11	20	1
411-0721-17		2	385	1
		11	-1	RF
411-0721-19		11	-5	1
411-0724-1		2	315	2
411-0725-1		2	320	2
411-0733-25		2	735	1
411-0745-1		10	5	1
411-0748-3		10	30	1
411-0748-7		10	25	1
411-0926-1		2	620	2
412-0129-7		2	555	1
416-0068-1		2	-112	1
416-0088-1		2	255	1
416-0139-3		2	-160A	1
		2	575	1
416-0396-17		3	20	1
416-0396-19		3	15	1
416-0463-1		2	205	1
416-0628-13		2	-265	1
423-0078-1		2	425	1
423-0078-3		2	435	1
425-0173-3		2	160	1
425-0702-1		2	345	1
		2	365	1
428-0074-1		6	170	1
42CBZSSCD960053		6	-20A	1
440-0150-3		6	145	3
445-0908-1		7	20	1
445-0909-1		7	5	1
445-0910-11		6	-140A	1
		7	-1A	RF
445-0910-9		6	140	1
		7	1	RF
445-0912-1		7	25	1
445-0915-1		7	35	1
445-0916-1		7	10	1
467-0865-1		10	100	1
467-0912-1		10	90	1
47-10-302-50		2	690	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
47439-000LF		4	-40A	3
		11	-15	2
47712		4	-40	3
4BZSSC		5	110	1
4TZSS		2	100	2
		5	105	1
500-0731-1		6	155	1
50FA440		4	90	2
50FK-632		4	-20	DEL
		2	550	1
		3	115	1
		3	140	1
		10	20	4
51863-1		2	-65A	1
		2	-140A	3
		2	-415A	3
		4	-95A	3
51864-3		4	-100A	2
532		3	55	1
55-0207-29		10	60	2
55-0207-30		2	560	1
		2	605	1
55-0207-45		10	45	2
55-0207-46		2	55	8
55-0392-1		2	530	1
55-0392-3		2	-530A	1
55-0393-7		2	760	1
55-0396-1		3	10	1
55-0396-7		3	30	1
55-0397-1		6	55	1
55-0397-3		6	-55A	1
5710		3	40	1
5720		3	50	1
5820		3	45	2
60-2001-1		2	650	1
60-2010-1		2	480	1
60-2011-1		2	750	2
60-2013-1		2	710	1
61-0814-1		4	-260	2
61-0817-1		2	30	1
61-1675-11		4	170	1
62-0007-1		6	40	1
62-0008-1		9	5	1
62-0010-1		2	170	1
62-0011-1		6	35	1
62-0013-1		9	10	1
62-0014-1		2	165	1
		9	1	RF
62-0018-1		2	445	1
62-0020-1		6	-25	1
62-0024-1		2	400	1

- ITEM NOT ILLUSTRATED

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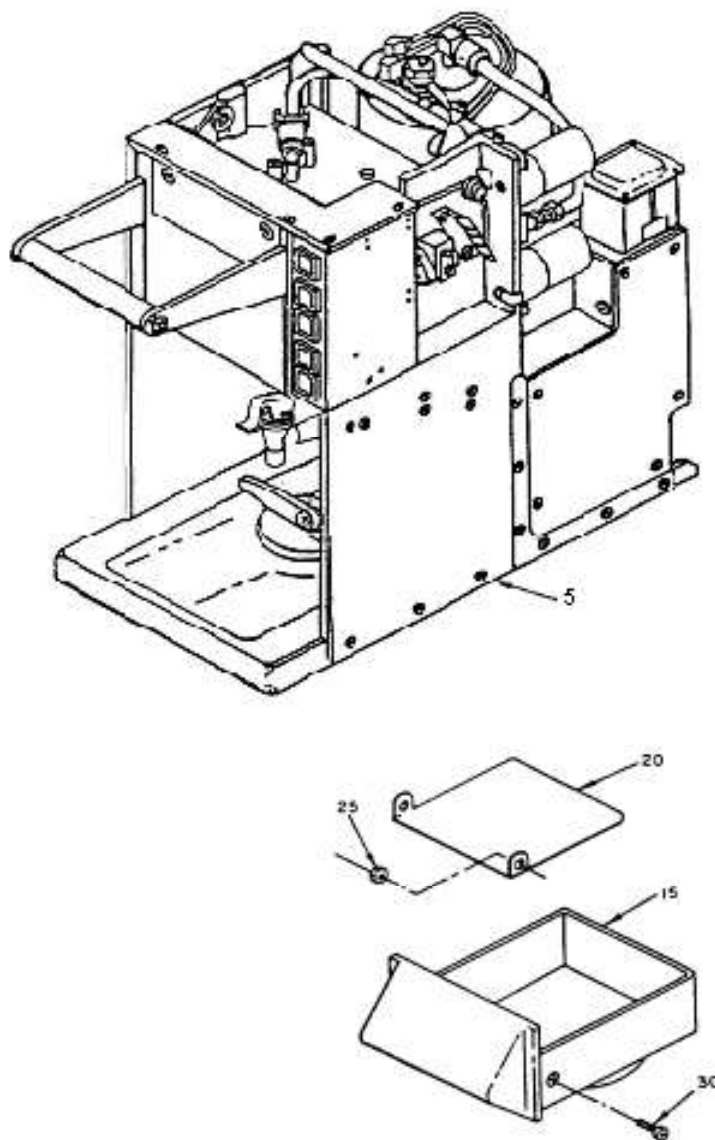
PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
62-0025-1		2	645	1
62-0037-1		2	470	1
62-0040-1		2	740	1
62-0061-1		2	-625	1
62-0066-1		5	95	1
		10	120	1
62-0077-3		2	545	1
62-0087-3		2	10	1
62-0118-11		2	485	1
62-0125-1		2	-765	1
62-0129-1		2	-555A	1
62-0131-1		2	420	1
62-0132-1		6	82	1
62-0134-1		6	-80A	1
62-0135-1		6	-60B	1
62-0136-11		6	10	1
62-0136-7		6	-5	1
		6	135	1
62-0136-9		6	15	1
62-0144-1		2	-57	1
62-0156-1		6	-65	1
62-0156-3		6	75	1
62-0196-1		2	-85	1
62-0196-3		2	90	1
62-0210-1		2	695	1
62-0210-2		2	700	1
62-0211-1		2	675	1
62-0223-1		6	-95	1
62-0223-11		6	130	1
62-0223-3		6	-100	1
62-0223-5		6	105	1
62-0223-9		6	110	1
62-0234-1		4	50	DEL
62-0248-17		2	180	1
		10	-1	RF
62-0250-11		5	55	1
62-0250-15		5	60	1
62-0250-17		5	65	1
62-0250-19		5	70	1
62-0250-21		5	75	1
62-0250-25		5	-30	2
62-0250-3		5	35	1
62-0250-5		5	40	1
62-0250-7		5	45	1
62-0250-9		5	50	1
65846-006		3	85	2
65846-006LF		3	-85A	2
665-0300-11		1	15	1
665-0343-3		1	-10	1
665-0423-3		4	5	1

- ITEM NOT ILLUSTRATED

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
7.0-505R		2 10	200 110	1 1
- ITEM NOT ILLUSTRATED				

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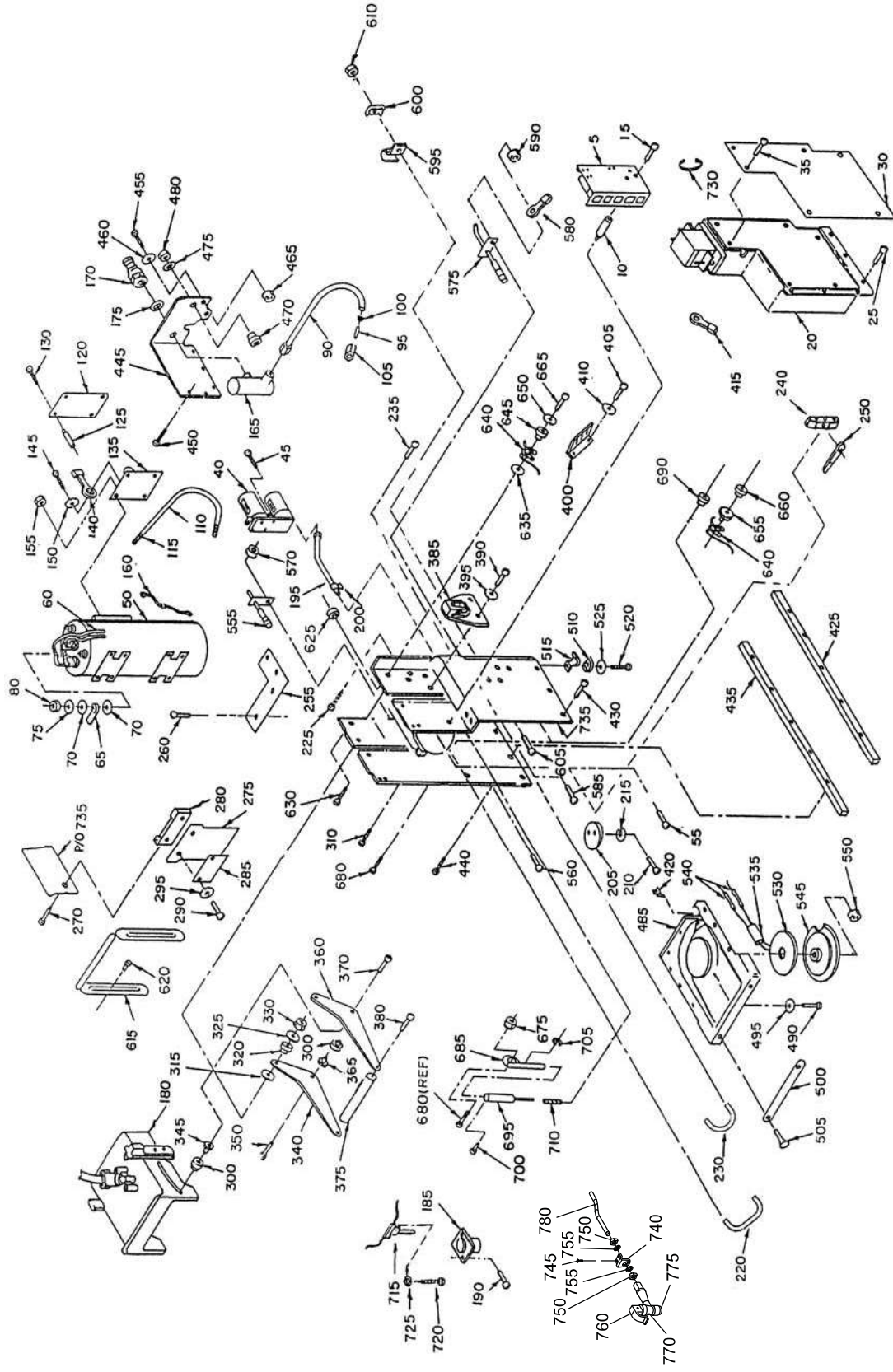


IPL Figure 1
Coffee Maker/Water Boiler Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-1	3510-0052-01		COFFEE MAKER/WATER BOILER		RF
5	3520-0275-01		• COFFEE MAKER/WATER BOILER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)		1
-10	665-0343-3		• BREW CUP ASSEMBLY		1
15	665-0300-11		• • CUP, BREW		1
20	3575-0010-01		• • PLATE, SPACER		1
25	3575-0388-01		• • NUT, HEX		2
30	NAS1189E06P5L		• • SCREW		2

- ITEM NOT ILLUSTRATED



IPL Figure 2
 Coffee Maker/Water Boiler Sub Assembly - Exploded View

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-1	3520-0275-01		COFFEE MAKER/WATER BOILER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)		RF
5	3520-0241-03		• SWITCH ASSEMBLY (SEE IPL FIGURE 3 FOR DET BKDN) (SUPSD BY ITEM 5A)		1
-5A	3520-0357-01		• SWITCH ASSEMBLY (SEE IPL FIGURE 3 FOR DET BKDN) (SUPSDS ITEM 5)		1
			ATTACHING PARTS		
10	62-0087-3		• STUD, SWITCH HOUSING		1
15	MS24693C24		• SCREW, FLAT HEAD		1
			* * *		
20	3520-0235-05		• POWER MODULE ASSEMBLY (SEE IPL FIGURE 4 FOR DET BKDN) (SUPSD BY ITEM 20A)		1
-20A	3520-0343-01		• POWER MODULE ASSEMBLY (SEE IPL FIGURE 4 FOR DET BKDN) (SUPSDS ITEM 20)		1
			ATTACHING PARTS		
25	MS24693C48		• SCREW, FLAT HEAD		6
			* * *		
30	61-0817-1		• COVER, POWER MODULE		1
			ATTACHING PARTS		
35	MS24693C25		• SCREW, FLAT HEAD		6
			* * *		
40	3520-0276-01		• SOLENOID VALVE ASSEMBLY (SEE IPL FIGURE 5 FOR DET BKDN)		1
			ATTACHING PARTS		
45	MS51957-46		• SCREW, PAN HEAD (SUPSD BY ITEM 45A)		2

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-45A	COMMERCIAL		• SCREW, FILLISTER HEAD (B/E AEROSPACE STOCK CODE SC-50241) (SUPSDS ITEM 45) * * *		2
50	3520-0177-07		• TANK ASSEMBLY (SEE IPL FIGURE 6 FOR DET BKDN) (SUPSD BY ITEM 50A)		1
-50A	3520-0177-11		• TANK ASSEMBLY (SEE IPL FIGURE 6 FOR DET BKDN) (SUPSDS ITEM 50) ATTACHING PARTS		1
55	55-0207-46		• SCREW, SLOTTED, PAN HEAD (SUPSD BY ITEM 55A)		8
-55A	MS51957-46		• SCREW, PAN HEAD (SUPSDS ITEM 55) * * *		8
-57	62-0144-1		• CLIP		1
60	COMMERCIAL		• TUBING (V1EN65) 0.12 IN. ID X 0.03 IN. WALL X 9 IN. LONG (B/E AEROSPACE STOCK CODE TU-379) (SUPSD BY ITEM 60A)		2
-60A	TENT001TTV03512		• TUBING (V1EN65) 0.12 IN. ID X 0.03 IN. WALL X 13.5 IN. LONG (B/E AEROSPACE STOCK CODE TU-379) (SUPSDS ITEM 60)		2
65	XR26343N		• TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-096) (SUPSD BY ITEM 65A)		1
-65A	51863-1		• TERMINAL, RING (V00779) (B/E AEROSPACE STOCK CODE TE-096) (SUPSDS ITEM 65) ATTACHING PARTS		1
70	AN960C6L		• WASHER, FLAT (SUPSD BY ITEM 70A)		2
-70A	NAS1149CN616R		• WASHER, FLAT (SUPSDS ITEM 70)		2
75	MS35338-136		• WASHER, LOCK		1
80	MS35649-264		• NUT, HEX * * *		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-85	62-0196-1		• TUBE ASSEMBLY, WATER SUPPLY		1
90	62-0196-3		• • TUBE (FORMED)		1
95	SS405-3		• • FITTING (V02570) (B/E AEROSPACE STOCK CODE PF-202)		2
100	4TZSS		• • FERRULE (V30780) (B/E AEROSPACE STOCK CODE PF-50002) (TPN 4TZ-SS)		2
105	4-BZSSC		• • NUT, HEX (V50948) (B/E AEROSPACE STOCK CODE PF-143) (TPN 4-BZ-SSC)		2
-110	TYPE PFA		DELETED		DEL
-112	416-0068-1		• TUBE ASSEMBLY		1
-114	4-4-4JBZSSC		• FITTING (V02570) (B/E AEROSPACE STOCK CODE PF-227) (TPN 4-4-4JBZ-SS-C)		1
115	SS405-3		• FITTING (V02570) (B/E AEROSPACE STOCK CODE PF-202)		1
120	352-0980-1		• COVER, THERMAL SWITCH		1
			ATTACHING PARTS		
125	352-0981-1		• STANDOFF, THERMAL SWITCH		4
130	MS51957-33		• SCREW, PAN HEAD		4
			* * *		
135	352-0036-3		• THERMAL SWITCH ASSEMBLY (SEE IPL FIGURE 8 FOR DET BKDN) (SUPSD BY ITEM 135A)		1
-135A	3520-0314-01		• THERMAL SWITCH ASSEMBLY, OVER TEMPERATURE (S6) (SEE IPL FIGURE 8 FOR DET BKDN) (SUPSDS ITEM 135)		1
140	XR26343N		• TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-096) (SUPSD BY ITEM 140A)		3
-140A	51863-1		• TERMINAL, RING (V00779) (B/E AEROSPACE STOCK CODE TE-096) (SUPSDS ITEM 140)		3
			ATTACHING PARTS		
145	MS51957-27		• SCREW, PAN HEAD		6
150	MS35338-136		• WASHER, LOCK		6

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
155	MS35649-264		• NUT, HEX * * *		6
160	425-0173-3		• WIRE ASSEMBLY (S6 GND) (SUPSD BY ITEM 160A)		1
-160A	416-0139-3		• WIRE ASSEMBLY (S6 GND) (SUPSDS ITEM 160)		1
165	62-0014-1		• VALVE, DRAIN ASSEMBLY (SEE IPL FIGURE 9 FOR DET BKDN)		1
170	62-0010-1		• PLUG, SHUTOFF, WATER		1
175	110-3077		• O-RING (B/E AEROSPACE STOCK CODE OR-0662) (SUPSD BY ITEM 175A)		1
-175A	S11382-110		• O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-0662) (SUPSDS ITEM 175)		1
180	62-0248-17		• BREW HEAD ASSEMBLY (SEE IPL FIGURE 10 FOR DET BKDN)		1
185	3520-0236-01		• TEA TUBE ASSEMBLY ATTACHING PARTS		1
190	MS24693C1		• SCREW, FLAT HEAD		2
195	3575-0608-01		• FORMED TUBE ASSEMBLY		1
200	7.0-505R		• CLAMP (V25281) (B/E AEROSPACE STOCK CODE CL-50010) (SUPSD BY ITEM 200A)		1
-200A	16700983		• CLAMP (V25281) (B/E AEROSPACE STOCK CODE CL-50010) (SUPSDS ITEM 200) * * *		1
205	416-0463-1		• RETAINER, CUP ATTACHING PARTS		1
210	MS51957-15		• SCREW, PAN HEAD		2
215	MS35338-135		• WASHER, LOCK * * *		2
220	3575-0692-01		• ROD, GUIDE ATTACHING PARTS		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
225	MS24693C48		• SCREW, FLAT HEAD * * *		2
230	3575-0693-01		• ROD, GUIDE ATTACHING PARTS		1
235	MS51957-43		• SCREW, PAN HEAD (SUPSD BY ITEM 235A)		2
-235A	MS24693C49		• SCREW, FLAT HEAD (SUPSDS ITEM 235) * * *		2
240	3575-0694-01		• BLOCK, SUPPORT ATTACHING PARTS		1
245	MS24693C49		• DELETED * * *		DEL
250	A35581-4R		• RETAINER, QUARTER TURN (V52607) (B/E AEROSPACE STOCK CODE LAT-50064)		1
255	416-0088-1		• PLATE, MOUNTING BRACE ATTACHING PARTS		1
260	MS24693C25		• SCREW, FLAT HEAD * * *		4
-265	416-0628-13		• FRONT PLATE ASSEMBLY ATTACHING PARTS		1
270	MS24693C25		• SCREW, FLAT HEAD * * *		4
275	411-0701-1		• • PLATE, FRONT		1
280	411-0715-1		• • BRACE		1
285	3530-0069-01		• • PLATE, INSTRUCTION ATTACHING PARTS		1
290	MS51957-14		• • SCREW, PAN HEAD		2

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
295	MS35338-135		• • WASHER, LOCK * * *		2
300	411-0692-1		• BUSHING, HANDLE		2
-305	3520-0216-01		• HANDLE ASSEMBLY, BREW ATTACHING PARTS		1
310	MS24693C53		• SCREW, FLAT HEAD		2
315	411-0724-1		• WASHER, HANDLE		2
320	411-0725-1		• BUSHING		2
325	MS15795-807		• WASHER, FLAT		2
330	22FT832		• NUT, LOCK (V56878) (B/E AEROSPACE STOCK CODE SN-431) (TPN 22FT-832) * * *		2
-335	3520-0215-01		• • ARM ASSEMBLY		1
340	3520-0215-03		• • • ARM		1
345	425-0702-1		• • • GUIDE BUSHING ASSEMBLY		1
350	MS24693C48		• • • SCREW, FLAT HEAD		1
-355	3520-0215-02		• • ARM ASSEMBLY		1
360	3520-0215-04		• • • ARM		1
365	425-0702-1		• • • GUIDE BUSHING ASSEMBLY		1
370	MS24693C48		• • • SCREW, FLAT HEAD		1
375	411-0695-1		• • HANDLE ATTACHING PARTS		1
380	MS51958-61		• • SCREW * * *		2
385	411-0721-17		• MICRO SWITCH ASSEMBLY (S7) (SEE IPL FIGURE 11 FOR DET BKDN) ATTACHING PARTS		1
390	MS51957-29		• SCREW, PAN HEAD		2
395	MS35338-136		• WASHER, LOCK * * *		2

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
400	62-0024-1		• BRACKET, SPRING ATTACHING PARTS		1
405	MS51957-26		• SCREW, PAN HEAD		2
410	MS35338-136		• WASHER, LOCK * * *		2
415	XR26344N		• TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-118) (SUPSD BY ITEM 415A)		3
-415A	51863-1		• TERMINAL, RING (V00779) (B/E AEROSPACE STOCK CODE TE-096) (SUPSDS ITEM 415)		3
420	62-0131-1		• SCREEN ASSEMBLY		1
425	423-0078-1		• RAIL, GUIDE ATTACHING PARTS		1
430	MS24693C48		• SCREW, FLAT HEAD * * *		3
435	423-0078-3		• RAIL, GUIDE ATTACHING PARTS		1
440	MS24693C48		• SCREW, FLAT HEAD * * *		3
445	62-0018-1		• SUPPORT, CONNECTOR ATTACHING PARTS		1
450	MS24693C48		• SCREW, FLAT HEAD		5
455	MS51957-46		• SCREW, PAN HEAD		1
460	MS15795-807		• WASHER, FLAT		1
465	22FT832		• NUT, LOCK (V56878) (B/E AEROSPACE STOCK CODE SN-431) (TPN 22FT-832) * * *		1
470	62-0037-1		• GUIDE, ALIGNMENT PIN ATTACHING PARTS		1 1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
475	1726-03		• WASHER, LOCK (V78189) (B/E AEROSPACE STOCK CODE LW-163)		1
480	60-2010-1		• NUT, HEX * * *		1
485	62-0118-11		• PLATEN ASSEMBLY ATTACHING PARTS		1
490	MS51957-26		• SCREW, PAN HEAD		8
495	MS35338-136		• WASHER, LOCK * * *		8
500	3530-0070-01		• PLACARD ATTACHING PARTS		1
505	MS20604AD3W1		• RIVET, BLIND * * *		2
510	WA-120		• WASHER (SUPSD BY ITEM 510A)		1
-510A	COMMERCIAL		• WASHER (V0KHJ5) (B/E AEROSPACE STOCK CODE WA-120) (SUPSDS ITEM 510)		1
515	F4NY-125BK		• CLAMP (V85480) (B/E AEROSPACE STOCK CODE CL-315)(SUPSD BY ITEM 515A)		1
-515A	W4NY125-0		• CLAMP (V85480) (B/E AEROSPACE STOCK CODE CL-315) (TPN W4NY-125-0) (SUPSDS ITEM 515) ATTACHING PARTS		1
520	MS51957-27		• SCREW, PAN HEAD		1
525	MS35338-136		• WASHER, LOCK * * *		1
530	55-0392-1		• HEATER, SILICONE RUBBER (HR4) (50W) (SUPSD BY ITEM 530A)		1
-530A	55-0392-3		• HEATER, SILICONE RUBBER (HR4) (35W) (SUPSDS ITEM 530)		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
535	TYPE HO, SIZE 9		• TUBING, VARGLASS, (V79074) (B/E AEROSPACE STOCK CODE TU-50051)		1
540	B1870BN		• SPLICE, BUTT (BS3, BS4) (V14726) (B/E AEROSPACE STOCK CODE TE-280) (SUPSD BY ITEM 540A)		2
-540A	B1870BNT		• SPLICE, BUTT (BS3, BS4) (V14726) (B/E AEROSPACE STOCK CODE TE-280) (SUPSDS ITEM 540)		2
545	62-0077-3		• RETAINER, HEATER		1
			ATTACHING PARTS		
550	50FK632		• NUT, LOCK (V85480) (B/E AEROSPACE STOCK CODE SN-428) (TPN 50FK-632)		1
			* * *		
555	412-0129-7		• WIRE ASSEMBLY, SENSOR (SUPSD BY ITEM 555A)		1
-555A	62-0129-1		• WIRE ASSEMBLY, SENSOR (SUPSDS ITEM 555)		1
			ATTACHING PARTS		
560	55-0207-30		• SCREW, SLOTTED, PAN HEAD (SUPSD BY ITEM 560A)		1
-560A	MS51957-30		• SCREW, PAN HEAD (SUPSDS ITEM 560)		1
565	AN960D6		DELETED		DEL
570	MS21043-06		• NUT, LOCK		1
			* * *		
575	416-0139-3		• WIRE ASSEMBLY, SENSOR		1
			ATTACHING PARTS		
580	XR26343N		• TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-096)		4
585	MS51957-26		• SCREW, PAN HEAD (SUPSD BY ITEM 585A)		4
-585A	MS51957-27		• SCREW, PAN HEAD (SUPSDS ITEM 585)		4
590	MS21043-06		• NUT, LOCK		4
			* * *		

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
595	F4NY-250BK		• CLAMP, WIRE (V85480) (B/E AEROSPACE STOCK CODE CL-3201) (SUPSD BY ITEM 595A)		1
-595A	F4NY250-0		• CLAMP, WIRE (V85480) (B/E AEROSPACE STOCK CODE CL-3201) (SUPSDS ITEM 595)		1
			ATTACHING PARTS		
600	WA-120		• WASHER (SUPSD BY ITEM 600A)		1
-600A	COMMERCIAL		• WASHER (V0KHJ5) (B/E AEROSPACE STOCK CODE WA-120) (SUPSDS ITEM 600)		1
605	55-0207-30		• SCREW, SLOTTED PAN HEAD (SUPSD BY ITEM 605A)		1
-605A	MS51957-30		• SCREW, PAN HEAD (SUPSDS ITEM 605)		1
610	MS21043-06		• NUT, LOCK		1
			* * *		
615	352-0134-1		• HANDLE, CARRYING		1
			ATTACHING PARTS		
620	411-0926-1		• SCREW, GUIDE		2
-625	62-0061-1		• BUSHING, HANDLE		1
630	MS51958-68		• SCREW, PAN HEAD		1
635	MS35338-138		• WASHER, LOCK		1
640	411-0676-3		• SPRING, TENSION		2
645	62-0025-1		• SPRING, RETAINER		1
650	60-2001-1		• WASHER		1
655	411-0686-1		• BUSHING, SPRING		1
660	3575-0633-01		• BUSHING, CARRYING HANDLE		1
665	MS24693C53		• SCREW, FLAT HEAD		1
			* * *		
-670	3575-0650-01		• LEVER ASSEMBLY, LOCKING		1
			ATTACHING PARTS		

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
675	62-0211-1		• PIN, PIVOT		1
680	MS24693C48		• SCREW, FLAT HEAD		1
			* * *		
685	3575-0650-03		• • LEVER		1
690	47-10-302-50		• • FASTENER (V94222) (B/E AEROSPACE STOCK CODE FA-50131)		1
695	62-0210-1		• PIN		1
700	62-0210-2		• PIN		1
705	MS16633-4012		• RING, RETAINING		1
710	60-2013-1		• SPRING, HELICAL		1
715	349-0544-11		• PROBE ASSEMBLY (U1)		1
			ATTACHING PARTS		
720	MS51957-28		• SCREW, PAN HEAD		2
725	MS35338-136		• WASHER, LOCK (SUPSD BY ITEM 725A)		2
-725A	MS35338-135		• WASHER, LOCK (SUPSDS ITEM 725)		2
-727	NAS620C6		• WASHER, FLAT		2
			* * *		
730	MS3367-4-9		• TIE, WIRE		14
735	411-0733-25		• HOUSING ASSEMBLY		1
-736	MS21209CO810		• • HELICOIL INSERT		1
-737	MS21209CO815		• • HELICOIL INSERT		7
740	62-0040-1		• BRACKET ASSEMBLY, FAUCET		1
			ATTACHING PARTS		
745	MS24693C27		• SCREW, FLAT HD		2
			* * *		
750	60-2011-1		• NUT, HEX		2

- ITEM NOT ILLUSTRATED

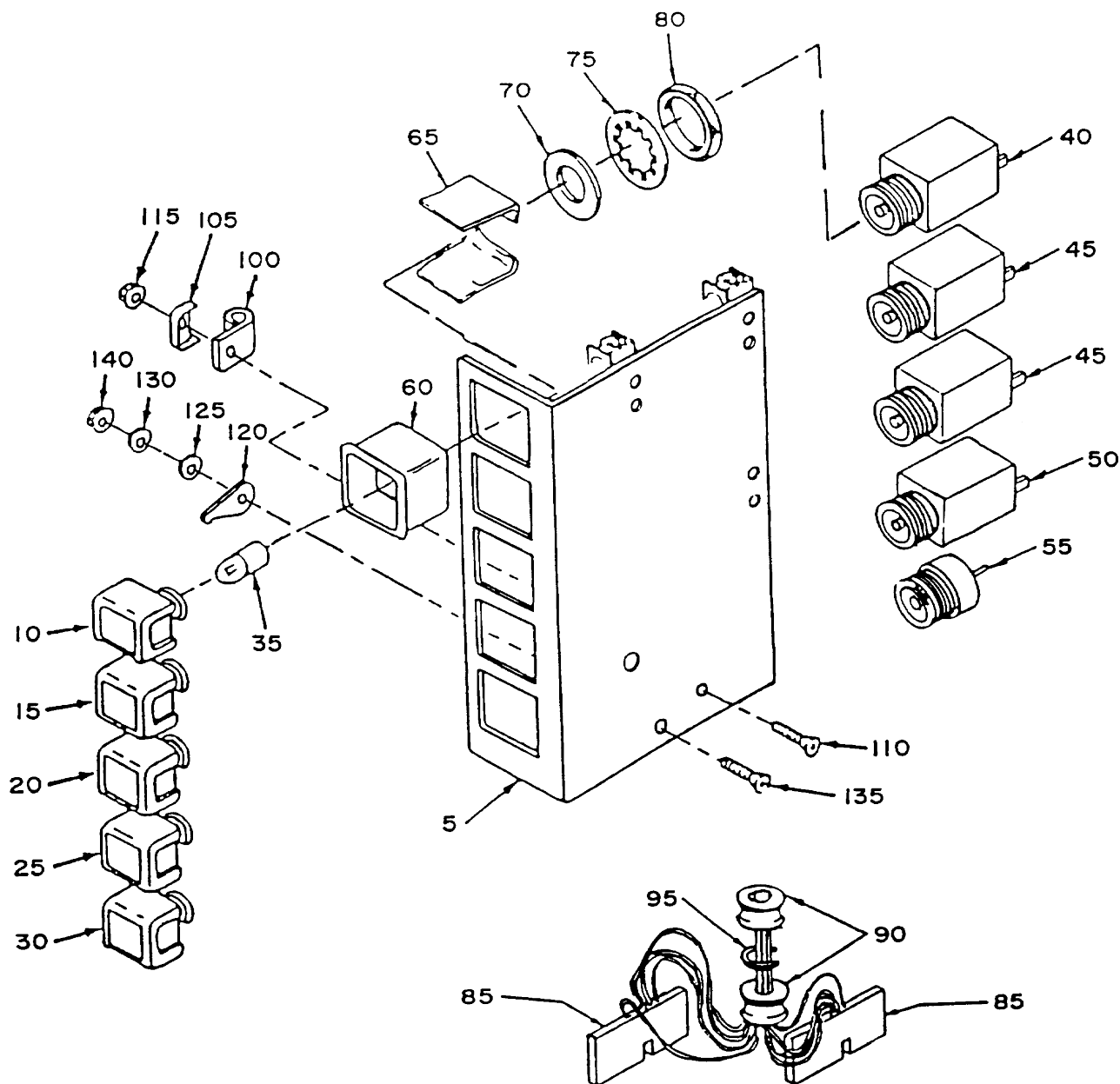
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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
755	1724-02		• LOCK WASHER, INTERNAL (V78189) (B/E AEROSPACE STOCK CODE LW-158)		2
760	55-0393-7		• CAP AND HANDLE ASSEMBLY		1
-765	62-0125-1		• YOKE, HANDLE FAUCET		1
770	1011088		• BODY, FAUCET (V87924) (B/E AEROSPACE STOCK CODE FAU-50010)		1
775	100KB		• AERATOR (V95124) (B/E AEROSPACE STOCK CODE MISC-013)		1
780	3520-0319-01		• TUBE, FAUCET		1
-785	3575-0893-01		• TUBE, FAUCET		1
-790	16700983		• CLAMP (V25281) (B/E AEROSPACE STOCK CODE CL-50010)		1

- ITEM NOT ILLUSTRATED

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IPL Figure 3
 Control Switch Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
-1	3520-0241-03		SWITCH ASSEMBLY (SEE IPL FIGURE 2 FOR NHA) (SUPSD BY ITEM 1A)		RF
-1A	3520-0357-01		SWITCH ASSEMBLY (SEE IPL FIGURE 2 FOR NHA) (SUPSDS ITEM 1)		RF
5	405-0044-13		• HOUSING, SWITCH		1
10	55-0396-1		• CAP, SWITCH (ON/OFF, BLUE)		1
15	416-0396-19		• CAP, SWITCH (COFFEE BREW, WHITE)		1
20	416-0396-17		• CAP, SWITCH (TEA, GREEN)		1
25	374-0872-5		• CAP, SWITCH (HOT PLATE ON, ORANGE)		1
30	55-0396-7		• CAP, SWITCH (NO WATER, WHITE/RED)		1
35	MS25237-387		• LAMP, INCANDESCENT (SUPSD BY ITEM 35A) (PRE SIL 01015)		5
-35A	FF200OCW028B		• LED (V8Z410) (B/E AEROSPACE STOCK CODE LP-50095) (SUPSDS ITEM 35) (POST SIL 01015) (TPN FF200-OCW-028B)		5
40	5710		• SWITCH (S1) (V76854) (B/E AEROSPACE STOCK CODE SW-50089)		1
45	5820		• SWITCH (S2, S3) (V29238) (B/E AEROSPACE STOCK CODE SW-50088)		2
50	5720		• SWITCH (S4) (V76854) (B/E AEROSPACE STOCK CODE SW-50087)		1
55	532		• INDICATOR (DS1) (V76854) (B/E AEROSPACE STOCK CODE DA-155)		1
60	3-3767-638		• BEZEL, SWITCH (V76854) (B/E AEROSPACE STOCK CODE SW-810)		5
65	352-0282-1		• BRACKET, SWITCH MOUNTING (SUPSD BY ITEM 65A)		5
-65A	3575-0191-01		• BRACKET, SWITCH MOUNTING (SUPSDS ITEM 65)		5
70	349-0551-1		• WASHER, FLAT		5
75	2-1174-144		• WASHER, LOCK (V76854) (B/E AEROSPACE STOCK CODE SW-812)		5
80	2-2718-620		• NUT, PLAIN (V76854) (B/E AEROSPACE STOCK CODE SW-813)		5
85	65846-006		• CONNECTOR (P3, P4) (V22526) (B/E AEROSPACE STOCK CODE CO-50269) (SUPSD BY ITEM 85A)		2

- ITEM NOT ILLUSTRATED

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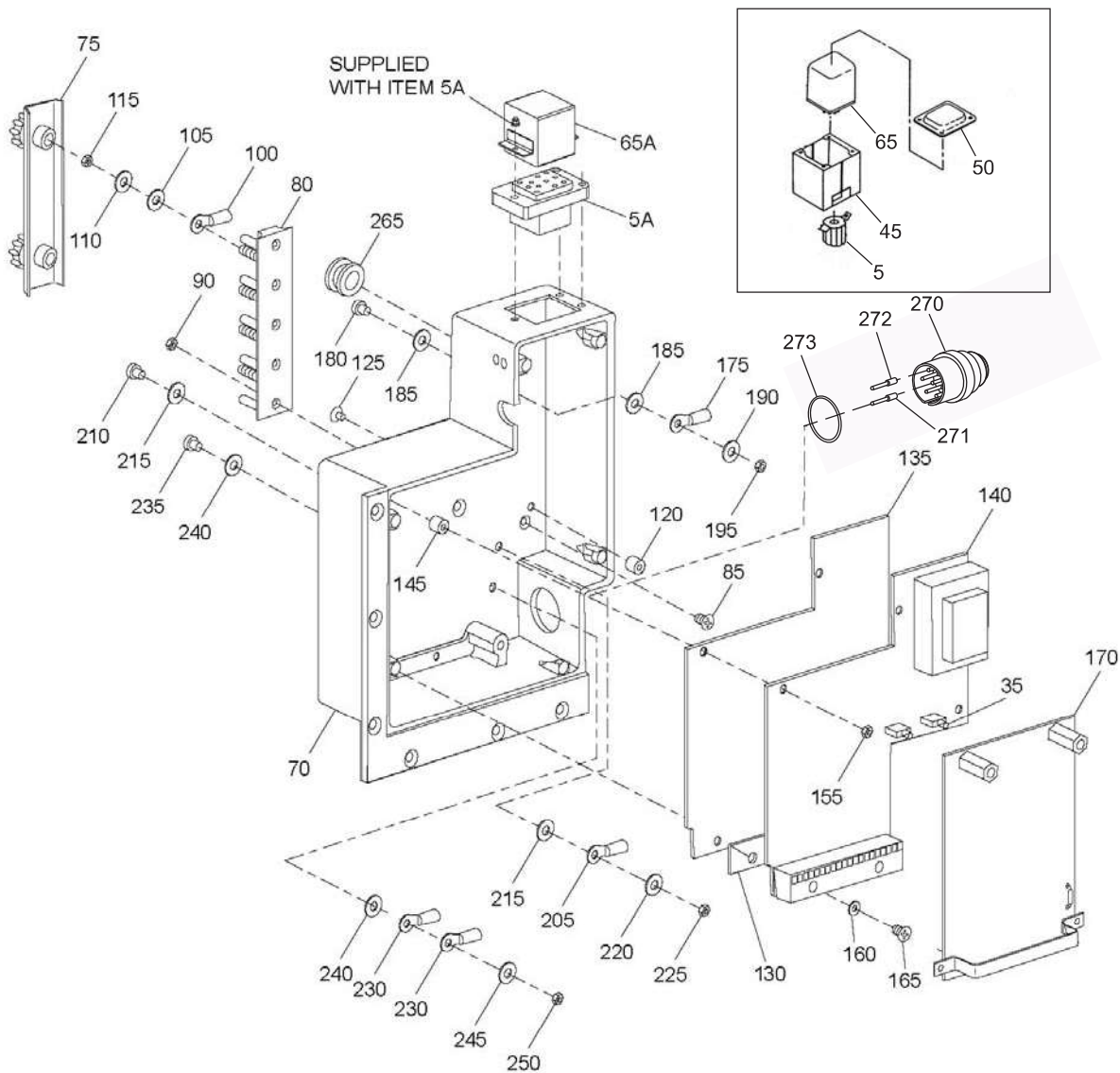
FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
-85A	65846-006LF		• CONNECTOR (P3, P4) (V30682) (B/E AEROSPACE STOCK CODE CO-50269) (SUPSDS ITEM 85)		2
90	2903Z6012		• GROMMET (V70485) (B/E AEROSPACE STOCK CODE GR-098) (SUPSD BY ITEM 90A)		2
-90A	2903-26012		• GROMMET (V0MT56) (B/E AEROSPACE STOCK CODE GR-098) (SUPSDS ITEM 90)		2
95	MS3367-4-9		STRAP, TIEDOWN		5
100	F4NY-250BK		• CLAMP (V85480) (B/E AEROSPACE STOCK CODE CL-3201) (SUPSD BY ITEM 100A)		1
-100A	F4NY250-0		• CLAMP (V85480) (B/E AEROSPACE STOCK CODE CL-3201) (SUPSDS ITEM 100)		1
			ATTACHING PARTS		
105	WA120		• WASHER, CLAMP (V0KHJ5) (B/E AEROSPACE STOCK CODE WA-120) (TPN WA-120)		1
110	MS24693C26		• SCREW, FLAT HEAD (TPN MS24693-C26)		1
115	50FK632		• NUT, LOCK (V56878) (B/E AEROSPACE STOCK CODE SN-428) (TPN 50FK-632)		1
			* * *		
120	1416-6		• TERMINAL, SOLDER (GND) (V83330) (B/E AEROSPACE STOCK CODE TE-4565)		1
			ATTACHING PARTS		
125	AN960D6L		• WASHER, FLAT		1
130	AN960-6L		• WASHER, FLAT		1
135	MS24693C26		• SCREW, FLAT HEAD (TPN MS24693-C26)		1
140	50FK632		• NUT, LOCK (V56878) (B/E AEROSPACE STOCK CODE SN-428) (TPN 50FK-632)		1

- ITEM NOT ILLUSTRATED

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IPL Figure 4
Power Module Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
-1	3520-0235-05		POWER MODULE ASSEMBLY (SEE IPL FIGURE 2 FOR NHA) (SUPSD BY ITEM 1A)		RF
-1A	3520-0343-01		POWER MODULE ASSEMBLY (SEE IPL FIGURE 2 FOR NHA) (SUPSDS ITEM 1)		RF
5	665-0423-3		• SOCKET, RELAY (SUPSD BY ITEM 5A BEFORE 31 MAR 2003)		1
5A	CL1-2107-267		• SOCKET, RELAY (V30528) (B/E AEROSPACE STOCK CODE RL-50017) (SUPSDS ITEM 5 AFTER 31 MAR 2003)		1
			ATTACHING PARTS		
-10	MS51957-29		DELETED		DEL
-15	MS15795-706		DELETED		DEL
-20	50FK-632		DELETED		DEL
			* * *		
-25	02-09-1206		DELETED		DEL
-30	02-09-1202		DELETED		DEL
35	BB8141-1		• TERMINAL, SPADE (V98410) (B/E AEROSPACE STOCK CODE TE-3473) (SUPSD BY ITEM 35A)		2
-35A	BB8141		• RECEPTACLE, QUICK DISCONNECT (V98410) (B/E AEROSPACE STOCK CODE TE-3473) (SUPSDS ITEM 35) (TPN BB-8141)		2
-40	47712		• RECEPTACLE, MINI PV (V22526) (B/E AEROSPACE STOCK CODE TE-50052) (SUPSD BY ITEM 40A)		3
-40A	47439-000LF		• RECEPTACLE, MINI PV (V30682) (B/E AEROSPACE STOCK CODE TE-50052) (SUPSDS ITEM 40)		3
45	3575-0021-01		• RELAY, HOUSING (DELETED AFTER 31 MARCH 2003)		DEL
50	62-0234-1		• COVER, RELAY (DELETED AFTER 31 MARCH 2003)		DEL
-55	MS35338-136		DELETED		DEL
-60	MS51957-28		DELETED		DEL

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4			* * *		
65	KR-90693-1		RELAY (K1) (V77342) (B/E AEROSPACE STOCK CODE RL-119) (SUPSD BY ITEM 65A)		1
-65A	KR-90693-2		RELAY (K1) (V77342) (PRE SIL 3510-0052-SIL-03) (B/E AEROSPACE STOCK CODE RL-119) (SUPSDS ITEM 65) (SUPSD BY ITEM 65B BEFORE 31 MAR 2003)		1
65B	M83536/33-003		RELAY (K1) (V78290) (POST SIL 3510-0052-SIL-03) (B/E AEROSPACE STOCK CODE RL-50035) (SUPSDS ITEM 65A AFTER 31 MAR 2003)		1
70	3520-0244-03		HOUSING ASSEMBLY (ROUND HOLE CONFIGURATION - NOT AVAILABLE FOR PURCHASE AFTER 3/31/2003) (SUPSD BY ITEM 70A)		1
-70A	3520-0244-01		HOUSING ASSEMBLY (SUPSDS ITEM 70)		1
75	MS18029-6S4		COVER, TERMINAL BOARD (SUPSD BY ITEM 75A) (MS18023-6S-4)		1
-75A	MS18029-1S5		COVER, TERMINAL BOARD (SUPSDS ITEM 75) (TPN MS18029-1S-5)		1
80	MS27212-6-4		BOARD, TERMINAL (TB3) (SUPSD BY ITEM 80A)		1
-80A	MS27212-1-5		BOARD, TERMINAL (TB3) (SUPSDS ITEM 80)		1
			ATTACHING PARTS		
85	MS24693C5		SCREW, FLAT HEAD		2
90	50FA440		NUT, LOCK (V56878) (B/E AEROSPACE STOCK CODE SN-424) (TPN 50FA-440)		2
			* * *		
-95	XR26344N		TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-118) (SUPSD BY ITEM 95A)		1
-95A	51863-1		TERMINAL, RING (V00779) (B/E AEROSPACE STOCK CODE TE-096) (SUPSDS ITEM 95)		3

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
100	XR1902SNT		• TERMINAL, RING (V14726) (B/E AEROSPACE STOCK CODE TE-50371) (SUPSD BY ITEM 100A)		4
-100A	51864-3		• TERMINAL, RING (V00779) (B/E AEROSPACE STOCK CODE TE-1905) (SUPSDS ITEM 100)		2
			ATTACHING PARTS		
105	AN960-8L		• WASHER, FLAT (SUPSD BY ITEM 105A)		4
-105A	NAS1149FN616B		• WASHER, FLAT (SUPSDS ITEM 105)		5
110	MS35338-42		• WASHER, LOCK (SUPSD BY ITEM 110A)		4
-110A	MS35338-41		• WASHER, LOCK (SUPSDS ITEM 110)		5
115	MS35649-282		• NUT, HEX (SUPSD BY ITEM 115A)		4
-115A	MS35649-262		• NUT, HEX (SUPSDS ITEM 115)		5
			* * *		
120	3575-0604-01		• SPACER		1
			ATTACHING PARTS		
125	MS24693C25		• SCREW, FLAT HEAD		1
			* * *		
130	3575-0610-01		• INSULATOR		1
135	3575-0611-01		• INSULATOR		1
140	3530-0059-05		• BOARD, PRINTED CIRCUIT (SEE CMM 25-33-35 FOR DET BKDN)		1
			ATTACHING PARTS		
145	3575-0604-01		• SPACER		2
-150	MS51859-3		• WASHER, FLAT		2
155	MS21043-06		• NUT, LOCK		2
160	MS15795-803		• WASHER, FLAT		2
165	MS51957-15		• SCREW, PAN HEAD		2
			* * *		

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
170	61-1675-11		• BOARD, PRINTED CIRCUIT (SEE CMM 25-33-36 FOR DET BKDN)		1
175	XR26345N		• TERMINAL, RING (DC) (V14726) (B/E AEROSPACE STOCK CODE TE-142) (SUPSD BY ITEM 175A)		1
-175A	XR1891SNT		• TERMINAL, RING (DC) (V14726) (B/E AEROSPACE STOCK CODE TE-142) (SUPSDS ITEM 175)		1
			ATTACHING PARTS		
180	NAS1801-3-7		• SCREW, FLAT HEAD		1
185	AN960D10L		• WASHER, FLAT (SUPSD BY ITEM 185A)		2
-185A	NAS1149D0316H		• WASHER, FLAT (SUPSDS ITEM 185)		2
190	AN960-10L		• WASHER, FLAT (SUPSD BY ITEM 190A)		1
-190A	NAS1149F0332P		• WASHER, FLAT (SUPSDS ITEM 190)		1
195	MS21042L3		• NUT, LOCK		1
			* * *		
-200	B1871BNT		• SPLICE, BUTT (BS5) (V14726) (B/E AEROSPACE STOCK CODE TE-50354)		1
205	X1903SNT		• TERMINAL, RING (SG1) (V14726) (B/E AEROSPACE STOCK CODE TE-215)		1
			ATTACHING PARTS		
210	NAS1801-3-7		• SCREW, FLAT HEAD		1
215	AN960D10L		• WASHER, FLAT (SUPSD BY ITEM 215A)		2
-215A	NAS1149D0316H		• WASHER, FLAT (SUPSDS ITEM 215)		2
220	AN960-10L		• WASHER, FLAT (SUPSD BY ITEM 220A)		1
-220A	NAS1149F0332P		• WASHER, FLAT (SUPSDS ITEM 220)		1
225	MS21042L3		• NUT, LOCK		1
			* * *		
230	XR1903SNT		• TERMINAL, RING (SG2) (V14726) (B/E AEROSPACE STOCK CODE TE-215)		2
			ATTACHING PARTS		

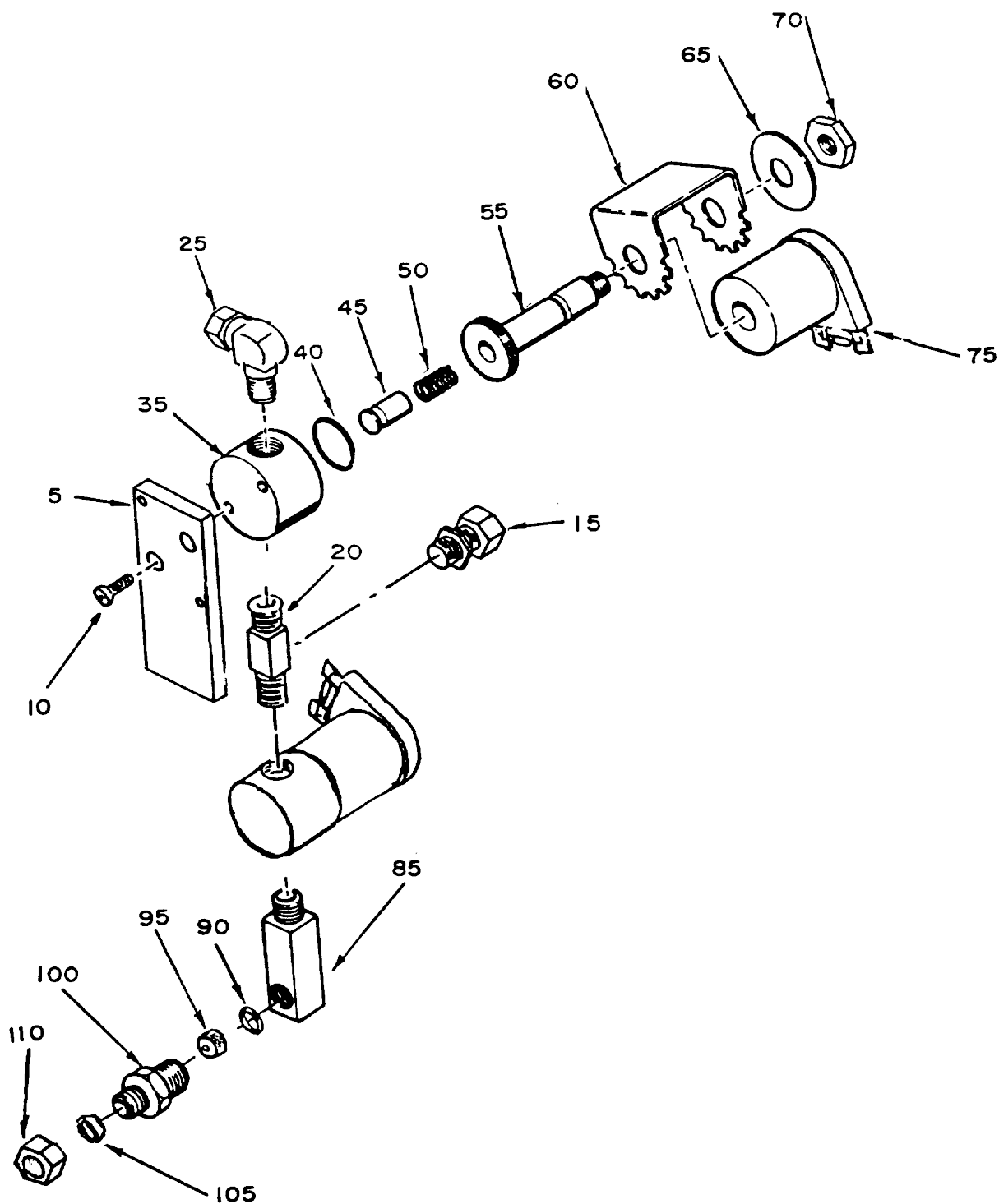
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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
235	NAS1801-3-8		• SCREW, FLAT HEAD		1
240	AN960D10L		• WASHER, FLAT (SUPSD BY ITEM 240A)		2
-240A	NAS1149D0316H		• WASHER, FLAT (SUPSDS ITEM 240)		2
245	AN960-10L		• WASHER, FLAT (SUPSD BY ITEM 245A)		1
-245A	NAS1149F0332P		• WASHER, FLAT (SUPSDS ITEM 245)		1
250	MS21042L3		• NUT, LOCK		1
			* * *		
-255	COMMERCIAL		• PIN (V76902) (B/E AEROSPACE STOCK CODE PN-5855) (SUPSD BY ITEM 255A)		1
-255A	COMMERCIAL		• PIN (V83131) (B/E AEROSPACE STOCK CODE PN-5855) (SUPSDS ITEM 255)		1
-260	61-0814-1		• GUIDE		2
265	2903Z6012		• GROMMET (V70485) (B/E AEROSPACE STOCK CODE GR-098) (SUPSD BY ITEM 265A)		1
-265A	2903-26012		• GROMMET (V0MT56) (B/E AEROSPACE STOCK CODE GR-098) (SUPSDS ITEM 265)		1
270	CDC06-16S1P		• CONNECTOR (P1) (V71771) (B/E AEROSPACE STOCK CODE CO-6505) (SUPSD BY ITEM 270A) (TPN CDC06-16S-1P)		1
-270A	COMMERCIAL		• CONNECTOR (P1) (B/E AEROSPACE STOCK CODE CO-6505) (SUPSDS ITEM 270)		1
271	CO6505-01		• • PIN (TPN CO-6505-01)		1
272	CO6505-03		• • PIN (TPN CO-6505-03)		6
273	CO6505-05		• • O-RING (TPN CO-6505-05)		1
-275	MS25226-2-2		• BUSS BAR		1

- ITEM NOT ILLUSTRATED

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IPL Figure 5
Solenoid Valve Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
5					
-1	3520-0276-01		SOLENOID VALVE ASSEMBLY (SEE IPL FIGURE 2 FOR NHA)		RF
5	405-0215-1		• SUPPORT		1
			ATTACHING PARTS		
10	MS24693C49		• SCREW, FLAT HEAD		2
			* * *		
15	4-2FBZSSC		• FITTING (V50948) (B/E AEROSPACE STOCK CODE PF-110) (TPN 4-2FBZ-SS-C)		1
20	405-0437-3		• FITTING		1
25	4-2CBZ-SS		• FITTING (V50948) (B/E AEROSPACE STOCK CODE PF-102) (SUPSD BY ITEM 25A)		1
-25A	4-2CBZSSCD960053		• FITTING (V50948) (B/E AEROSPACE STOCK CODE PF-102) (SUPSDS ITEM 25) (TPN 4-2C-BZ-SS-C-D960053)		1
-30	62-0250-25		• VALVE		2
35	62-0250-3		• • BODY DETAIL		1
40	62-0250-5		• • O-RING		1
45	62-0250-7		• • PLUNGER ASSEMBLY		1
50	62-0250-9		• • SPRING		1
55	62-0250-11		• • PLUNGER GUIDE-ASSEMBLY		1
60	62-0250-15		• • STRAP HOUSING		1
65	62-0250-17		• • NAMEPLATE		1
70	62-0250-19		• • NUT, HEX		1
75	62-0250-21		• • COIL ASSEMBLY (W/DIODE)		1
-80	1N4007		• • • DIODE (B/E AEROSPACE STOCK CODE RC-50105)		1
85	3575-0432-01		• FITTING		1
90	MS16625-4031		• RING, RETAINING (SUPSD BY ITEM 90A)		1
-90A	VHL33QPAS02		• RING, RETAINING (V51814) (B/E AEROSPACE STOCK CODE RR-50009) (SUPSDS ITEM 90) (TPN VHL-33-Q-PA-S02)		1
95	62-0066-1		• WASHER, FLOW		1

- ITEM NOT ILLUSTRATED

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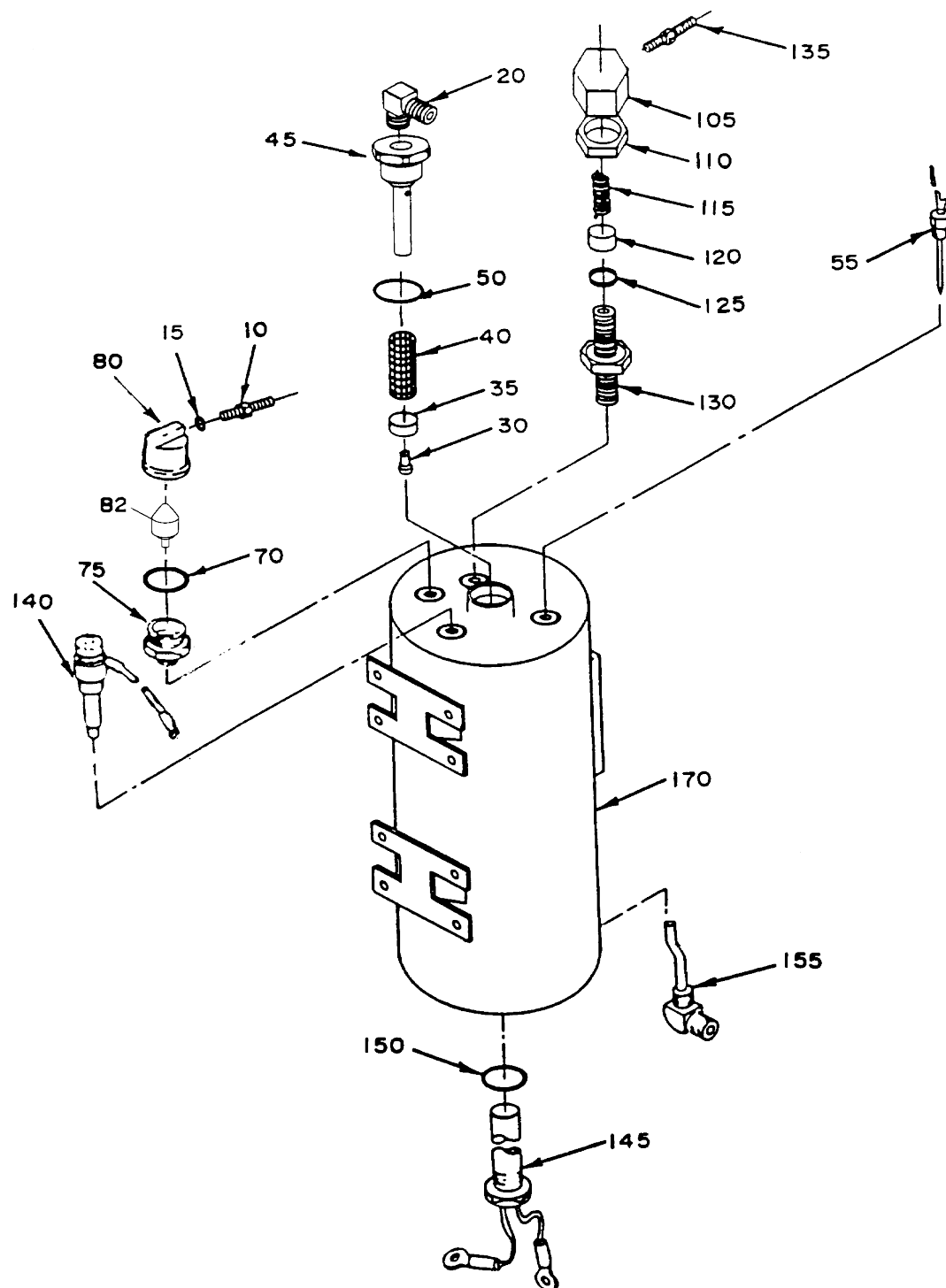
FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
5					
100	3575-0185-01		• FITTING		1
105	4TZSS		• FERRULE (V30780) (B/E AEROSPACE STOCK CODE PF-50002) (TPN 4TZ-SS)		1
110	4BZSSC		• NUT, FITTING (V50948) (B/E AEROSPACE STOCK CODE PF-143) (TPN 4-BZ-SSC)		1

- ITEM NOT ILLUSTRATED

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IPL Figure 6
 Tank Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
6					
-1	3520-0177-07		WATER TANK ASSEMBLY (SEE IPL FIGURE 2 FOR NHA) (SUPSD BY ITEM 1A)		RF
-1A	3520-0177-11		WATER TANK ASSEMBLY (SEE IPL FIGURE 2 FOR NHA) (SUPSDS ITEM 1)		RF
-5	62-0136-7		• FITTING ASSEMBLY		1
10	62-0136-11		• • FITTING		1
15	62-0136-9		• • GASKET (SUPPLIED WITH ITEM 10)		1
20	4-2CBZ-SS		• ELBOW (V50948) (B/E AEROSPACE STOCK CODE PF-102)		1
-20A	42CBZSSCD960053		• ELBOW, MALE (V50948) (B/E AEROSPACE STOCK CODE PF-102) (TPN 4-2CBZ-SS-C-D960053)		1
-25	62-0020-1		• FILTER ASSEMBLY		1
30	MS51957-26		• • SCREW, PAN HEAD		1
-32	MS35338-136		• • WASHER, LOCK		1
35	62-0011-1		• • RETAINER		1
40	62-0007-1		• • SCREEN, FILTER		1
45	3575-0647-01		• • JET, WATER		1
50	115-3077		• O-RING (V25184) (B/E AEROSPACE STOCK CODE OR-0941) (SUPSD BY ITEM 50A)		1
-50A	S11382-115		• O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-0941) (SUPSDS ITEM 50) (TPN S1138 2-115)		1
55	55-0397-1		• THERMISTOR ASSEMBLY (SUPSD BY ITEM 55A)		1
-55A	55-0397-3		• THERMISTOR ASSEMBLY (SUPSDS ITEM 55)		1
-60	3540-0001-01		• VENT VALVE ASSEMBLY (PRE SIL 3510-0052-SIL-01) (SUPSD BY ITEM 60A)		1
-60A	3520-0523-01		• VENT VALVE ASSEMBLY (POST SIL 3510-0052-SIL-01) (SUPSDS ITEM 60) (SUPSD BY ITEM 60B)		
-60B	62-0135-1		• VENT VALVE ASSEMBLY (SUPSDS ITEMS 60 AND 60A)		1

- ITEM NOT ILLUSTRATED

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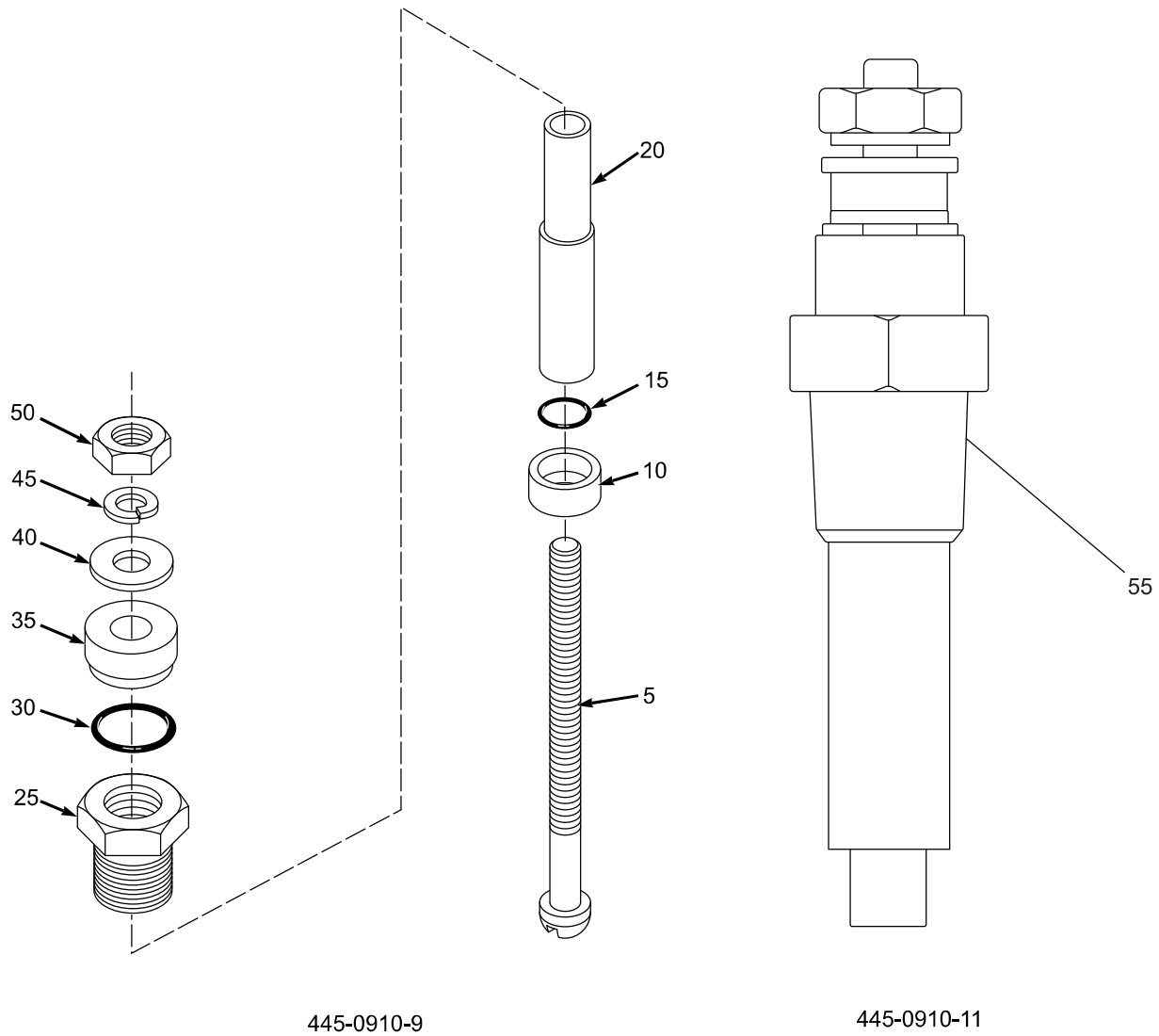
FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
6					
-65	62-0156-1		• • BODY ASSEMBLY		1
70	115-3077		• • • O-RING (V25184) (B/E STOCK CODE OR-0941) (SUPSD BY ITEM 70A)		1
-70A	S11382-115		• • • O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-0941) (SUPSDS ITEM 70) (TPN S1138 2-115)		1
75	62-0156-3		• • • BODY SUBASSEMBLY		1
80	3575-0584-01		• • HOUSING (SUPSD BY ITEM 80A)		1
-80A	62-0134-1		• • HOUSING (SUPSDS ITEM 80)		1
82	62-0132-1		• • FLOAT ASSEMBLY		1
-85	COMMERCIAL		DELETED		DEL
-90	010-3077		DELETED		DEL
-95	62-0223-1		• RELIEF VALVE ASSEMBLY		1
-100	62-0223-3		• • RELIEF VALVE SUBASSEMBLY		1
105	62-0223-5		• • • ADJUSTING CAP		1
110	62-0223-9		• • • LOCKNUT		1
115	1410-01		• • • SPRING (BRASS) (V43990) (B/E AEROSPACE STOCK CODE VA-1004) (SUPSD BY ITEM 115A)		1
-115A	1410-02-523CC7		• • • SPRING (STAINLESS STEEL) (V43990) (B/E AEROSPACE STOCK CODE VA-1004) (SUPSDS ITEM 115)		1
120	1534-01		• • • REST, SPRING (BRASS) (V43990) (B/E AEROSPACE STOCK CODE VA-1003) (SUPSD BY ITEM 120A)		1
-120A	1534-05-523CC7		• • • REST, SPRING (STAINLESS STEEL) (V43990) (B/E AEROSPACE STOCK CODE VA-1003) (SUPSDS ITEM 120)		
125	1497-03		• • • VALVE ASSEMBLY (BRASS) (V43990) (B/E AEROSPACE STOCK CODE VA-1031) (SUPSD BY ITEM 125A)		1
-125A	1497-06-523CC7		• • • VALVE ASSEMBLY (STAINLESS STEEL) (V43990)(B/E AEROSPACE STOCK CODE VA-1031) (SUPSDS ITEM 125)		1
130	62-0223-11		• • • VALVE BODY		1
135	62-0136-7		• • FITTING ASSEMBLY		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
6					
140	445-0910-9		• WATER SENSOR ASSEMBLY (U2) (SEE IPL FIGURE 7 FOR DET BKDN) (SUPSD BY ITEM 140A)		1
-140A	445-0910-11		• LOW WATER SENSOR ASSEMBLY (U2) (SEE IPL FIGURE 7 FOR DET BKDN) (SUPSDS ITEM 140)		1
145	440-0150-3		• HEATER ASSEMBLY (HR1 -HR3) (PRE SB 3510-0052-25-052) (SUPSD BY ITEM 145A)		3
-145A	3575-1251-07		• HEATER ASSEMBLY (900 WATT) (POST SB 3510-0052-25-052) (SUPSDS ITEM 145)		3
150	115-3077		• O-RING (V25184) (B/E AEROSPACE STOCK CODE OR-0941) (SUPSD BY ITEM 150A)		3
-150A	S11382-115		• O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-0941) (SUPSDS ITEM 150) (TPN S1138 2-115)		3
155	500-0731-1		• TUBE ASSEMBLY		1
170	428-0074-1		• TANK WELDED ASSEMBLY		1

- ITEM NOT ILLUSTRATED

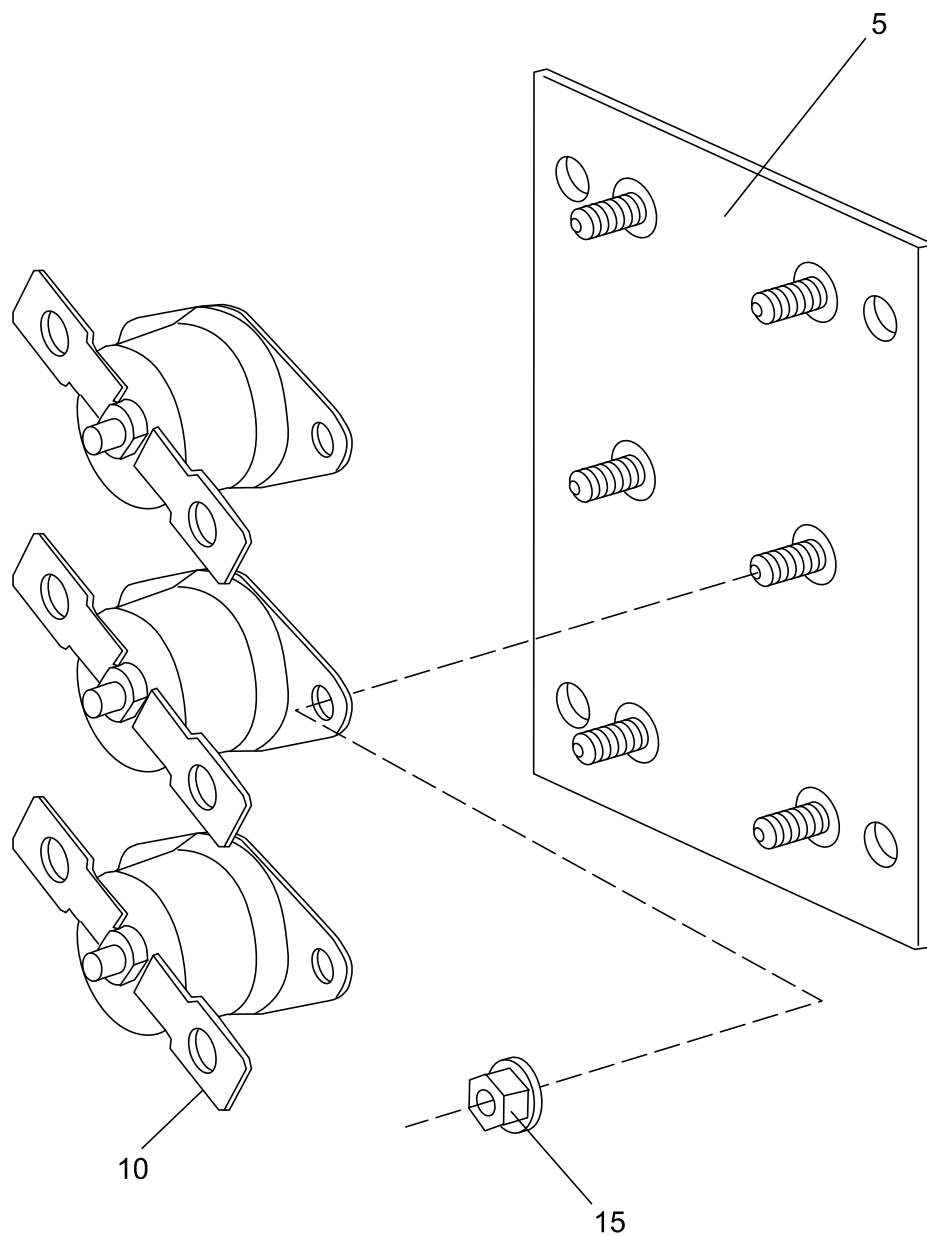


IPL Figure 7
Low Water Sensor Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
1	445-0910-9		LOW WATER SENSOR ASSEMBLY (U2) (SEE IPL FIGURE 6 FOR NHA) (SUPSD BY ITEM 1A)		RF
-1A	445-0910-11		LOW WATER SENSOR ASSEMBLY (U2) (SEE IPL FIGURE 6 FOR NHA) (SUPSDS ITEM 1)		RF
5	445-0909-1		• SHAFT (SUPSD BY ITEM 55)		1
10	445-0916-1		• CAP (SUPSD BY ITEM 55)		1
15	006-3077		• O-RING (V25184) (B/E AEROSPACE STOCK CODE OR-0361) (SUPSD BY ITEM 55)		1
20	445-0908-1		• SLEEVE (SUPSD BY ITEM 55)		1
25	445-0912-1		• ADAPTER (SUPSD BY ITEM 55)		1
30	010-3077		• O-RING (V25184) (B/E AEROSPACE STOCK CODE OR-0571) (SUPSD BY ITEM 55)		1
35	445-0915-1		• SPACER (SUPSD BY ITEM 55)		1
40	AN960C6		• WASHER, FLAT (SUPSD BY ITEM 55)		1
45	MS35338-136		• WASHER, LOCK (SUPSD BY ITEM 55)		1
50	MS35649-264		• NUT, HEX (SUPSD BY ITEM 55)		1
55	0500-0243		• SENSOR, LOW WATER (B/E AEROSPACE STOCK CODE MISC-50389) (SUPSDS ITEMS 5, 10, 15, 20, 25, 30, 35, 40, 45, 50)		1

- ITEM NOT ILLUSTRATED

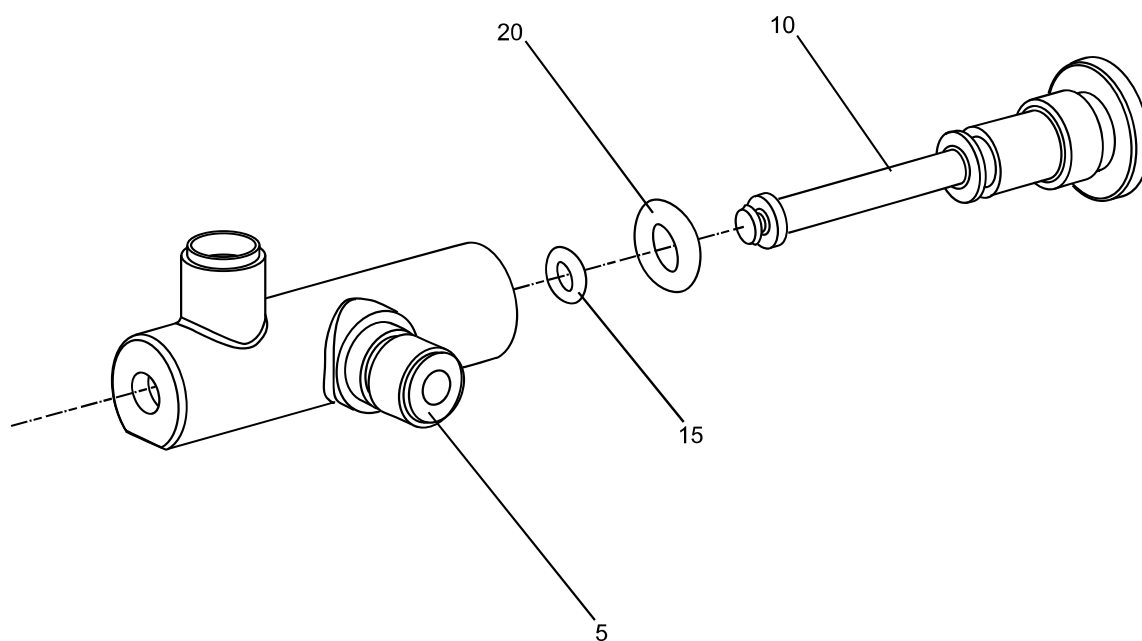


IPL Figure 8
Thermal Switch, Over Temperature Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
8					
-1	352-0036-3		ASSEMBLY, THERMAL SWITCH (SEE IPL FIGURE 2 FOR NHA) (SUPSD BY ITEM 1A)		RF
-1A	3520-0314-01		ASSEMBLY, THERMAL SWITCH, OVER TEMPERATURE (SEE IPL FIGURE 2 FOR NHA) (SUPSDS ITEM 1)		RF
5	352-0984-1		• MOUNTING PLATE ASSEMBLY		1
10	352-0983-1		• THERMAL SWITCH		3
			ATTACHING PARTS		
15	MS21043-04		• NUT, LOCK		6
			* * *		
-20	352-0183-1		DELETED		DEL
-25	MS51957-26		DELETED		DEL
-30	MS35338-136		DELETED		DEL
-35	MS35649-264		DELETED		DEL

- ITEM NOT ILLUSTRATED



IPL Figure 9
Drain Valve Assembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
9					
1	62-0014-1		DRAIN VALVE ASSEMBLY (SEE IPL FIGURE 2 FOR NHA)		RF
5	62-0008-1		• BODY, VALVE		1
10	62-0013-1		• STEM ASSEMBLY		1
15	005-3077		• O-RING (V25184) (B/E AEROSPACE STOCK CODE OR-0341) (SUPSD BY ITEM 15A)		1
-15A	S1138-702-005		• O-RING (V4J413) (B/E AEROSPACE STOCK CODE OR-0341) (SUPSDS ITEM 15) (TPN S1138-70 2-005)		1
20	010-3077		• O-RING (V25184) (B/E AEROSPACE STOCK CODE OR-0571) (SUPSD BY ITEM 20A)		1
-20A	S11382-010		• O-RING (V02697) (B/E AEROSPACE STOCK CODE OR-0571) (SUPSDS ITEM 20) (TPN S1138 2-010)		1

- ITEM NOT ILLUSTRATED

Exploded perspective view of a mechanical assembly. The assembly includes a base plate (10) with mounting holes, a central vertical shaft (20) with a threaded section (25), and various mounting brackets and fasteners. A circular inset labeled "FLOW WASHER VIEW" shows a top-down view of the central shaft assembly. The exploded view shows the following components and their assembly sequence:

- Base plate (10) with mounting holes.
- Mounting bracket (15) with mounting holes.
- Mounting bracket (30) with mounting holes.
- Mounting bracket (45B) with mounting holes.
- Mounting bracket (45C) with mounting holes.
- Mounting bracket (50) with mounting holes.
- Mounting bracket (55) with mounting holes.
- Mounting bracket (60) with mounting holes.
- Mounting bracket (65) with mounting holes.
- Mounting bracket (70) with mounting holes.
- Mounting bracket (75) with mounting holes.
- Mounting bracket (80) with mounting holes.
- Mounting bracket (90) with mounting holes.
- Mounting bracket (95) with mounting holes.
- Mounting bracket (100) with mounting holes.
- Mounting bracket (105) with mounting holes.
- Mounting bracket (110) with mounting holes.
- Mounting bracket (115) with mounting holes.

25-30-50

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
10					
-1	62-0248-17		BREW HEAD ASSEMBLY (SEE IPL FIGURE 2 FOR NHA)		RF
5	411-0745-1		• HEAD, BREW		1
10	411-0693-1		• GASKET		1
15	411-0709-1		• RETAINER ASSEMBLY		1
			ATTACHING PARTS		
20	50FK632		• NUT (V56878) (B/E AEROSPACE STOCK CODE SN-428) (TPN 50FK-632) (SUPSD BY ITEM 20A)		4
-20A	MS21043-06		• NUT, LOCKING (SUPSDS ITEM 20)		4
			* * *		
25	411-0748-7		• GUIDE (LEFT)		1
30	411-0748-3		• GUIDE (RIGHT)		1
			ATTACHING PARTS		
35	MS15795-807		• WASHER, FLAT		2
40	MS35338-137		• WASHER, LOCK		2
45	55-0207-45		• SCREW, PAN HEAD (SUPSD BY ITEM 45A)		2
-45A	MS51957-45		• SCREW, PAN HEAD (SUPSDS ITEM 45)		2
45B	MS51957-28		• SCREW, PAN HEAD		1
45C	3575-0634-01		• BUSHING, SPRING		1
50	MS15795-805		• WASHER, FLAT		3
55	M35338-136		• WASHER, LOCK (SUPSD BY ITEM 55A)		2
-55A	MS35338-136		• WASHER, LOCK (SUPSDS ITEM 55)		2
60	55-0207-29		• SCREW, PAN HEAD (SUPSD BY ITEM 60A)		2
-60A	MS51957-29		• SCREW, PAN HEAD (SUPSDS ITEM 60)		2
			* * *		
65	3575-0189-01		• FITTING, BREW		1
			ATTACHING PARTS		
70	MS24693C31		• SCREW		2
75	NAS620C6		• WASHER, FLAT		2

- ITEM NOT ILLUSTRATED

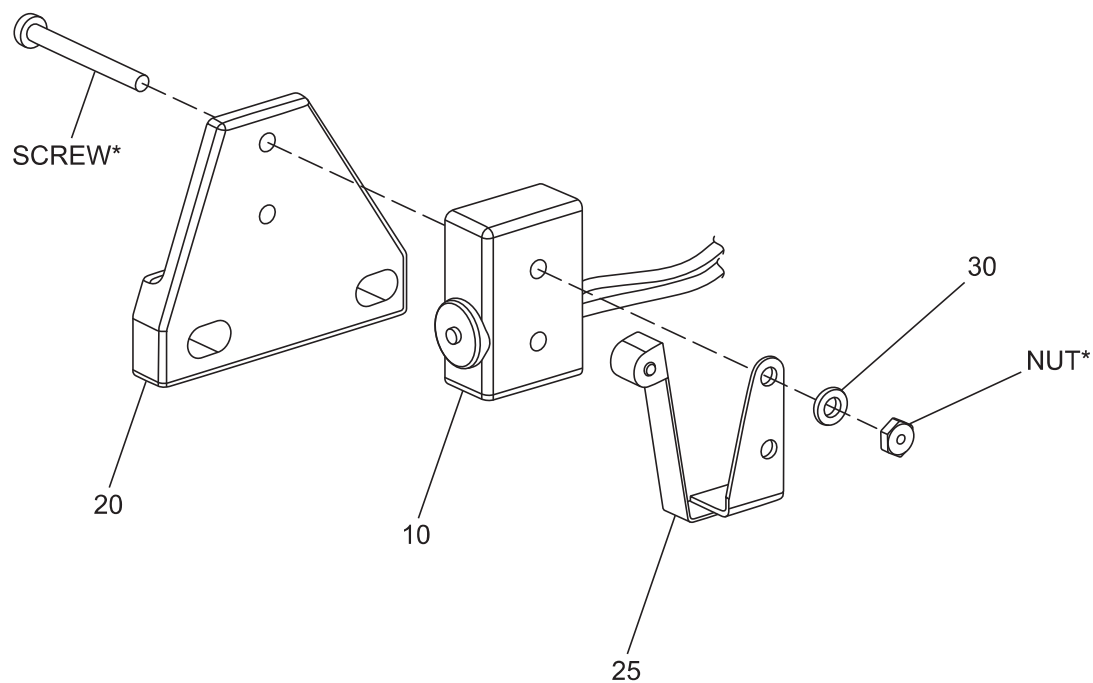
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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
10					
80	MS21043-06		• NUT, LOCK * * *		2
-85	3520-0228-01		• TUBE ASSEMBLY ATTACHING PARTS		1
90	467-0912-1		• CLIP * * *		1
95	010-3077		• • O-RING (V25184) (B/E AEROSPACE STOCK CODE OR-0571) (SUPSD BY ITEM 95A)		2
-95A	S11382-010		• • O-RING (V4J413) (B/E AEROSPACE STOCK CODE OR-0571) (SUPSDS ITEM 95) (TPN S1138 2-010)		2
100	467-0865-1		• • FITTING, BREW		1
105	3520-0228-03		• • TUBE		1
110	7.0-505R		• • CLAMP (V25281) (B/E AEROSPACE STOCK CODE CL-50010) (SUPSD BY ITEM 110A)		1
-110A	16700983		• • CLAMP (V25281) (B/E AEROSPACE STOCK CODE CL-50010) (SUPSDS ITEM 110)		1
115	SS405-3		• • INSERT (V02570) (B/E AEROSPACE STOCK CODE PF-202)		1
120	62-0066-1		• WASHER, FLOW		1

- ITEM NOT ILLUSTRATED

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*SUPPLIED WITH ITEM 25

IPL Figure 11
Micro Switch Assembly

25-30-50

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
11					
-1	411-0721-17		MICRO SWITCH ASSEMBLY (S7) (SEE IPL FIGURE 2 FOR NHA)		RF
-5	411-0721-19		• SWITCH SUBASSEMBLY		1
10	1SE3-3		• • SWITCH (V91929) (B/E AEROSPACE STOCK CODE SW-520)		1
-15	47439-000LF		• • TERMINAL STRIP, MINI PV (V30682) (B/E AEROSPACE STOCK CODE TE-50052)		2
20	411-0720-1		• SWITCH, MOUNTING PLATE		1
25	JE5		• SWITCH, ACTUATOR (0.560 LG) (V83803) (TPN JE-5) (B/E AEROSPACE STOCK CODE SW-737)		1
			ATTACHING PARTS		
30	MS35338-134		• WASHER, LOCK (NO. 2 SPLIT CRESCENT)		2
			* * *		
			<u>NOTE:</u> SCREW AND NUT ARE SUPPLIED WITH ITEM 25. STAR WASHER SUPPLIED WITH ITEM 25 TO BE REPLACED BY ITEM 30.		

- ITEM NOT ILLUSTRATED

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